

Master Codebase Document (All Files)

Project: T&C; Red Flag Scanner | Team: CorruptX

File: backend/app.py

Size: 9620 bytes

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1 | """
2 | T&C Red Flag Scanner - Flask Server
3 | Team: CorruptX
4 | Cybersecurity Hackathon Platform
5 | """
6 |
7 | import os
8 | import sys
9 | import json
10 | from flask import Flask, request, jsonify, send_from_directory, render_template_string
11 | from flask_cors import CORS
12 | from bs4 import BeautifulSoup
13 |
14 | from analyzer import ClauseAnalyzer
15 |
16 | # Setup Flask application
17 | BASE_DIR = os.path.dirname(os.path.abspath(__file__))
18 | FRONTEND_DIR = os.path.abspath(os.path.join(BASE_DIR, "..", "frontend"))
19 |
20 | app = Flask(__name__, static_folder=FRONTEND_DIR, static_url_path="")
21 | CORS(app) # Allow cross-origin requests for local dev / dual-mode hosting
22 |
23 | # Initialize Analyzer engine
24 | analyzer = ClauseAnalyzer(os.path.join(BASE_DIR, "risk_rules.json"))
25 |
26 | # Load Sample Policies
27 | SAMPLE_POLICIES_PATH = os.path.join(BASE_DIR, "sample_policies.json")
28 | try:
29 |     with open(SAMPLE_POLICIES_PATH, "r", encoding="utf-8") as f:
30 |         SAMPLE_POLICIES = json.load(f).get("samples", [])
31 | except Exception as e:
32 |     print(f"Warning: Could not load sample policies: {e}")
33 |     SAMPLE_POLICIES = []
34 |
35 |
36 | # -----
37 | # FRONTEND STATIC ROUTES
38 | # -----
39 |
40 | @app.route("/")
41 | def serve_index():
42 |     return send_from_directory(FRONTEND_DIR, "index.html")
43 |
44 | @app.route("/scanner")
45 | def serve_scanner():
46 |     return send_from_directory(FRONTEND_DIR, "scanner.html")
47 |
48 | @app.route("/results")
49 | def serve_results():
50 |     return send_from_directory(FRONTEND_DIR, "results.html")
51 |
52 | @app.route("/history")
53 | def serve_history():
54 |     return send_from_directory(FRONTEND_DIR, "history.html")
55 |
56 | @app.route("/about")
57 | def serve_about():
58 |     return send_from_directory(FRONTEND_DIR, "about.html")
59 |
60 |

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61 | # -----
62 | # REST API ENDPOINTS
63 | # -----
64 |
65 | @app.route("/api/health", methods=["GET"])
66 | def health_check():
67 |     """Health & capability probe."""
68 |     return jsonify({
69 |         "status": "healthy",
70 |         "app": "T&C Red Flag Scanner",
71 |         "team": "CorruptX",
72 |         "version": "1.0.0",
73 |         "offline_capable": True,
74 |         "engine": "Modular Cybersecurity Rule-Based NLP",
75 |         "active_categories": len(analyzer.categories)
76 |     })
77 |
78 | @app.route("/api/rules", methods=["GET"])
79 | def get_rules():
80 |     """Return active risk taxonomy and scoring rules."""
81 |     return jsonify(analyzer.rules_data)
82 |
83 | @app.route("/api/demo", methods=["GET"])
84 | def get_demo_policies():
85 |     """Return realistic demo policies for instant hackathon evaluation."""
86 |     return jsonify({
87 |         "status": "success",
88 |         "count": len(SAMPLE_POLICIES),
89 |         "samples": SAMPLE_POLICIES
90 |     })
91 |
92 | @app.route("/api/scan", methods=["POST"])
93 | def scan_policy():
94 |     """
95 |     Main analysis endpoint.
96 |     Accepts JSON body: { "text": "...", "document_name": "..." }
97 |     or multipart file upload with key 'file' and optional 'document_name'.
98 |     """
99 |     try:
100 |         raw_text = ""
101 |         doc_name = "Submitted Document"
102 |
103 |         # 1. Check if multipart file upload
104 |         if "file" in request.files:
105 |             uploaded_file = request.files["file"]
106 |
107 |             filename = uploaded_file.filename or "uploaded_policy.txt"
108 |             doc_name = request.form.get("document_name", filename)
109 |
110 |             content_bytes = uploaded_file.read()
111 |             # Try utf-8 decoding, fallback to latin-1
112 |             try:
113 |                 file_content = content_bytes.decode("utf-8")
114 |             except UnicodeDecodeError:
115 |                 file_content = content_bytes.decode("latin-1", errors="ignore")
116 |
117 |             # If HTML, strip tags cleanly with BeautifulSoup
118 |             if filename.lower().endswith((".html", ".htm")):
119 |                 soup = BeautifulSoup(file_content, "html.parser")
120 |                 # Remove scripts and styles
121 |                 for script in soup(["script", "style", "meta", "noscript"]):
122 |                     script.extract()
123 |                 raw_text = soup.get_text(separator="\n\n")
124 |             else:
125 |                 raw_text = file_content
126 |
127 |         # 2. Check if JSON payload
128 |         elif request.is_json:
129 |             data = request.get_json(silent=True) or {}
130 |             raw_text = data.get("text", "")

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130 |         doc_name = data.get("document_name", "Pasted Policy")
131 |
132 |     # 3. Check form data
133 |     elif request.form:
134 |         raw_text = request.form.get("text", "")
135 |         doc_name = request.form.get("document_name", "Submitted Policy")
136 |
137 |     # Edge case: Empty input
138 |     if not raw_text or len(raw_text.strip()) == 0:
139 |         return jsonify({
140 |             "error": "Input text is empty. Please provide Terms & Conditions text or upload a document.",
141 |             "risk_score": 0,
142 |             "risk_level": "SAFE",
143 |             "clauses_analyzed": 0,
144 |             "red_flags": 0
145 |         }), 400
146 |
147 |     # Edge case: Extremely short text
148 |     if len(raw_text.strip()) < 15:
149 |         return jsonify({
150 |             "error": "The supplied text is too short to perform a meaningful Terms & Conditions risk assessment.",
151 |             "risk_score": 0,
152 |             "risk_level": "SAFE",
153 |             "clauses_analyzed": 0,
154 |             "red_flags": 0
155 |         }), 400
156 |
157 |     # Safety limit for hackathon demo (e.g. max 500,000 chars)
158 |     if len(raw_text) > 500000:
159 |         raw_text = raw_text[:500000]
160 |
161 |     # Perform analysis
162 |     result = analyzer.scan(raw_text, doc_name)
163 |     return jsonify(result), 200
164 |
165 | except Exception as e:
166 |     print(f"Error during scan: {str(e)}", file=sys.stderr)
167 |     return jsonify({
168 |         "error": f"Internal scan engine exception: {str(e)}",
169 |         "risk_score": 0,
170 |         "risk_level": "ERROR"
171 |     }), 500
172 |
173 |
174 | @app.route("/api/chat", methods=["POST"])
175 | def chat_assistant():
176 |     """
177 |     AI Privacy Assistant Chatbot Endpoint.
178 |     Accepts JSON body: { "message": "...", "active_scan": {...} }
179 |     """
180 |     try:
181 |         data = request.get_json(silent=True) or {}
182 |         user_msg = data.get("message", "").strip()
183 |         active_scan = data.get("active_scan")
184 |
185 |         if not user_msg:
186 |             return jsonify({"reply": "Please ask a question about Terms & Conditions or digital privacy."}), 400
187 |
188 |         msg_lower = user_msg.lower()
189 |
190 |         # Contextual question about active scan
191 |         if active_scan and any(w in msg_lower for w in ["my document", "my policy", "scan", "scanned", "results", "red f
192 |             doc_name = active_scan.get("document_name", "Uploaded Policy")
193 |             score = active_scan.get("risk_score", 0)
194 |             level = active_scan.get("risk_level", "SAFE")
195 |             flagged = active_scan.get("flagged_clauses", [])
196 |
197 |             if "sell" in msg_lower or "sharing" in msg_lower:
198 |                 data_sell = [c for c in flagged if any(m.get("category_id") in ["DATA_SELLING", "THIRD_PARTY_SHARING"] f

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199 |         if data_sell:
200 |             snippet = data_sell[0].get("text", "")[:150]
201 |             reply = f"■ &lt;b&gt;Data Selling Warning for '{doc_name}':&lt;/b&gt;&lt;br&gt;&lt;br&gt;Clause ma
202 |         else:
203 |             reply = f"■ &lt;b&gt;No explicit Data Selling clauses&lt;/b&gt; detected in &lt;b&gt;'{doc_name}'&lt;/b&gt;
204 |             return jsonify({"reply": reply})
205 |
206 |         reply = f"■ &lt;b&gt;Analysis Summary for '{doc_name}':&lt;/b&gt;&lt;br&gt;• &lt;b&gt;Risk Score:&lt;/b&gt;
207 |         return jsonify({"reply": reply})
208 |
209 |     # General Knowledge Answers
210 |     if "arbitration" in msg_lower or "sue" in msg_lower or "court" in msg_lower:
211 |         reply = "■ &lt;b&gt;Mandatory Arbitration Clauses&lt;/b&gt; strip away your constitutional right to take a
212 |     elif "sell" in msg_lower or "monetiz" in msg_lower or "third party" in msg_lower:
213 |         reply = "■ &lt;b&gt;Data Selling Clauses&lt;/b&gt; allow companies to package and monetize your personal det
214 |     elif "gdpr" in msg_lower or "ccpa" in msg_lower or "right" in msg_lower:
215 |         reply = "■ &lt;b&gt;Digital Privacy Laws (GDPR/CCPA):&lt;/b&gt; You have the legal right to access your da
216 |     else:
217 |         reply = f"I reviewed your question regarding &lt;b&gt;'{user_msg}'&lt;/b&gt;.&lt;br&gt;&lt;br&gt;Always look
218 |
219 |     return jsonify({"reply": reply}), 200
220 |
221 | except Exception as e:
222 |     return jsonify({"reply": f"Sorry, I encountered an error processing your query: {str(e)}"}), 500
223 |
224 |
225 | # -----
226 | # ENTRY POINT
227 | # -----
228 |
229 | if __name__ == "__main__":
230 |     port = int(os.environ.get("PORT", 5000))
231 |     print("=" * 60)
232 |     print(" CORRUPTX - T&C RED FLAG SCANNER")
233 |     print(" Cybersecurity Hackathon Engine")
234 |     print(f" Serving on http://127.0.0.1:{port}")
235 |     print(" Offline capable: Yes (zero external cloud dependencies)")
236 |     print("=" * 60)
237 |     app.run(host="0.0.0.0", port=port, debug=True)

```

File: backend/analyzer.py

Size: 17659 bytes

```

1 | """
2 | T&C Red Flag Scanner - Analysis Engine
3 | Team: CorruptX
4 | Modular rule-based NLP and risk taxonomy engine for Terms & Conditions, Privacy Policies, and EULAs.
5 | Designed for 100% offline capability and drop-in extensibility for future LLM pipelines.
6 | """
7 |
8 | import re
9 | import json
10 | import os
11 | from datetime import datetime
12 | from typing import Dict, List, Any, Optional
13 |
14 | class ClauseAnalyzer:
15 |     """
16 |     Modular analysis engine that segments legal text into clauses,
17 |     matches them against a cybersecurity risk taxonomy, and computes
18 |     a deterministic 0-100 risk score with plain-English security explanations.
19 |     """
20 |
21 |     def __init__(self, rules_path: Optional[str] = None):
22 |         if rules_path is None:
23 |             # Default to adjacent risk_rules.json
24 |             current_dir = os.path.dirname(os.path.abspath(__file__))
25 |             rules_path = os.path.join(current_dir, "risk_rules.json")
26 |
27 |             self.rules_path = rules_path
28 |             self.rules_data = self._load_rules()
29 |             self.categories = self.rules_data.get("categories", [])
30 |             self._compile_patterns()
31 |
32 |     def _load_rules(self) -> Dict[str, Any]:
33 |         """Load the cybersecurity taxonomy from JSON."""
34 |         if not os.path.exists(self.rules_path):
35 |             raise FileNotFoundError(f"Risk rules taxonomy file not found: {self.rules_path}")
36 |
37 |         with open(self.rules_path, "r", encoding="utf-8") as f:
38 |             return json.load(f)
39 |
40 |     def _compile_patterns(self):
41 |         """Pre-compile regex patterns for high-throughput clause evaluation."""
42 |         self.compiled_rules = []
43 |         for cat in self.categories:
44 |             compiled_patterns = []
45 |             for pattern_str in cat.get("patterns", []):
46 |                 try:
47 |                     compiled_patterns.append(re.compile(pattern_str, re.IGNORECASE))
48 |                 except re.error as e:
49 |                     print(f"Warning: Invalid regex in category {cat.get('id')}: {pattern_str} ({e})")
50 |
51 |             self.compiled_rules.append({
52 |                 "id": cat.get("id"),
53 |                 "name": cat.get("name"),
54 |                 "severity": cat.get("severity", "LOW"),
55 |                 "weight": cat.get("weight", 5),
56 |                 "color": cat.get("color", "#38bdf8"),
57 |                 "explanation": cat.get("explanation", ""),
58 |                 "why_it_matters": cat.get("why_it_matters", ""),
59 |                 "recommendation": cat.get("recommendation", ""),
60 |                 "keywords": [k.lower() for k in cat.get("keywords", [])],
61 |                 "patterns": compiled_patterns
62 |             })
63 |
64 |     def clean_text(self, raw_text: str) -> str:
65 |         """Sanitize and normalize text formatting without losing legal sentence boundaries."""
66 |         if not raw_text:
67 |             return ""

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68 |         # Replace smart quotes and dashes
69 |         cleaned = raw_text.replace("'", "\"").replace("\"", "\"")
70 |         cleaned = cleaned.replace("'", "'").replace("'", "'")
71 |         cleaned = cleaned.replace("-", " - ").replace("-", " - ")
72 |
73 |         # Normalize carriage returns and excessive tabs
74 |         cleaned = cleaned.replace("\r\n", "\n").replace("\r", "\n")
75 |
76 |         # Remove null bytes or non-printable control characters
77 |         cleaned = re.sub(r"[\x00-\x08\x0b\x0c\x0e-\x1f]", "", cleaned)
78 |
79 |         return cleaned.strip()
80 |
81 | def segment_clauses(self, text: str) -> List[str]:
82 |     """
83 |     Segment policy text into distinct clauses or sentences while preserving
84 |     numbered sections (e.g. '1.1', 'Section 4(a)') and common abbreviations.
85 |     """
86 |     cleaned = self.clean_text(text)
87 |     if not cleaned:
88 |         return []
89 |
90 |     # Mask known abbreviations that contain periods
91 |     abbr_map = {
92 |         r"\be\.g\.\.b": "EG_TEMP_MARKER",
93 |         r"\bi\.e\.\.b": "IE_TEMP_MARKER",
94 |         r"\bU\.S\.\.b": "US_TEMP_MARKER",
95 |         r"\betc\.\.b": "ETC_TEMP_MARKER",
96 |         r"\bInc\.\.b": "INC_TEMP_MARKER",
97 |         r"\bLtd\.\.b": "LTD_TEMP_MARKER",
98 |         r"\bLLC\.\.b": "LLC_TEMP_MARKER",
99 |         r"\bSec\.\.b": "SEC_TEMP_MARKER",
100 |         r"\bNo\.\.b": "NO_TEMP_MARKER",
101 |         r"\bv\.\.b": "V_TEMP_MARKER",
102 |         r"\bvs\.\.b": "VS_TEMP_MARKER"
103 |     }
104 |
105 |     masked = cleaned
106 |     for regex_pat, replacement in abbr_map.items():
107 |         masked = re.sub(regex_pat, replacement, masked, flags=re.IGNORECASE)
108 |
109 |     # Protect section numbering like 1.1 or 3.2.1
110 |     masked = re.sub(r"(<=d)\.(<=d)", "DECIMAL_DOT_MARKER", masked)
111 |
112 |     # Split on double newlines (paragraphs/clauses) or sentence-ending punctuation followed by space/newline
113 |     # or numbered list items like \n1. or \n(a)
114 |     chunks = re.split(r"(?:\n\s*\n|(<=[.!?])\s+(?=[A-Z0-9\"'])|(<=\\n)\\s*(?=[0-9]+\\.\\s|[A-Z]\\.\\s|\\([a-z0-9]+)+)",
115 |
116 |     clauses = []
117 |     for chunk in chunks:
118 |         # Restore markers
119 |         restored = chunk
120 |         for regex_pat, replacement in abbr_map.items():
121 |             marker_target = replacement
122 |             original_sample = regex_pat.replace(r"\b", "").replace("\\", "")
123 |             restored = restored.replace(marker_target, original_sample)
124 |
125 |             restored = restored.replace("DECIMAL_DOT_MARKER", ".")
126 |             restored = re.sub(r"\s+", " ", restored).strip()
127 |
128 |             # Avoid single stray numbers or trivial fragments
129 |             if len(restored) >= 12 or re.search(r"[a-zA-Z]{3,}", restored):
130 |                 clauses.append(restored)
131 |
132 |     # If splitting resulted in no valid clauses but text was present, use full text
133 |     if not clauses and len(cleaned) > 0:
134 |         clauses = [cleaned]
135 |
136 |     return clauses

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137 |
138 | def is_negated_protection(self, clause_text: str, category_id: str) -> bool:
139 |     """
140 |     Detect if the clause explicitly promises protection via negation
141 |     (e.g., 'we do not sell your data', 'we do not use device fingerprinting', 'never track').
142 |     Protective clauses should not be flagged as red flags.
143 |     """
144 |     clause_lower = clause_text.lower()
145 |
146 |     # Specific negative assertions by category that represent protections rather than risks
147 |     negation_rules = {
148 |         "data_sharing": [
149 |             r"(?:we|company|our\s+service)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:sell|share|disclose|rent|
150 |             r"no\s+(?:third-party\s+sale|selling|sharing\s+with\s+third\s+parties)",
151 |             r"(?:we\s+do\s+not|never)\s+disclose\s+personal\s+data"
152 |         ],
153 |         "tracking": [
154 |             r"(?:we|company|our\s+service)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:track|use\s+tracking|recor
155 |             r"(?:no|zero)\s+(?:tracking|fingerprinting|telemetry|session\s+replay)",
156 |             r"do\s+not\s+use\s+(?:tracking\s+cookies|device\s+fingerprinting|analytics\s+pixels)"
157 |         ],
158 |         "data_collection": [
159 |             r"(?:we|company)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+collect\s+(?:biometric|location|sensitive|
160 |             r"do\s+not\s+collect\s+location\s+data"
161 |         ],
162 |         "location_tracking": [
163 |             r"(?:we|company)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:collect|track)\s+(?:your\s+)?location",
164 |             r"do\s+not\s+collect\s+location\s+data"
165 |         ],
166 |         "advertising_profiling": [
167 |             r"no\s+advertising\s+networks\s+are\s+embedded",
168 |             r"(?:we|company)\s+(?:do\s+not|will\s+not|never)\s+(?:serve\s+targeted|profile|sell\s+ads)"
169 |         ],
170 |         "arbitration": [
171 |             r"(?:we|company)\s+(?:do\s+not|will\s+not)\s+enforce\s+(?:mandatory\s+)?(?:binding\s+)?arbitration",
172 |             r"do\s+not\s+enforce\s+mandatory\s+binding\s+arbitration"
173 |         ]
174 |     }
175 |
176 |     cat_negations = negation_rules.get(category_id, [])
177 |     for pat in cat_negations:
178 |         if re.search(pat, clause_lower):
179 |             return True
180 |     return False
181 |
182 | def analyze_clause(self, clause_text: str, index: int) -> Dict[str, Any]:
183 |     """
184 |     Evaluate a single clause against the cybersecurity rule taxonomy.
185 |     Returns matched risk details or clean status.
186 |     """
187 |     clause_lower = clause_text.lower()
188 |     matched_categories = []
189 |
190 |     for rule in self.compiled_rules:
191 |         # First check if this clause is explicitly a privacy protection for this category
192 |         if self.is_negated_protection(clause_text, rule["id"]):
193 |             continue
194 |
195 |         matched_flag = False
196 |         matched_pattern_str = None
197 |
198 |         # 1. Test regex patterns
199 |         for pattern in rule["patterns"]:
200 |             match = pattern.search(clause_text)
201 |             if match:
202 |                 matched_flag = True
203 |                 matched_pattern_str = match.group(0)
204 |                 break
205 |

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206 |         # 2. Test keywords if regex didn't trigger
207 |         if not matched_flag:
208 |             for kw in rule["keywords"]:
209 |                 if kw in clause_lower:
210 |                     matched_flag = True
211 |
212 |                     matched_pattern_str = kw
213 |                     break
214 |
215 |         if matched_flag:
216 |             matched_categories.append({
217 |                 "category_id": rule["id"],
218 |                 "category_name": rule["name"],
219 |                 "severity": rule["severity"],
220 |                 "weight": rule["weight"],
221 |                 "color": rule["color"],
222 |                 "explanation": rule["explanation"],
223 |                 "why_it_matters": rule["why_it_matters"],
224 |                 "recommendation": rule["recommendation"],
225 |                 "snippet": matched_pattern_str
226 |             })
227 |
228 |     severity_rank = {"HIGH": 3, "MEDIUM": 2, "LOW": 1}
229 |
230 |     if matched_categories:
231 |         # Sort by severity weight descending
232 |         matched_categories.sort(
233 |             key=lambda x: (severity_rank.get(x["severity"], 0), x["weight"]),
234 |             reverse=True
235 |         )
236 |         top_match = matched_categories[0]
237 |
238 |         return {
239 |             "index": index,
240 |             "text": clause_text,
241 |             "is_flagged": True,
242 |             "category_id": top_match["category_id"],
243 |             "category_name": top_match["category_name"],
244 |             "severity": top_match["severity"],
245 |             "weight": top_match["weight"],
246 |             "color": top_match["color"],
247 |             "explanation": top_match["explanation"],
248 |             "why_it_matters": top_match["why_it_matters"],
249 |             "recommendation": top_match["recommendation"],
250 |             "all_matched_categories": [m["category_name"] for m in matched_categories]
251 |         }
252 |     else:
253 |         return {
254 |             "index": index,
255 |             "text": clause_text,
256 |             "is_flagged": False,
257 |             "category_id": None,
258 |             "category_name": "Standard Clause",
259 |             "severity": "SAFE",
260 |             "weight": 0,
261 |             "color": "#00ff88",
262 |             "explanation": "No elevated risk indicators detected in this clause.",
263 |             "why_it_matters": "Standard service provision or routine operational statement.",
264 |             "recommendation": "Clause appears customary; verify context if unsure.",
265 |             "all_matched_categories": []
266 |         }
267 |
268 | def calculate_risk_score(self, total_clauses: int, flagged_clauses: List[Dict[str, Any]]) -> int:
269 |     """
270 |     Compute a reproducible 0-100 risk score based on:
271 |     - Quantity and severity weights of detected red flags
272 |     - Risk density relative to total document clauses
273 |     - Guardrail thresholds (e.g. presence of multiple HIGH risks)
274 |     """
275 |     if total_clauses <= 0 or not flagged_clauses:

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275 |         return 5 # Baseline minimal score for empty/neutral policies
276 |
277 |     high_count = sum(1 for c in flagged_clauses if c["severity"] == "HIGH")
278 |     medium_count = sum(1 for c in flagged_clauses if c["severity"] == "MEDIUM")
279 |     low_count = sum(1 for c in flagged_clauses if c["severity"] == "LOW")
280 |
281 |     # Base weighted sum
282 |     weighted_sum = (high_count * 16) + (medium_count * 8) + (low_count * 4)
283 |
284 |     # Density factor: proportion of document containing red flags (up to 20 pts)
285 |     density = len(flagged_clauses) / max(total_clauses, 1)
286 |     density_bonus = min(20, int(density * 35))
287 |
288 |     # Composite score
289 |     raw_score = int((weighted_sum * 0.82) + density_bonus)
290 |
291 |     # Guardrails for high-severity issues
292 |     if high_count >= 3:
293 |         raw_score = max(raw_score, 68)
294 |     elif high_count >= 1:
295 |         raw_score = max(raw_score, 42)
296 |     elif medium_count >= 2:
297 |         raw_score = max(raw_score, 35)
298 |
299 |     # Final bounding to 0-100
300 |     score = max(0, min(100, raw_score))
301 |     return score
302 |
303 | def determine_risk_level(self, score: int) -> str:
304 |     """Map 0-100 numerical score to formal risk tier."""
305 |     if score <= 30:
306 |         return "SAFE"
307 |     elif score <= 60:
308 |         return "CAUTION"
309 |     else:
310 |         return "HIGH RISK"
311 |
312 | def scan(self, text: str, document_name: str = "Submitted Policy") -> Dict[str, Any]:
313 |     """
314 |     Execute full end-to-end policy scan.
315 |     Returns complete structured analysis ready for JSON transmission.
316 |     """
317 |     if not text or len(text.strip()) == 0:
318 |         return {
319 |             "error": "Input text is empty. Please paste or upload a policy to scan.",
320 |             "risk_score": 0,
321 |             "risk_level": "SAFE",
322 |             "clauses_analyzed": 0,
323 |             "red_flags": 0
324 |         }
325 |
326 |     clauses = self.segment_clauses(text)
327 |     total_clauses = len(clauses)
328 |
329 |     analyzed_clauses = []
330 |     flagged_clauses = []
331 |
332 |     category_counts: Dict[str, Dict[str, Any]] = {}
333 |
334 |     for idx, clause in enumerate(clauses):
335 |         res = self.analyze_clause(clause, idx)
336 |         analyzed_clauses.append(res)
337 |         if res["is_flagged"]:
338 |             flagged_clauses.append(res)
339 |             cat_id = res["category_id"]
340 |             if cat_id not in category_counts:
341 |                 category_counts[cat_id] = {
342 |                     "category_id": cat_id,
343 |                     "category_name": res["category_name"],

```

```

344 |         "severity": res["severity"],
345 |         "color": res["color"],
346 |         "count": 0,
347 |         "clauses": []
348 |     }
349 |     category_counts[cat_id]["count"] += 1
350 |     category_counts[cat_id]["clauses"].append(res["index"])
351 |
352 |     score = self.calculate_risk_score(total_clauses, flagged_clauses)
353 |     level = self.determine_risk_level(score)
354 |
355 |     severity_counts = {
356 |         "HIGH": sum(1 for c in flagged_clauses if c["severity"] == "HIGH"),
357 |         "MEDIUM": sum(1 for c in flagged_clauses if c["severity"] == "MEDIUM"),
358 |         "LOW": sum(1 for c in flagged_clauses if c["severity"] == "LOW")
359 |     }
360 |
361 |     # Format sorted categories for breakdown charts
362 |     sorted_categories = sorted(
363 |         list(category_counts.values()),
364 |         key=lambda x: (x["count"], 3 if x["severity"] == "HIGH" else (2 if x["severity"] == "MEDIUM" else 1)),
365 |         reverse=True
366 |     )
367 |
368 |     summary_msg = self._generate_summary(score, level, severity_counts, total_clauses)
369 |
370 |     return {
371 |         "document_name": document_name,
372 |         "analyzed_at": datetime.now().isoformat(),
373 |         "risk_score": score,
374 |         "risk_level": level,
375 |         "summary": summary_msg,
376 |         "clauses_analyzed": total_clauses,
377 |         "red_flags": len(flagged_clauses),
378 |         "severity_counts": severity_counts,
379 |         "categories": sorted_categories,
380 |         "flagged_clauses": flagged_clauses,
381 |         "all_clauses": analyzed_clauses,
382 |         "disclaimer": "This tool provides automated cybersecurity risk analysis for informational purposes and is not intended for legal advice."
383 |     }
384 |
385 | def _generate_summary(self, score: int, level: str, severities: Dict[str, int], total_clauses: int) -> str:
386 |     """Construct a high-level executive security summary."""
387 |     high = severities.get("HIGH", 0)
388 |     med = severities.get("MEDIUM", 0)
389 |     low = severities.get("LOW", 0)
390 |     total_flags = high + med + low
391 |
392 |     if level == "HIGH RISK":
393 |         return (f"CRITICAL PRIVACY & LEGAL EXPOSURE DETECTED. Analysis identified {total_flags} red flags "
394 |                 f"({high} High-Risk clauses) across {total_clauses} clauses. This policy contains terms that may "
395 |                 f"severely compromise personal data privacy, waive legal remedies, or grant unilateral vendor control.")
396 |     elif level == "CAUTION":
397 |         return (f"MODERATE USER EXPOSURE DETECTED. Identified {total_flags} flagged clauses ({high} High, {med} Medium, "
398 |                 f"{low} Low) out of {total_clauses} total clauses. Contains clauses such as recurring billing, data logging, "
399 |                 f"or liability disclaimers that warrant conscious review before agreeing.")
400 |     else:
401 |         return (f"ACCEPTABLE RISK PROFILE. Minimal legal or data exposure detected across {total_clauses} clauses. "
402 |                 f"Only {total_flags} minor items noted. Policy adheres to standard or user-friendly privacy practices.")

```

File: backend/test_server.py

Size: 4935 bytes

```

1 | """
2 | Automated Test Suite for T&C Red Flag Scanner
3 | Tests Flask API endpoints, static serving, input validation, and scoring consistency.
4 | """
5 |
6 | import unittest
7 | import json
8 | import os
9 | from app import app, analyzer
10 |
11 | class TestTCRedFlagScanner(unittest.TestCase):
12 |     @classmethod
13 |     def setUpClass(cls):
14 |         cls.client = app.test_client()
15 |
16 |     def test_health_check(self):
17 |         """Verify /api/health returns healthy and correct taxonomy size."""
18 |         response = self.client.get("/api/health")
19 |         self.assertEqual(response.status_code, 200)
20 |         data = response.get_json()
21 |         self.assertEqual(data.get("status"), "healthy")
22 |         self.assertEqual(data.get("team"), "CorruptX")
23 |         self.assertEqual(data.get("offline_capable"), True)
24 |         self.assertEqual(data.get("active_categories"), 14)
25 |
26 |     def test_rules_endpoint(self):
27 |         """Verify /api/rules returns all 14 categories."""
28 |         response = self.client.get("/api/rules")
29 |         self.assertEqual(response.status_code, 200)
30 |         data = response.get_json()
31 |         self.assertIn("categories", data)
32 |         self.assertEqual(len(data["categories"]), 14)
33 |
34 |     def test_demo_endpoint(self):
35 |         """Verify /api/demo returns preloaded policies."""
36 |
37 |         response = self.client.get("/api/demo")
38 |         self.assertEqual(response.status_code, 200)
39 |         data = response.get_json()
40 |         self.assertEqual(data.get("status"), "success")
41 |         self.assertGreaterEqual(len(data.get("samples", [])), 3)
42 |
43 |     def test_scan_empty_input(self):
44 |         """Verify /api/scan rejects empty input with friendly error."""
45 |         response = self.client.post("/api/scan", json={"text": "", "document_name": "Empty"})
46 |         self.assertEqual(response.status_code, 400)
47 |         data = response.get_json()
48 |         self.assertIn("error", data)
49 |
50 |     def test_scan_too_short_input(self):
51 |         """Verify /api/scan rejects input that is too short."""
52 |         response = self.client.post("/api/scan", json={"text": "Hello world", "document_name": "Short"})
53 |         self.assertEqual(response.status_code, 400)
54 |         data = response.get_json()
55 |         self.assertIn("error", data)
56 |
57 |     def test_scan_high_risk_policy(self):
58 |         """Verify scan on high-risk clauses detects categories and assigns HIGH RISK."""
59 |         high_risk_text = (
60 |             "We reserve the right to share your personal information with third parties. "
61 |             "We collect information about your activity across third-party websites to build demographic profiles. "
62 |             "We track your physical location even when the app is closed. "
63 |             "Subscriptions automatically renew unless cancelled. "
64 |             "All disputes shall be resolved exclusively through confidential binding arbitration. "
65 |             "You waive any right to participate in a class action. "
66 |             "We reserve the right to modify these terms at any time without prior notice."
67 |         )
68 |         response = self.client.post("/api/scan", json={

```

```

68 |         "text": high_risk_text,
69 |         "document_name": "Test High Risk"
70 |     })
71 |     self.assertEqual(response.status_code, 200)
72 |     data = response.get_json()
73 |     self.assertEqual(data.get("risk_level"), "HIGH RISK")
74 |     self.assertGreaterEqual(data.get("risk_score"), 68)
75 |     self.assertGreaterEqual(data.get("red_flags"), 5)
76 |     self.assertGreaterEqual(data.get("severity_counts", {}).get("HIGH", 0), 3)
77 |
78 |     def test_scan_safe_policy_with_negation(self):
79 |         """Verify scan on privacy-protective clauses with negation returns SAFE."""
80 |         safe_text = (
81 |             "AegisGuard operates as a local-first software utility. "
82 |             "We do not use tracking cookies, device fingerprinting, analytics pixels, or session replay scripts. "
83 |             "We do not collect location data, contact lists, or background sensor information. "
84 |             "We do not sell, rent, monetize, or disclose personal data to advertisers. "
85 |             "You retain full ownership and intellectual property rights in all data. "
86 |             "We do not enforce mandatory binding arbitration or prohibit collective remedies."
87 |         )
88 |         response = self.client.post("/api/scan", json={
89 |             "text": safe_text,
90 |             "document_name": "Test Safe"
91 |         })
92 |         self.assertEqual(response.status_code, 200)
93 |         data = response.get_json()
94 |         self.assertEqual(data.get("risk_level"), "SAFE")
95 |         self.assertLessEqual(data.get("risk_score"), 30)
96 |
97 |     def test_static_routes(self):
98 |         """Verify static pages are served correctly."""
99 |         routes = ["/", "/scanner", "/results", "/history", "/about"]
100 |         for route in routes:
101 |             response = self.client.get(route)
102 |             self.assertEqual(response.status_code, 200, f"Failed on route: {route}")
103 |             self.assertIn("CorruptX", response.get_data(as_text=True))
104 |
105 | if __name__ == "__main__":
106 |     unittest.main()

```

File: backend/test_analyzer.py**Size:** 768 bytes

```
1 | from analyzer import ClauseAnalyzer
2 | import json
3 |
4 | analyzer = ClauseAnalyzer()
5 | with open('sample_policies.json', 'r', encoding='utf-8') as f:
6 |     samples = json.load(f)['samples']
7 |
8 | print("--- TESTING CLAUSE ANALYZER ---")
9 | for sample in samples:
10 |     res = analyzer.scan(sample['content'], sample['title'])
11 |     print(f"[{res['risk_level']}] {sample['title']}")
12 |     print(f"  Score: {res['risk_score']}/100")
13 |     print(f"  Clauses: {res['clauses_analyzed']}, Red Flags: {res['red_flags']}")
14 |     print(f"  Severity: High={res['severity_counts']['HIGH']}, Med={res['severity_counts']['MEDIUM']}, Low={res['severity_counts']['LOW']}")
15 |     print(f"  Top categories: {[c['category_name'] for c in res['categories'][:3]]}")
16 |     print()
17 |
18 | print("ALL SAMPLES TESTED SUCCESSFULLY!")
```

File: backend/test_client_js.py

Size: 1401 bytes

```
1 | import js2py
2 |
3 | # Polyfill minimal browser environment
4 | context = js2py.EvalJs()
5 | context.execute("""
6 | var window = {
7 |   TC_RISK_RULES: null,
8 |   clientClauseAnalyzer: null
9 | };
10 | var console = {
11 |   warn: function() {},
12 |   log: function() {},
13 |   error: function() {}
14 | };
15 | """)
16 |
17 | with open('../frontend/js/rules_data.js', 'r', encoding='utf-8') as f:
18 |     context.execute(f.read())
19 |
20 | with open('../frontend/js/client_analyzer.js', 'r', encoding='utf-8') as f:
21 |     context.execute(f.read())
22 |
23 | sample_text = (
24 |     "We reserve the right to share your personal information with third parties. "
25 |     "We track your browsing activity and device fingerprinting across external websites. "
26 |     "All disputes shall be resolved exclusively through confidential binding arbitration. "
27 |     "You waive any right to participate in a class action. "
28 |     "We may modify these terms at any time without prior notice."
29 | )
30 |
31 | context.set('sample_text', sample_text)
32 | res = context.eval('window.clientClauseAnalyzer.scan(sample_text, "JS Offline Test")')
33 |
34 | print("--- CLIENT-SIDE JS ENGINE TEST RESULT ---")
35 | print("Document:", res['document_name'])
36 |
37 | print("Risk Score:", res['risk_score'], "/ 100")
38 | print("Risk Level:", res['risk_level'])
39 | print("Clauses Analyzed:", res['clauses_analyzed'])
40 | print("Red Flags Detected:", res['red_flags'])
41 | print("Engine Mode:", res['engine_mode'])
41 | print("SUCCESS: 100% OFFLINE CLIENT-SIDE JS ENGINE IS OPERATIONAL!")
```

File: backend/test_chatbot_and_storage.py

Size: 1546 bytes

```

1 | import unittest
2 | import json
3 | import os
4 | import sys
5 |
6 | BASE_DIR = os.path.dirname(os.path.abspath(__file__))
7 | sys.path.insert(0, BASE_DIR)
8 |
9 | from app import app
10 |
11 | class TestChatbotAndStorage(unittest.TestCase):
12 |     def setUp(self):
13 |         self.client = app.test_client()
14 |         self.client.testing = True
15 |
16 |     def test_chat_general_query(self):
17 |         response = self.client.post("/api/chat", json={
18 |             "message": "What is forced arbitration?"
19 |         })
20 |         self.assertEqual(response.status_code, 200)
21 |         data = json.loads(response.data)
22 |         self.assertIn("reply", data)
23 |         self.assertIn("Arbitration", data["reply"])
24 |
25 |     def test_chat_scanned_doc_context(self):
26 |         active_scan = {
27 |             "document_name": "Test Social App Policy",
28 |             "risk_score": 85,
29 |             "risk_level": "HIGH RISK",
30 |             "flagged_clauses": [
31 |                 {
32 |                     "text": "We may sell and monetize user data to third party advertisers.",
33 |                     "matches": [{"category_id": "DATA_SELLING", "category_name": "Data Selling"}]
34 |                 }
35 |             ]
36 |         }
37 |
38 |         response = self.client.post("/api/chat", json={
39 |             "message": "Does my scanned document sell data?",
40 |             "active_scan": active_scan
41 |         })
42 |         self.assertEqual(response.status_code, 200)
43 |         data = json.loads(response.data)
44 |         self.assertIn("Data Selling", data["reply"])
45 |         self.assertIn("Test Social App Policy", data["reply"])
46 |
47 | if __name__ == "__main__":
48 |     unittest.main()

```

File: frontend/index.html

Size: 21301 bytes

```

1 | <!DOCTYPE html>;
2 | <html lang="en">;
3 | <head>;
4 |   <meta charset="UTF-8">;
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">;
6 |   <title>CorruptX – T&C Red Flag Scanner | Cybersecurity Hackathon</title>;
7 |   <link rel="stylesheet" href="css/style.css">;
8 |   <link rel="stylesheet" href="css/responsive.css">;
9 |   <link rel="stylesheet" href="css/chatbot.css">;
10 | </style>;
11 |   /* Landing-specific styles */
12 |   .hero-section {
13 |     padding: 72px 0 64px;
14 |     position: relative;
15 |   }
16 |   .hero-grid {
17 |     display: grid;
18 |     grid-template-columns: 1.15fr 0.85fr;
19 |     gap: 48px;
20 |     align-items: center;
21 |   }
22 |   .hero-tagline {
23 |     font-size: 1.15rem;
24 |     font-weight: 700;
25 |     color: var(--accent-cyan);
26 |     font-family: var(--font-mono);
27 |     letter-spacing: 0.05em;
28 |     margin-bottom: 12px;
29 |     text-transform: uppercase;
30 |   }
31 |   .hero-title {
32 |     font-size: 3.4rem;
33 |     font-weight: 900;
34 |     line-height: 1.1;
35 |     margin-bottom: 20px;
36 |     background: linear-gradient(135deg, #ffffff 0%, #cbd5e1 50%, #38bdf8 100%);
37 |     -webkit-background-clip: text;
38 |     -webkit-text-fill-color: transparent;
39 |   }
40 |   .hero-subtitle {
41 |     font-size: 1.18rem;
42 |     line-height: 1.7;
43 |     color: #94a3b8;
44 |     margin-bottom: 32px;
45 |     max-width: 580px;
46 |   }
47 |   .hero-cta-group {
48 |     display: flex;
49 |     align-items: center;
50 |     gap: 16px;
51 |     flex-wrap: wrap;
52 |   }
53 |
54 |   /* Mockup Visual Card */
55 |   .mockup-card {
56 |     background: rgba(13, 19, 32, 0.85);
57 |     border: 1px solid var(--border-focus);
58 |     box-shadow: 0 0 40px rgba(0, 242, 255, 0.15), 0 20px 50px rgba(0,0,0,0.8);
59 |     border-radius: var(--radius-lg);
60 |     padding: 24px;
61 |     position: relative;
62 |   }
63 |   .mockup-header {
64 |     display: flex;
65 |     align-items: center;
66 |     justify-content: space-between;
67 |     margin-bottom: 20px;

```

```
68 |     padding-bottom: 12px;
69 |     border-bottom: 1px solid var(--border-card);
70 | }
71 | .mockup-title {
72 |     font-size: 0.82rem;
73 |     font-family: var(--font-mono);
74 |     color: var(--accent-cyan);
75 |     text-transform: uppercase;
76 |     letter-spacing: 0.08em;
77 | }
78 | .mockup-score-box {
79 |     display: flex;
80 |     align-items: center;
81 |     justify-content: space-between;
82 |     background: rgba(255, 51, 102, 0.08);
83 |     border: 1px solid rgba(255, 51, 102, 0.3);
84 |     border-radius: var(--radius-md);
85 |     padding: 16px 20px;
86 |     margin-bottom: 20px;
87 | }
88 | .mockup-score-num {
89 |     font-size: 2.8rem;
90 |     font-weight: 900;
91 |     color: var(--accent-crimson);
92 |     line-height: 1;
93 | }
94 | .mockup-tags-grid {
95 |     display: grid;
96 |     grid-template-columns: repeat(2, 1fr);
97 |     gap: 10px;
98 | }
99 | .mockup-tag {
100 |     background: rgba(15, 23, 42, 0.7);
101 |     border: 1px solid var(--border-card);
102 |     border-radius: var(--radius-sm);
103 |     padding: 8px 12px;
104 |     font-size: 0.76rem;
105 |     font-family: var(--font-mono);
106 |     color: #cbd5e1;
107 |     display: flex;
108 |     align-items: center;
109 |     justify-content: space-between;
110 | }
111 |
112 | /* Section Styles */
113 | .section-title-wrap {
114 |     text-align: center;
115 |     max-width: 720px;
116 |     margin: 0 auto 56px;
117 | }
118 | .section-title {
119 |     font-size: 2.2rem;
120 |     margin-bottom: 14px;
121 | }
122 | .features-grid {
123 |     display: grid;
124 |     grid-template-columns: repeat(3, 1fr);
125 |     gap: 28px;
126 |     margin-bottom: 80px;
127 | }
128 | .feature-card {
129 |     padding: 32px 28px;
130 | }
131 | .feature-icon-box {
132 |     width: 52px;
133 |     height: 52px;
134 |     border-radius: 12px;
135 |     background: rgba(0, 242, 255, 0.08);
136 |     border: 1px solid var(--border-focus);
```

```
137 |     display: flex;
138 |     align-items: center;
139 |     justify-content: center;
140 |     color: var(--accent-cyan);
141 |
142 |     margin-bottom: 20px;
143 |     box-shadow: 0 0 16px var(--accent-cyan-glow);
144 | }
145 | .feature-title {
146 |     font-size: 1.35rem;
147 |     margin-bottom: 12px;
148 |     color: #fff;
149 | }
150 | .feature-desc {
151 |     font-size: 0.94rem;
152 |     line-height: 1.65;
153 | }
154 | /* Workflow Steps */
155 | .workflow-steps-wrap {
156 |     display: grid;
157 |     grid-template-columns: repeat(5, 1fr);
158 |     gap: 16px;
159 |     margin-bottom: 80px;
160 | }
161 | .workflow-step {
162 |     background: var(--bg-card);
163 |     border: 1px solid var(--border-card);
164 |     border-radius: var(--radius-md);
165 |     padding: 20px 16px;
166 |     text-align: center;
167 |     position: relative;
168 | }
169 | .workflow-num {
170 |     font-size: 1.4rem;
171 |     font-weight: 800;
172 |     color: var(--accent-cyan);
173 |     font-family: var(--font-mono);
174 |     margin-bottom: 8px;
175 | }
176 | .workflow-label {
177 |     font-size: 0.88rem;
178 |     font-weight: 600;
179 |     color: #fff;
180 | }
181 |
182 | /* Team Section */
183 | .team-grid {
184 |     display: grid;
185 |     grid-template-columns: repeat(3, 1fr);
186 |     gap: 24px;
187 |     margin-bottom: 80px;
188 | }
189 | .team-member-card {
190 |     text-align: center;
191 |     padding: 28px 20px;
192 | }
193 | .team-avatar {
194 |     width: 72px;
195 |     height: 72px;
196 |     border-radius: 50%;
197 |     background: linear-gradient(135deg, #00d2df 0%, #ff3366 100%);
198 |     margin: 0 auto 16px;
199 |     display: flex;
200 |     align-items: center;
201 |     justify-content: center;
202 |     font-size: 1.4rem;
203 |     font-weight: 800;
204 |     color: #fff;
205 |     box-shadow: 0 0 20px var(--accent-cyan-glow);
```

```

206 |     }
207 |
208 |     @media (max-width: 900px) {
209 |         .hero-grid { grid-template-columns: 1fr; }
210 |         .features-grid { grid-template-columns: 1fr; }
211 |         .workflow-steps-wrap { grid-template-columns: 1fr 1fr; }
212 |         .team-grid { grid-template-columns: 1fr; }
213 |         .hero-title { font-size: 2.5rem; }
214 |     }
215 |     </style>
216 | </head>
217 | <body>
218 |
219 |     <!-- Site Header -->
220 |     <header class="site-header">
221 |         <div class="container header-inner">
222 |             <a href="index.html" class="brand-wrapper">
223 |                 <div class="brand-shield-icon">
224 |                     <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
225 |                         </div>
226 |                     <div class="brand-meta">
227 |                         <span class="brand-team">Team CorruptX</span>
228 |                         <span class="brand-name">T&C RED FLAG <span>SCANNER</span></span>
229 |                     </div>
230 |                 </a>
231 |
232 |                 <nav class="main-nav">
233 |                     <a href="index.html" class="nav-link">Home</a>
234 |                     <a href="scanner.html" class="nav-link">Scanner</a>
235 |                     <a href="results.html" class="nav-link">Results</a>
236 |                     <a href="history.html" class="nav-link">History</a>
237 |                     <a href="about.html" class="nav-link">About Team</a>
238 |                 </nav>
239 |
240 |                 <div class="header-actions">
241 |                     <div class="system-status" id="system-status-indicator">
242 |                         <span class="status-dot"></span>
243 |                         <span id="system-status-text">CYBER ENGINE READY</span>
244 |                     </div>
245 |                     <button class="mobile-menu-btn" aria-label="Toggle navigation">
246 |                         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
247 |                             </svg>
248 |                     </button>
249 |                 </div>
250 |             </header>
251 |
252 |     <!-- Main Content -->
253 |     <main>
254 |         <!-- Hero Section -->
255 |         <section class="hero-section">
256 |             <div class="container hero-grid">
257 |                 <div class="hero-text-block">
258 |                     <div class="hero-tagline">Before you click Agree, know what you're agreeing to.</div>
259 |                     <h1 class="hero-title">T&C RED FLAG SCANNER</h1>
260 |                     <p class="hero-subtitle">
261 |                         AI-powered analysis that finds risky clauses hidden inside Terms & Conditions, Privacy Policies, and EULAs
262 |                     </p>
263 |                     <div class="hero-cta-group">
264 |                         <a href="scanner.html" class="btn btn-primary btn-lg">Scan a Document</a>
265 |                         <a href="scanner.html?demo=high_risk_social" class="btn btn-secondary btn-lg">Try Demo Scan</a>
266 |                         <a href="#how-it-works" class="btn btn-outline">See How It Works</a>
267 |                     </div>
268 |                 </div>
269 |
270 |                 <!-- Visual Cybersecurity Dashboard Mockup -->
271 |                 <div class="mockup-card">
272 |                     <div class="mockup-header">
273 |                         <span class="mockup-title">DEMO SECURITY AUDIT MOCKUP</span>
274 |                         <span class="badge badge-high">LIVE ENGINE READY</span>

```

```

275 |         </div>;
276 |
277 |         <div class="mockup-score-box">
278 |             <div>
279 |                 <div style="font-size: 0.8rem; font-family: var(--font-mono); color: var(--text-dim); text-transform: uppercase;">
280 |                     <div style="font-size: 1.1rem; font-weight: 800; color: #fff; margin-top: 2px;">HIGH RISK DETECTED</div>
281 |                 </div>
282 |                 <div class="mockup-score-num">72</div><span style="font-size: 1.2rem; color: var(--text-dim);">/100</span>
283 |             </div>
284 |
285 |         <div style="font-size: 0.78rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 10px;">
286 |             Detected Critical Categories:
287 |         </div>
288 |
289 |         <div class="mockup-tags-grid">
290 |             <div class="mockup-tag">
291 |                 <span>DATA SHARING</span>
292 |                 <span class="badge badge-high" style="padding: 2px 6px; font-size: 0.68rem;">HIGH</span>
293 |             </div>
294 |             <div class="mockup-tag">
295 |                 <span>TRACKING</span>
296 |                 <span class="badge badge-high" style="padding: 2px 6px; font-size: 0.68rem;">HIGH</span>
297 |             </div>
298 |             <div class="mockup-tag">
299 |                 <span>AUTO-RENEWAL</span>
300 |                 <span class="badge badge-medium" style="padding: 2px 6px; font-size: 0.68rem;">MED</span>
301 |             </div>
302 |             <div class="mockup-tag">
303 |                 <span>ARBITRATION</span>
304 |                 <span class="badge badge-high" style="padding: 2px 6px; font-size: 0.68rem;">HIGH</span>
305 |             </div>
306 |             <div class="mockup-tag">
307 |                 <span>LIABILITY</span>
308 |                 <span class="badge badge-medium" style="padding: 2px 6px; font-size: 0.68rem;">MED</span>
309 |             </div>
310 |             <div class="mockup-tag">
311 |                 <span>UNILATERAL CHANGES</span>
312 |                 <span class="badge badge-high" style="padding: 2px 6px; font-size: 0.68rem;">HIGH</span>
313 |             </div>
314 |         </div>
315 |
316 |         <div style="margin-top: 18px; text-align: center;">
317 |             <a href="scanner.html?demo=high_risk_social" style="font-size: 0.85rem; font-family: var(--font-mono);">
318 |                 Execute Live Analysis on this Policy &arr;
319 |             </a>
320 |         </div>
321 |     </div>
322 | </div>
323 | </section>
324 |
325 | <!-- The Problem Statement Section -->
326 | <section style="padding: 60px 0; border-top: 1px solid var(--border-card); background: rgba(12, 16, 24, 0.4);">
327 |     <div class="container" style="text-align: center; max-width: 800px;">
328 |         <span class="badge badge-medium" style="margin-bottom: 16px;">THE INVISIBLE VULNERABILITY</span>
329 |         <h2 style="font-size: 2.2rem; margin-bottom: 18px; color: #fff;">
330 |             "Nobody reads the fine print. That's exactly where the risk hides."
331 |         </h2>
332 |         <p style="font-size: 1.05rem; line-height: 1.7; color: #94a3b8;">
333 |             The average Terms of Service agreement contains over 4,000 words—requiring roughly 30 minutes of legal deciphering.
334 |         </p>
335 |     </div>
336 | </section>
337 |
338 | <!-- 3 Feature Cards: DETECT, SCORE, EXPLAIN -->
339 | <section class="container" style="padding: 60px 0;">
340 |     <div class="features-grid">
341 |         <div class="card card-cyber-accent feature-card">
342 |             <div class="feature-icon-box">
343 |                 <svg viewBox="0 0 24 24" width="28" height="28" fill="none" stroke="currentColor" stroke-width="2">

```

```

344 |         </div>;
345 |         <h3 class="feature-title">DETECT</h3>;
346 |         <p class="feature-desc">
347 |             Automatically identify potentially risky clauses across 14 cybersecurity and privacy categories, including c
348 |         </p>;
349 |     </div>;
350 |
351 |     <div class="card card-cyber-accent feature-card">;
352 |         <div class="feature-icon-box" style="color: var(--accent-green); border-color: rgba(0, 255, 136, 0.3);">;
353 |             <svg viewBox="0 0 24 24" width="28" height="28" fill="none" stroke="currentColor" stroke-width="2"><!--
354 |             </div>;
355 |             <h3 class="feature-title">SCORE</h3>;
356 |             <p class="feature-desc">
357 |                 Generate a deterministic 0-100 risk score and tier classification (SAFE, CAUTION, HIGH RISK) based on clause
358 |             </p>;
359 |         </div>;
360 |
361 |     <div class="card card-cyber-accent feature-card">;
362 |         <div class="feature-icon-box" style="color: var(--accent-amber); border-color: rgba(255, 183, 0, 0.3);">;
363 |             <svg viewBox="0 0 24 24" width="28" height="28" fill="none" stroke="currentColor" stroke-width="2"><!--
364 |             </div>;
365 |             <h3 class="feature-title">EXPLAIN</h3>;
366 |             <p class="feature-desc">
367 |                 Translate confusing legalese into plain English. Understand exactly why a clause matters from an information
368 |             </p>;
369 |         </div>;
370 |     </div>;
371 | </section>;
372 |
373 | <!-- How It Works Section -->;
374 | <section id="how-it-works" class="container" style="padding: 40px 0 70px;">;
375 |     <div class="section-title-wrap">;
376 |         <span class="badge badge-safe" style="margin-bottom: 12px;">SYSTEM PIPELINE</span>;
377 |         <h2 class="section-title">How It Works</h2>;
378 |         <p>A fast, 5-stage automated cybersecurity inspection pipeline designed for 100% offline local privacy.</p>;
379 |     </div>;
380 |
381 |     <div class="workflow-steps-wrap">;
382 |         <div class="workflow-step">;
383 |             <div class="workflow-num">01</div>;
384 |             <div class="workflow-label">Paste or Upload Policy</div>;
385 |             <p style="font-size: 0.78rem; margin-top: 6px; color: var(--text-dim);">Accepts raw text, .txt, .md, or
386 |         </div>;
387 |         <div class="workflow-step">;
388 |             <div class="workflow-num">02</div>;
389 |             <div class="workflow-label">Extract Clauses</div>;
390 |             <p style="font-size: 0.78rem; margin-top: 6px; color: var(--text-dim);">Segments sentences while protect
391 |         </div>;
392 |         <div class="workflow-step">;
393 |             <div class="workflow-num">03</div>;
394 |             <div class="workflow-label">Analyze Risks</div>;
395 |             <p style="font-size: 0.78rem; margin-top: 6px; color: var(--text-dim);">Evaluates clauses against 14 sec
396 |         </div>;
397 |         <div class="workflow-step">;
398 |             <div class="workflow-num">04</div>;
399 |             <div class="workflow-label">Generate Score</div>;
400 |             <p style="font-size: 0.78rem; margin-top: 6px; color: var(--text-dim);">Calculates weighted 0-100 exposu
401 |         </div>;
402 |         <div class="workflow-step">;
403 |             <div class="workflow-num">05</div>;
404 |             <div class="workflow-label">Explain Red Flags</div>;
405 |             <p style="font-size: 0.78rem; margin-top: 6px; color: var(--text-dim);">Delivers plain English translati
406 |         </div>;
407 |     </div>;
408 | </section>;
409 |
410 | <!-- Cybersecurity Relevance Banner -->;
411 | <section style="background: rgba(15, 23, 42, 0.6); border-top: 1px solid var(--border-card); border-bottom: 1px s
412 |     <div class="container" style="display: flex; align-items: center; justify-content: space-between; flex-wrap: wr

```

```

413 |         <div style="max-width: 650px;">
414 |             <span class="badge badge-high" style="margin-bottom: 12px;">CYBERSECURITY PERSPECTIVE</span>
415 |             <h3 style="font-size: 1.65rem; color: #fff; margin-bottom: 12px;">
416 |                 Consent can become a critical data-security blind spot.
417 |             </h3>
418 |             <p style="font-size: 0.95rem; line-height: 1.6; color: #94a3b8;">
419 |                 When users blindly click "I Agree", they grant legal permission for background tracking, credential sharing,
420 |             </p>
421 |         </div>
422 |     </div>
423 |     <a href="scanner.html" class="btn btn-primary btn-lg">Launch Scanner Now</a>
424 | </div>
425 | </div>
426 | </section>
427 |
428 | <!-- Team Section -->
429 | <section class="container" style="padding: 80px 0 40px;">
430 |     <div class="section-title-wrap">
431 |         <span class="badge badge-safe" style="margin-bottom: 12px;">HACKATHON TEAM</span>
432 |         <h2 class="section-title">Team CorruptX</h2>
433 |         <p>Built for the Cybersecurity Hackathon to protect user privacy rights in the digital age.</p>
434 |     </div>
435 |
436 |     <div class="team-grid">
437 |         <div class="card team-member-card">
438 |             <div class="team-avatar">VK</div>
439 |             <h3 style="font-size: 1.2rem; color: #fff; margin-bottom: 4px;">Vyom Khanna</h3>
440 |             <p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 10px">
441 |                 <p style="font-size: 0.88rem; color: var(--text-dim);">System Architecture & Full-Stack Security Eng
442 |             </div>
443 |
444 |             <div class="card team-member-card">
445 |                 <div class="team-avatar" style="background: linear-gradient(135deg, #00f2ff 0%, #00ff88 100%);">NS</div>
446 |                 <h3 style="font-size: 1.2rem; color: #fff; margin-bottom: 4px;">Nalin Shakher</h3>
447 |                 <p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 10px">
448 |                     <p style="font-size: 0.88rem; color: var(--text-dim);">Rule Engine, NLP Taxonomy & Offline Pipeline&
449 |                 </div>
450 |
451 |                 <div class="card team-member-card">
452 |                     <div class="team-avatar" style="background: linear-gradient(135deg, #ffb700 0%, #ff3366 100%);">AV</div>
453 |                     <h3 style="font-size: 1.2rem; color: #fff; margin-bottom: 4px;">Aayush Verma</h3>
454 |                     <p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 10px">
455 |                         <p style="font-size: 0.88rem; color: var(--text-dim);">Cybersecurity Interface & Interactive Clause
456 |                     </div>
457 |                 </div>
458 |             </div>
459 |         </main>
460 |
461 | <!-- Site Footer -->
462 | <footer class="site-footer">
463 |     <div class="container">
464 |         <div class="footer-top">
465 |             <div class="footer-brand">
466 |                 <div class="brand-wrapper">
467 |                     <div class="brand-shield-icon">
468 |                         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 22s8-4 8-8" /></svg>
469 |                     </div>
470 |                     <div class="brand-meta">
471 |                         <span class="brand-team">Team CorruptX</span>
472 |                         <span class="brand-name">T&C RED FLAG SCANNER</span>
473 |                     </div>
474 |                 </div>
475 |                 <p class="footer-desc">
476 |                     A cybersecurity hackathon project dedicated to de-obfuscating legal fine print and exposing privacy-invasive
477 |                 </p>
478 |             </div>
479 |
480 |             <div class="footer-links">
481 |

```

```
482 |         <div class="footer-group-title">Navigation</div>;
483 |         <ul class="footer-link-list">;
484 |             <li><a href="index.html">Home</a></li>;
485 |             <li><a href="scanner.html">Scanner</a></li>;
486 |             <li><a href="results.html">Results Dashboard</a></li>;
487 |             <li><a href="history.html">Scan History</a></li>;
488 |             <li><a href="about.html">About & Team</a></li>;
489 |         </ul>;
490 |     </div>;
491 |
492 |     <div class="footer-group-title">Demo Policies</div>;
493 |     <ul class="footer-link-list">;
494 |         <li><a href="scanner.html?demo=high_risk_social">Social App (High Risk)</a></li>;
495 |         <li><a href="scanner.html?demo=moderate_risk_saas">NovaDesk SaaS (Caution)</a></li>;
496 |         <li><a href="scanner.html?demo=safe_open_source">AegisGuard (Safe)</a></li>;
497 |     </ul>;
498 | </div>;
499 | </div>;
500 | </div>;
501 |
502 | <!-- Legal Disclaimer -->;
503 | <div class="footer-disclaimer">;
504 |     <strong>Important Cybersecurity & Legal Disclaimer:</strong> This tool provides automated cybers
505 | </div>;
506 |
507 | <div class="footer-bottom">;
508 |     <span>&copy; 2026 Team CorruptX — Cybersecurity Hackathon Project</span>;
509 |     <span>Offline-Ready Local Architecture</span>;
510 | </div>;
511 | </div>;
512 | </footer>;
513 |
514 | <script src="js/rules_data.js"></script>;
515 | <script src="js/sample_data.js"></script>;
516 | <script src="js/client_analyzer.js"></script>;
517 | <script src="js/app.js"></script>;
518 | <script src="js/chatbot.js" defer></script>;
519 | </body>;
520 | </html>;
```

File: frontend/scanner.html

Size: 8563 bytes

```

1 | <!DOCTYPE html>;
2 | <html lang="en">;
3 | <head>;
4 |   <meta charset="UTF-8">;
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">;
6 |   <title>Scanner – CorruptX T&C Red Flag Scanner</title>;
7 |   <link rel="stylesheet" href="css/style.css">;
8 |   <link rel="stylesheet" href="css/scanner.css">;
9 |   <link rel="stylesheet" href="css/responsive.css">;
10 |  <link rel="stylesheet" href="css/chatbot.css">;
11 | </head>;
12 | <body>;
13 |
14 |   <!-- Site Header -->;
15 |   <header class="site-header">;
16 |     <div class="container header-inner">;
17 |       <a href="index.html" class="brand-wrapper">;
18 |         <div class="brand-shield-icon">;
19 |           <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">;
20 |             </svg>;
21 |         <div class="brand-meta">;
22 |           <span class="brand-team">Team CorruptX</span>;
23 |           <span class="brand-name">T&C RED FLAG </span>SCANNER</span></div>;
24 |         </div>;
25 |       </a>;
26 |
27 |       <nav class="main-nav">;
28 |         <a href="index.html" class="nav-link">Home</a>;
29 |         <a href="scanner.html" class="nav-link active">Scanner</a>;
30 |         <a href="results.html" class="nav-link">Results</a>;
31 |         <a href="history.html" class="nav-link">History</a>;
32 |         <a href="about.html" class="nav-link">About Team</a>;
33 |       </nav>;
34 |
35 |       <div class="header-actions">;
36 |         <div class="system-status" id="system-status-indicator">;
37 |           <span class="status-dot"></span>;
38 |           <span id="system-status-text">CYBER ENGINE READY</span>;
39 |         </div>;
40 |         <button class="mobile-menu-btn" aria-label="Toggle navigation">;
41 |           <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">;
42 |             </svg>;
43 |         </button>;
44 |       </div>;
45 |     </header>;
46 |
47 |   <!-- Scanner Workspace -->;
48 |   <main class="container">;
49 |     <div class="scanner-hero">;
50 |       <div class="scanner-badge">CYBERSECURITY AUDIT CONSOLE</div>;
51 |       <h1 class="scanner-title">Scan Terms & Conditions</h1>;
52 |       <p class="scanner-subtitle">;
53 |         Paste a policy below and let the scanner identify potential red flags, data sharing traps, and arbitration waivers.
54 |       </p>;
55 |     </div>;
56 |
57 |     <div class="scanner-workspace">;
58 |       <div class="scanner-card card-cyber-accent">;
59 |         <!-- Document Meta & Actions Bar -->;
60 |         <div class="doc-meta-bar">;
61 |           <div class="doc-name-input-group">;
62 |             <span class="doc-name-label">Document:</span>;
63 |             <input type="text" id="doc-name-input" class="doc-name-input" placeholder="e.g. HyperNet Social App Terms of Service">;
64 |           </div>;
65 |
66 |           <div class="scanner-actions-bar">;
67 |             <button type="button" id="btn-load-demo" class="btn btn-secondary btn-sm" title="Load realistic sample policy">Load Demo</button>;

```

```
68 |         <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-width="2">&
69 |         Load Demo Policy
70 |     </button>;
71 |     <button type="button" id="btn-paste" class="btn btn-secondary btn-sm" title="Paste text from clipboard">
72 |         <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-width="2">&
73 |         Paste
74 |     </button>;
75 |     <button type="button" id="btn-clear" class="btn btn-outline btn-sm" title="Clear workspace">
76 |         Clear
77 |     </button>;
78 | </div>;
79 | </div>;
80 |
81 | <!-- Hidden File Input & Dropzone -->;
82 | <input type="file" id="file-upload-input" accept=".txt,.md,.html,.htm" style="display: none;">;
83 | <div id="file-dropzone" class="file-dropzone">;
84 |     <div class="dropzone-inner">;
85 |         <div class="dropzone-icon">;
86 |             <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stro
87 |         </div>;
88 |         <div class="dropzone-text"><strong>Drag & Drop policy document here</strong> or click
89 |         <div class="dropzone-subtext">Supports .TXT, .MD, and .HTML files &bull; Processed 100% locally on
90 |     </div>;
91 | </div>;
92 |
93 | <!-- Textarea -->;
94 | <div class="textarea-wrapper">;
95 |     <textarea id="policy-text" class="policy-textarea" placeholder="Paste Terms & Conditions, Privacy Polic
96 | </div>;
97 |
98 | <!-- Stats Bar -->;
99 | <div class="textarea-stats-bar">;
100 |     <div class="stats-group">;
101 |         <div class="stat-item">&Characters: <span id="char-count">0</span></div>;
102 |         <div class="stat-item">&Words: <span id="word-count">0</span></div>;
103 |         <div class="stat-item">&Estimated Clauses: <span id="clause-estimate">0</span></div>;
104 |     </div>;
105 |     <div class="security-privacy-note">;
106 |         Zero Cloud Uploads &bull; Private & Secure
107 |     </div>;
108 | </div>;
109 |
110 | <!-- Launch Bar -->;
111 | <div class="scanner-launch-bar">;
112 |     <div class="launch-tips">;
113 |         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="12" cy="12" r="12"
114 |         <span>Tip: Click "Load Demo Policy" above for a preloaded high-risk test file.</span>;
115 |     </div>;
116 |
117 |     <button type="button" id="btn-start-scan" class="btn btn-primary btn-scan-main">
118 |         <svg viewBox="0 0 24 24" width="20" height="20" fill="none" stroke="currentColor" stroke-width="2">&
119 |         Start Security Scan
120 |     </button>;
121 | </div>;
122 | </div>;
123 | </div>;
124 | </main>;
125 |
126 | <!-- Scanning Modal Overlay -->;
127 | <div id="scanning-overlay" class="scanning-overlay">;
128 |     <div class="scanning-modal">;
129 |         <div class="radar-box">;
130 |             <div class="radar-ring ring-1"></div>;
131 |             <div class="radar-ring ring-2"></div>;
132 |             <div class="radar-ring ring-3"></div>;
133 |             <div class="radar-sweep"></div>;
134 |             <div class="radar-center-icon">;
135 |                 <svg viewBox="0 0 24 24" width="32" height="32" fill="none" stroke="currentColor" stroke-width="2">&
136 |             </div>;
```

```
137 |         </div>;
138 |
139 |         <h3 class="scan-progress-title">Scanning Policy</h3>;
140 |         <p class="scan-progress-sub">De-obfuscating legal text and analyzing risk indicators...</p>;
141 |
142 |         <div class="scan-steps-list">;
143 |             <div class="scan-step-item" id="step-1">;
144 |                 <div class="step-icon-indicator">•</div>;
145 |                 <span>Extracting and segmenting clauses...</span>;
146 |             </div>;
147 |             <div class="scan-step-item" id="step-2">;
148 |                 <div class="step-icon-indicator">•</div>;
149 |                 <span>Classifying risks against security taxonomy...</span>;
150 |             </div>;
151 |             <div class="scan-step-item" id="step-3">;
152 |                 <div class="step-icon-indicator">•</div>;
153 |                 <span>Calculating weighted risk score (0-100)...</span>;
154 |             </div>;
155 |             <div class="scan-step-item" id="step-4">;
156 |                 <div class="step-icon-indicator">•</div>;
157 |                 <span>Generating plain-English explanations...</span>;
158 |             </div>;
159 |         </div>;
160 |
161 |         <div class="progress-bar-track">;
162 |             <div id="scan-progress-fill" class="progress-bar-fill"></div>;
163 |         </div>;
164 |     </div>;
165 | </div>;
166 |
167 | <!-- Site Footer -->;
168 | <footer class="site-footer">;
169 |     <div class="container">;
170 |         <div class="footer-disclaimer">;
171 |             <strong>Disclaimer:</strong> This tool provides automated risk analysis for informational purposes a
172 |         </div>;
173 |         <div class="footer-bottom">;
174 |             <span>&copy; 2026 Team CorruptX – T&C Red Flag Scanner</span>;
175 |             <span>Offline-Capable Architecture</span>;
176 |         </div>;
177 |     </div>;
178 | </footer>;
179 |
180 | <script src="js/rules_data.js"></script>;
181 | <script src="js/sample_data.js"></script>;
182 | <script src="js/client_analyzer.js"></script>;
183 | <script src="js/app.js"></script>;
184 | <script src="js/scanner.js"></script>;
185 | <script src="js/chatbot.js" defer></script>;
186 | </body>;
187 | </html>;
```

File: frontend/results.html

Size: 10394 bytes

```

1 | <!DOCTYPE html>;
2 | <html lang="en">;
3 | <head>;
4 |   <meta charset="UTF-8">;
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">;
6 |   <title>Results Dashboard – CorruptX T&C Red Flag Scanner</title>;
7 |   <link rel="stylesheet" href="css/style.css">;
8 |   <link rel="stylesheet" href="css/results.css">;
9 |   <link rel="stylesheet" href="css/responsive.css">;
10 |  <link rel="stylesheet" href="css/chatbot.css">;
11 | </head>;
12 | <body>;
13 |
14 |   <!-- Site Header -->;
15 |   <header class="site-header">;
16 |     <div class="container header-inner">;
17 |       <a href="index.html" class="brand-wrapper">;
18 |         <div class="brand-shield-icon">;
19 |           <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"></svg>;
20 |         </div>;
21 |         <div class="brand-meta">;
22 |           <span class="brand-team">Team CorruptX</span>;
23 |           <span class="brand-name">T&C RED FLAG </span>;SCANNER</span>;</div>;
24 |         </div>;
25 |       </a>;
26 |
27 |       <nav class="main-nav">;
28 |         <a href="index.html" class="nav-link">Home</a>;
29 |         <a href="scanner.html" class="nav-link">Scanner</a>;
30 |         <a href="results.html" class="nav-link active">Results</a>;
31 |         <a href="history.html" class="nav-link">History</a>;
32 |         <a href="about.html" class="nav-link">About Team</a>;
33 |       </nav>;
34 |
35 |       <div class="header-actions">;
36 |         <div class="system-status" id="system-status-indicator">;
37 |           <span class="status-dot"></span>;
38 |           <span id="system-status-text">CYBER ENGINE READY</span>;
39 |         </div>;
40 |         <button class="mobile-menu-btn" aria-label="Toggle navigation">;
41 |           <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"></svg>;
42 |         </button>;
43 |       </div>;
44 |     </div>;
45 |   </header>;
46 |
47 |   <!-- Results Dashboard -->;
48 |   <main class="container">;
49 |     <section class="results-header-section">;
50 |       <div class="results-doc-title-row">;
51 |         <div class="results-title-group">;
52 |           <h1>;
53 |             <span style="color: var(--accent-green); font-size: 1.4rem;">&#10003</span>;</span>;
54 |             <span id="res-doc-title">Scan Results</span>;
55 |           </h1>;
56 |           <div class="results-doc-subtitle" id="res-doc-date">Scanned: Just now</div>;
57 |         </div>;
58 |
59 |         <div class="results-actions-group">;
60 |           <button type="button" id="btn-copy-summary" class="btn btn-secondary btn-sm">;
61 |             <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-width="2"></svg>;
62 |             Copy Summary
63 |           </button>;
64 |           <button type="button" id="btn-export-report" class="btn btn-secondary btn-sm">;
65 |             <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-width="2"></svg>;
66 |             Export JSON
67 |           </button>;

```

```

68 |         <button type="button" id="btn-rescan-new" class="btn btn-primary btn-sm">
69 |             <svg viewBox="0 0 24 24" width="16" height="16" fill="none" stroke="currentColor" stroke-width="2"><
70 |                 Scan Another Policy
71 |             </button>
72 |         </div>
73 |     </div>
74 |
75 |     <!-- Top Score Overview Banner -->
76 |     <div class="overview-card">
77 |         <!-- Circular Risk Meter -->
78 |         <div class="meter-wrapper">
79 |             <div class="circular-gauge">
80 |                 <svg class="gauge-svg" viewBox="0 0 190 190">
81 |                     <circle class="gauge-track" cx="95" cy="95" r="80"></circle>
82 |                     <circle class="gauge-fill" id="gauge-fill-path" cx="95" cy="95" r="80"></circle>
83 |                 </svg>
84 |                 <div class="gauge-content">
85 |                     <div class="gauge-score" id="gauge-score-value">0</div>
86 |                     <div class="gauge-max">OUT OF 100</div>
87 |                 </div>
88 |             </div>
89 |             <div class="gauge-label">
90 |                 <span class="risk-level-badge badge" id="res-risk-badge">CALCULATING</span>
91 |             </div>
92 |         </div>
93 |
94 |         <!-- Overview Content & Metrics Grid -->
95 |         <div class="overview-info-panel">
96 |             <div class="overview-status-line">
97 |                 <span style="font-family: var(--font-mono); font-size: 0.8rem; color: var(--accent-cyan); text-transform:
98 |                     SCAN AUDIT COMPLETE
99 |                 </span>
100 |             </div>
101 |
102 |             <p class="overview-summary-text" id="res-summary-text">
103 |                 Executing risk classification against cybersecurity taxonomy...
104 |             </p>
105 |
106 |             <!-- 5 Stats Counters -->
107 |             <div class="metrics-grid">
108 |                 <div class="metric-box">
109 |                     <span class="metric-label">Clauses</span>
110 |                     <span class="metric-value" id="metric-clauses">0</span>
111 |                 </div>
112 |                 <div class="metric-box">
113 |                     <span class="metric-label">Red Flags</span>
114 |                     <span class="metric-value high-val" id="metric-redflags">0</span>
115 |                 </div>
116 |                 <div class="metric-box">
117 |                     <span class="metric-label">High Risk</span>
118 |                     <span class="metric-value high-val" id="metric-high">0</span>
119 |                 </div>
120 |                 <div class="metric-box">
121 |                     <span class="metric-label">Medium Risk</span>
122 |                     <span class="metric-value med-val" id="metric-med">0</span>
123 |                 </div>
124 |                 <div class="metric-box">
125 |                     <span class="metric-label">Low Risk</span>
126 |                     <span class="metric-value low-val" id="metric-low">0</span>
127 |                 </div>
128 |             </div>
129 |         </div>
130 |     </div>
131 |
132 |     <!-- Tab View Switcher & Filters -->
133 |     <div class="results-tabs-bar">
134 |         <div class="tab-nav">
135 |             <button type="button" class="tab-btn active" data-tab="cards">
136 |                 <span>Detected Red Flags</span>

```

```

137 |         <span class="tab-badge" id="tab-count-flags">0</span>;
138 |     </button>;
139 |     <button type="button" class="tab-btn" data-tab="highlighter">;
140 |         <span>Interactive Policy Analysis</span>;
141 |     </button>;
142 |     <button type="button" class="tab-btn" data-tab="breakdown">;
143 |         <span>Risk Breakdown</span>;
144 |     </button>;
145 | </div>;
146 |
147 | <div class="filter-group">;
148 |     <span style="font-size: 0.76rem; font-family: var(--font-mono); color: var(--text-dim); margin-right: 4px;">;
149 |     <button type="button" class="filter-btn active" data-filter="all">All</button>;
150 |     <button type="button" class="filter-btn" data-filter="HIGH">High Risk</button>;
151 |     <button type="button" class="filter-btn" data-filter="MEDIUM">Medium Risk</button>;
152 |     <button type="button" class="filter-btn" data-filter="LOW">Low Risk</button>;
153 | </div>;
154 | </div>;
155 | </section>;
156 |
157 | <!-- VIEW 1: Red Flag Cards -->;
158 | <section id="view-red-flags" class="red-flags-view">;
159 |     <div id="red-flags-container">;
160 |         <!-- Populated via results.js -->;
161 |     </div>;
162 | </section>;
163 |
164 | <!-- VIEW 2: Interactive Policy Highlighter -->;
165 | <section id="view-policy-analysis" class="policy-analysis-view" style="display: none;">;
166 |     <div>;
167 |         <div style="display: flex; align-items: center; justify-content: space-between; margin-bottom: 14px;">;
168 |             <h3 style="font-size: 1.15rem; color: #fff;">Full Policy Document</h3>;
169 |             <span style="font-size: 0.8rem; font-family: var(--font-mono); color: var(--text-dim);">;
170 |                 Click any highlighted clause to inspect
171 |             </span>;
172 |         </div>;
173 |         <div id="policy-full-text-container" class="policy-doc-container">;
174 |             <!-- Text with highlighted spans populated via results.js -->;
175 |         </div>;
176 |     </div>;
177 |
178 | <!-- Sticky Inspector Panel -->;
179 | <aside class="sticky-inspector-panel">;
180 |     <div class="inspector-title">;
181 |         <svg viewBox="0 0 24 24" width="18" height="18" fill="none" stroke="currentColor" stroke-width="2"><!--
182 |         <span>Clause Inspector</span>;
183 |     </div>;
184 |
185 |     <div id="inspector-placeholder" class="inspector-placeholder">;
186 |         <svg viewBox="0 0 24 24" width="36" height="36" fill="none" stroke="currentColor" stroke-width="1.5"><!--
187 |         <p>Click on any clause in the policy view to inspect risk details, category weights, and security impact</p>;
188 |     </div>;
189 |
190 |     <div id="inspector-content-pane" class="inspector-content-pane" style="display: none;">;
191 |         <!-- Active clause details inserted dynamically -->;
192 |     </div>;
193 | </aside>;
194 | </section>;
195 |
196 | <!-- VIEW 3: Category Risk Breakdown -->;
197 | <section id="view-category-breakdown" class="category-breakdown-view" style="display: none;">;
198 |     <div class="breakdown-title-row">;
199 |         <div>;
200 |             <h3 style="font-size: 1.25rem; color: #fff; margin-bottom: 4px;">Top Risk Categories</h3>;
201 |             <p style="font-size: 0.88rem; color: var(--text-muted);">;
202 |                 Proportion and severity distribution of detected clauses across cybersecurity taxonomy.
203 |             </p>;
204 |         </div>;
205 |     </div>;

```

```
206 |
207 |     <div id="category-breakdown-container">
208 |         <!-- Populated via results.js -->
209 |     </div>
210 | </section>
211 | </main>
212 |
213 | <!-- Site Footer -->
214 | <footer class="site-footer">
215 |     <div class="container">
216 |         <div class="footer-disclaimer">
217 |             <strong>Important Cybersecurity & Legal Disclaimer:</strong> This tool provides automated risk a
218 |         </div>
219 |         <div class="footer-bottom">
220 |             <span>&copy; 2026 Team CorruptX — T&C Red Flag Scanner</span>
221 |             <span>Offline-Capable Architecture</span>
222 |         </div>
223 |     </div>
224 | </footer>
225 |
226 | <script src="js/rules_data.js"></script>
227 | <script src="js/sample_data.js"></script>
228 | <script src="js/client_analyzer.js"></script>
229 | <script src="js/app.js"></script>
230 | <script src="js/results.js"></script>
231 | <script src="js/chatbot.js" defer></script>
232 | </body>
233 | </html>
```

File: frontend/history.html

Size: 6489 bytes

```

1 | <!DOCTYPE html>;
2 | <html lang="en">;
3 | <head>;
4 |   <meta charset="UTF-8">;
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">;
6 |   <title>&Scan History – CorruptX T&C Red Flag Scanner</title>;
7 |   <link rel="stylesheet" href="css/style.css">;
8 |   <link rel="stylesheet" href="css/responsive.css">;
9 |   <link rel="stylesheet" href="css/chatbot.css">;
10 | </style>;
11 | .history-header {
12 |   padding: 48px 0 28px;
13 |   display: flex;
14 |   align-items: center;
15 |   justify-content: space-between;
16 |   flex-wrap: wrap;
17 |   gap: 20px;
18 | }
19 | .history-stats-bar {
20 |   display: grid;
21 |   grid-template-columns: repeat(3, 1fr);
22 |   gap: 20px;
23 |   margin-bottom: 32px;
24 | }
25 | .history-stat-card {
26 |   background: var(--bg-card);
27 |   border: 1px solid var(--border-card);
28 |   border-radius: var(--radius-md);
29 |   padding: 20px;
30 |   display: flex;
31 |   flex-direction: column;
32 | }
33 | .history-stat-label {
34 |   font-size: 0.78rem;
35 |   font-family: var(--font-mono);
36 |   color: var(--text-dim);
37 |   text-transform: uppercase;
38 |   letter-spacing: 0.05em;
39 |   margin-bottom: 6px;
40 | }
41 | .history-stat-num {
42 |   font-size: 2rem;
43 |   font-weight: 800;
44 |   color: #fff;
45 | }
46 | .empty-history-box {
47 |   text-align: center;
48 |   padding: 64px 20px;
49 |   background: var(--bg-card);
50 |   border: 1px dashed var(--border-subtle);
51 |   border-radius: var(--radius-lg);
52 | }
53 | @media (max-width: 640px) {
54 |   .history-stats-bar { grid-template-columns: 1fr; }
55 | }
56 | </style>;
57 | </head>;
58 | <body>;
59 |
60 | <!-- Site Header -->;
61 | <header class="site-header">;
62 |   <div class="container header-inner">;
63 |     <a href="index.html" class="brand-wrapper">;
64 |       <div class="brand-shield-icon">;
65 |         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-l
66 |       </div>;
67 |     <div class="brand-meta">;

```

```

68 |         <span class="brand-team">Team CorruptX</span>;
69 |         <span class="brand-name">T&C RED FLAG <span>SCANNER</span></span>;
70 |     </div>;
71 | </a>;
72 |
73 |     <nav class="main-nav">
74 |         <a href="index.html" class="nav-link">Home</a>;
75 |         <a href="scanner.html" class="nav-link">Scanner</a>;
76 |         <a href="results.html" class="nav-link">Results</a>;
77 |         <a href="history.html" class="nav-link active">History</a>;
78 |         <a href="about.html" class="nav-link">About Team</a>;
79 |     </nav>;
80 |
81 |     <div class="header-actions">
82 |         <div class="system-status" id="system-status-indicator">
83 |             <span class="status-dot"></span>;
84 |             <span id="system-status-text">CYBER ENGINE READY</span>;
85 |         </div>;
86 |         <button class="mobile-menu-btn" aria-label="Toggle navigation">
87 |             <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
88 |             </button>;
89 |         </div>;
90 |     </div>;
91 | </header>;
92 |
93 | <!-- History Container -->
94 | <main class="container" style="margin-bottom: 60px;">
95 |     <div class="history-header">
96 |         <div>
97 |             <h1 style="font-size: 2rem; color: #fff; margin-bottom: 6px;">Scan History</h1>;
98 |             <p style="font-size: 0.92rem; color: var(--text-muted);">
99 |                 Audit logs and previous risk evaluations stored privately in local browser storage.
100 |             </p>;
101 |         </div>;
102 |
103 |         <div style="display: flex; gap: 10px; align-items: center; flex-wrap: wrap;">
104 |             <button type="button" id="btn-export-all-history" class="btn btn-secondary btn-sm">
105 |                 Export History (JSON)
106 |             </button>;
107 |             <button type="button" id="btn-clear-history" class="btn btn-danger btn-sm">
108 |                 Clear History
109 |             </button>;
110 |         </div>;
111 |     </div>;
112 |
113 | <!-- Summary Metrics -->
114 | <div class="history-stats-bar">
115 |     <div class="history-stat-card">
116 |         <span class="history-stat-label">Total Documents Audited</span>;
117 |         <span class="history-stat-num" id="stat-total-scans">0</span>;
118 |     </div>;
119 |     <div class="history-stat-card">
120 |         <span class="history-stat-label">Average Risk Exposure</span>;
121 |         <span class="history-stat-num" id="stat-avg-score">0/100</span>;
122 |     </div>;
123 |     <div class="history-stat-card">
124 |         <span class="history-stat-label">High Risk Policies</span>;
125 |         <span class="history-stat-num" id="stat-high-risk-scans" style="color: var(--accent-crimson);">0</span>;
126 |     </div>;
127 | </div>;
128 |
129 | <!-- Scans List -->
130 | <div id="history-list-container">
131 |     <!-- Populated via history.js -->
132 | </div>;
133 |
134 | <!-- Empty State -->
135 | <div id="history-empty-state" class="empty-history-box" style="display: none;">
136 |     <svg viewBox="0 0 24 24" width="48" height="48" fill="none" stroke="currentColor" stroke-width="1.5" style="color: var(--text-muted);"/>

```

```
137 |         <path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8" />
138 |     </svg>
139 |     <h3 style="font-size: 1.25rem; color: #fff; margin-bottom: 8px;">No Scan Records Yet</h3>
140 |     <p style="color: var(--text-muted); max-width: 440px; margin: 0 auto 20px; font-size: 0.9rem;">
141 |         You haven't scanned any policies on this browser yet. Run your first analysis or load realistic demo data.
142 |     </p>
143 |     <div style="display: flex; gap: 12px; justify-content: center; flex-wrap: wrap;">
144 |         <a href="scanner.html" class="btn btn-primary btn-sm">Scan a Document</a>
145 |         <button type="button" id="btn-load-sample-history" class="btn btn-outline btn-sm">Load Demo Records</button>
146 |     </div>
147 | </div>
148 | </main>
149 |
150 | <!-- Site Footer -->
151 | <footer class="site-footer">
152 |     <div class="container">
153 |         <div class="footer-disclaimer">
154 |             <strong>Important Disclaimer:</strong> All scan records are stored locally on your device in browser.
155 |         </div>
156 |         <div class="footer-bottom">
157 |             <span>&copy; 2026 Team CorruptX – T&C Red Flag Scanner</span>
158 |             <span>Offline-Capable Architecture</span>
159 |         </div>
160 |     </div>
161 | </footer>
162 |
163 | <script src="js/rules_data.js"></script>
164 | <script src="js/sample_data.js"></script>
165 | <script src="js/client_analyzer.js"></script>
166 | <script src="js/app.js"></script>
167 | <script src="js/history.js"></script>
168 | <script src="js/chatbot.js" defer></script>
169 | </body>
170 | </html>
```

File: frontend/about.html

Size: 14237 bytes

```

1 | <!DOCTYPE html>;
2 | <html lang="en">;
3 | <head>;
4 |   <meta charset="UTF-8">;
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">;
6 |   <title>About — Team CorruptX | T&C Red Flag Scanner</title>;
7 |   <link rel="stylesheet" href="css/style.css">;
8 |   <link rel="stylesheet" href="css/responsive.css">;
9 |   <link rel="stylesheet" href="css/chatbot.css">;
10 | </style>;
11 | .about-hero {
12 |   padding: 60px 0 36px;
13 |   text-align: center;
14 |   max-width: 800px;
15 |   margin: 0 auto;
16 | }
17 | .about-hero-title {
18 |   font-size: 2.8rem;
19 |   font-weight: 800;
20 |   margin-bottom: 16px;
21 |   background: linear-gradient(135deg, #ffffff 0%, #cbd5e1 50%, #38bdf8 100%);
22 |   -webkit-background-clip: text;
23 |   -webkit-text-fill-color: transparent;
24 | }
25 | .about-card-section {
26 |   margin-bottom: 48px;
27 | }
28 | .roadmap-flowchart {
29 |   background: rgba(8, 12, 20, 0.8);
30 |   border: 1px solid var(--border-subtle);
31 |   border-radius: var(--radius-md);
32 |   padding: 24px;
33 |   margin-top: 20px;
34 |   display: flex;
35 |   flex-wrap: wrap;
36 |
37 |   align-items: center;
38 |   justify-content: center;
39 |   gap: 12px;
40 |   font-family: var(--font-mono);
41 |   font-size: 0.85rem;
42 | }
43 | .flow-node {
44 |   background: rgba(15, 23, 42, 0.9);
45 |   border: 1px solid var(--border-focus);
46 |   padding: 8px 14px;
47 |   border-radius: var(--radius-sm);
48 |   color: #fff;
49 | }
50 | .flow-arrow {
51 |   color: var(--accent-cyan);
52 |   font-weight: 800;
53 | }
54 | </style>;
55 | </head>;
56 | <body>;
57 |
58 | <!-- Site Header -->;
59 | <header class="site-header">;
60 |   <div class="container header-inner">;
61 |     <a href="index.html" class="brand-wrapper">;
62 |       <div class="brand-shield-icon">;
63 |         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">;
64 |       </div>;
65 |       <div class="brand-meta">;
66 |         <span class="brand-team">Team CorruptX</span>;
67 |         <span class="brand-name">T&C RED FLAG <span>SCANNER</span></span></div>;

```

```

68 |         </a>;
69 |
70 |         <nav class="main-nav">
71 |             <a href="index.html" class="nav-link">Home</a>;
72 |             <a href="scanner.html" class="nav-link">Scanner</a>;
73 |             <a href="results.html" class="nav-link">Results</a>;
74 |             <a href="history.html" class="nav-link">History</a>;
75 |             <a href="about.html" class="nav-link active">About Team</a>;
76 |         </nav>;
77 |
78 |         <div class="header-actions">
79 |             <div class="system-status" id="system-status-indicator">
80 |                 <span class="status-dot"></span>;
81 |                 <span id="system-status-text">CYBER ENGINE READY</span>;
82 |             </div>;
83 |             <button class="mobile-menu-btn" aria-label="Toggle navigation">
84 |                 <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-l
85 |                 </button>;
86 |             </div>;
87 |         </div>;
88 |     </header>;
89 |
90 |     <!-- About Content -->;
91 |     <main class="container" style="margin-bottom: 80px;">
92 |         <div class="about-hero">
93 |             <span class="badge badge-safe" style="margin-bottom: 12px;">CYBERSECURITY HACKATHON INITIATIVE</span>;
94 |             <h1 class="about-hero-title">About T&C Red Flag Scanner</h1>;
95 |             <p style="font-size: 1.1rem; line-height: 1.7; color: #94a3b8;">
96 |                 Empowering users to decode legal obfuscation, identify privacy-invasive terms, and preserve their digital soverei
97 |             </p>;
98 |         </div>;
99 |
100 |     <!-- The Problem & Solution -->;
101 |     <div class="about-card-section card-cyber-accent">
102 |         <div style="display: grid; grid-template-columns: 1fr 1fr; gap: 36px;">
103 |             <div>
104 |                 <span class="badge badge-high" style="margin-bottom: 12px;">THE PROBLEM</span>;
105 |                 <h2 style="font-size: 1.5rem; color: #fff; margin-bottom: 12px;">The Consent Fatigue Crisis</h2>;
106 |                 <p style="font-size: 0.94rem; line-height: 1.7; color: #cbd5e1; margin-bottom: 12px;">
107 |                     Over 95% of consumers accept legal terms and privacy policies without reading them. Companies intentionally
108 |                 </p>;
109 |                 <p style="font-size: 0.94rem; line-height: 1.7; color: #cbd5e1;">
110 |                     Buried inside these agreements are aggressive terms: mandatory binding arbitration that eliminates court acc
111 |                 </p>;
112 |             </div>;
113 |
114 |             <div>
115 |                 <span class="badge badge-safe" style="margin-bottom: 12px;">THE SOLUTION</span>;
116 |                 <h2 style="font-size: 1.5rem; color: #fff; margin-bottom: 12px;">Lightweight Automated De-Obfuscation</h2>;
117 |                 <p style="font-size: 0.94rem; line-height: 1.7; color: #cbd5e1; margin-bottom: 12px;">
118 |                     The <strong>T&C Red Flag Scanner</strong> is a specialized security utility designed to insp
119 |                 </p>;
120 |                 <p style="font-size: 0.94rem; line-height: 1.7; color: #cbd5e1;">
121 |                     Crucially, it is built with an <strong>offline-first local architecture</strong>, meaning your s
122 |                 </p>;
123 |             </div>;
124 |         </div>;
125 |     </div>;
126 |
127 |     <!-- Core Methodology: Detect, Score, Explain -->;
128 |     <div class="about-card-section card">
129 |         <h2 style="font-size: 1.6rem; color: #fff; margin-bottom: 24px; text-align: center;">Our Core Three-Pillar M
130 |         <div style="display: grid; grid-template-columns: repeat(3, 1fr); gap: 24px;">
131 |             <div style="background: rgba(15, 23, 42, 0.6); padding: 20px; border-radius: 10px; border: 1px solid var(--bc
132 |                 <div style="font-size: 1.15rem; font-weight: 700; color: var(--accent-cyan); margin-bottom: 8px;">1. DET
133 |                 <p style="font-size: 0.88rem; color: #cbd5e1; line-height: 1.6;">
134 |                     Rule-based and pattern-driven NLP segmentation scans text sentence-by-sentence against a taxonomy of 14 priv
135 |                 </p>;
136 |             </div>;

```

```

137 |         <div style="background: rgba(15, 23, 42, 0.6); padding: 20px; border-radius: 10px; border: 1px solid var(--bo
138 |         <div style="font-size: 1.15rem; font-weight: 700; color: var(--accent-green); margin-bottom: 8px;">2. SC
139 |         <p style="font-size: 0.88rem; color: #cbd5e1; line-height: 1.6;">
140 |             Computes a deterministic 0-100 score weighted by severity (High: 15-16 pts, Medium: 8-9 pts, Low: 4-5 pts) a
141 |         </p>
142 |     </div>
143 |     <div style="background: rgba(15, 23, 42, 0.6); padding: 20px; border-radius: 10px; border: 1px solid var(--bo
144 |     <div style="font-size: 1.15rem; font-weight: 700; color: var(--accent-amber); margin-bottom: 8px;">3. EX
145 |     <p style="font-size: 0.88rem; color: #cbd5e1; line-height: 1.6;">
146 |         Translates convoluted legal rhetoric into concise English explanations, details why the term matters from a
147 |     </p>
148 |     </div>
149 | </div>
150 | </div>
151 |
152 | <!-- Cybersecurity Relevance -->
153 | <div class="about-card-section card" style="border-left: 4px solid var(--accent-cyan);">
154 |     <h3 style="font-size: 1.35rem; color: #fff; margin-bottom: 12px;">Cybersecurity Relevance & Threat Model
155 |     <p style="font-size: 0.95rem; line-height: 1.7; color: #cbd5e1; margin-bottom: 12px;">
156 |         In cybersecurity, human users are often considered the weakest link. However, this weakness is exacerbated when
157 |     </p>
158 |     <blockquote style="padding: 16px 20px; background: rgba(0, 242, 255, 0.04); border-left: 3px solid var(--accent
159 |         "Consent can become a data-security blind spot when users unknowingly authorize collection, tracking or third-pa
160 |     </blockquote>
161 |     <p style="font-size: 0.95rem; line-height: 1.7; color: #cbd5e1;">
162 |         By scanning terms prior to consent, users can identify third-party sub-processor supply chain vulnerabilities, p
163 |     </p>
164 | </div>
165 |
166 | <!-- Future AI Architecture & Roadmap -->
167 | <div class="about-card-section card">
168 |     <h3 style="font-size: 1.35rem; color: #fff; margin-bottom: 8px;">Future AI Architecture & Roadmap</h3>
169 |     <p style="font-size: 0.9rem; color: var(--text-muted);">
170 |         The current version implements an efficient, zero-cost, local rule-based NLP pipeline. The architecture is engin
171 |     </p>
172 |
173 |     <div class="roadmap-flowchart">
174 |         <div class="flow-node">Website / Document</div>
175 |         <div class="flow-arrow">&rarr;</div>
176 |         <div class="flow-node">Text Extraction</div>
177 |         <div class="flow-arrow">&rarr;</div>
178 |         <div class="flow-node">Clause Segmentation</div>
179 |         <div class="flow-arrow">&rarr;</div>
180 |         <div class="flow-node" style="border-color: var(--accent-cyan); color: var(--accent-cyan);">AI/NLP Classif
181 |         <div class="flow-arrow">&rarr;</div>
182 |         <div class="flow-node">Risk Taxonomy</div>
183 |         <div class="flow-arrow">&rarr;</div>
184 |         <div class="flow-node">Weighted Risk Score</div>
185 |         <div class="flow-arrow">&rarr;</div>
186 |         <div class="flow-node">Plain-English Explanations</div>
187 |         <div class="flow-arrow">&rarr;</div>
188 |         <div class="flow-node" style="border-color: var(--accent-green); color: var(--accent-green);">User Dashboa
189 |     </div>
190 | </div>
191 |
192 | <!-- Hackathon Team Section -->
193 | <div class="about-card-section">
194 |     <div style="text-align: center; margin-bottom: 32px;">
195 |         <span class="badge badge-safe" style="margin-bottom: 10px;">CREDITS</span>
196 |         <h2 style="font-size: 2rem; color: #fff;">Team CorruptX</h2>
197 |     </div>
198 |
199 |     <div style="display: grid; grid-template-columns: repeat(3, 1fr); gap: 24px;">
200 |         <div class="card" style="text-align: center; padding: 32px 20px;">
201 |             <div style="width: 68px; height: 68px; border-radius: 50%; background: linear-gradient(135deg, #00d2df 0%,
202 |                 VK
203 |             </div>
204 |             <h3 style="font-size: 1.25rem; color: #fff; margin-bottom: 4px;">Vyom Khanna</h3>
205 |             <p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 12px"

```

```
206 |         &lt;p style="font-size: 0.88rem; color: var(--text-dim); line-height: 1.5;"&gt;
207 |             Architectural leadership, core Flask service implementation, threat taxonomy coordination, and hackathon pre
208 |         &lt;/p&gt;
209 |     &lt;/div&gt;
210 |
211 |     &lt;div class="card" style="text-align: center; padding: 32px 20px;"&gt;
212 |         &lt;div style="width: 68px; height: 68px; border-radius: 50%; background: linear-gradient(135deg, #00f2ff 0%,
213 |             NS
214 |         &lt;/div&gt;
215 |         &lt;h3 style="font-size: 1.25rem; color: #fff; margin-bottom: 4px;"&gt;Nalin Shakher&lt;/h3&gt;
216 |         &lt;p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 12px
217 |         &lt;p style="font-size: 0.88rem; color: var(--text-dim); line-height: 1.5;"&gt;
218 |             NLP rule-based parsing engine, regex optimization, negation handling for user protections, and offline runti
219 |         &lt;/p&gt;
220 |     &lt;/div&gt;
221 |
222 |     &lt;div class="card" style="text-align: center; padding: 32px 20px;"&gt;
223 |         &lt;div style="width: 68px; height: 68px; border-radius: 50%; background: linear-gradient(135deg, #ffb700 0%,
224 |             AV
225 |         &lt;/div&gt;
226 |         &lt;h3 style="font-size: 1.25rem; color: #fff; margin-bottom: 4px;"&gt;Aayush Verma&lt;/h3&gt;
227 |         &lt;p style="font-size: 0.85rem; font-family: var(--font-mono); color: var(--accent-cyan); margin-bottom: 12px
228 |         &lt;p style="font-size: 0.88rem; color: var(--text-dim); line-height: 1.5;"&gt;
229 |             Cybersecurity dashboard UI, interactive clause highlighting inspector, SVG circular gauge, and responsive fr
230 |         &lt;/p&gt;
231 |     &lt;/div&gt;
232 | &lt;/div&gt;
233 | &lt;/div&gt;
234 | &lt;/main&gt;
235 |
236 | &lt;!-- Site Footer --&gt;
237 | &lt;footer class="site-footer"&gt;
238 |     &lt;div class="container"&gt;
239 |         &lt;div class="footer-disclaimer"&gt;
240 |             &lt;strong&gt;Important Disclaimer:&lt;/strong&gt; This tool provides automated risk analysis for informational
241 |         &lt;/div&gt;
242 |         &lt;div class="footer-bottom"&gt;
243 |             &lt;span&gt;&amp;copy; 2026 Team CorruptX – T&amp;C Red Flag Scanner&lt;/span&gt;
244 |             &lt;span&gt;Offline-Capable Architecture&lt;/span&gt;
245 |         &lt;/div&gt;
246 |     &lt;/div&gt;
247 | &lt;/footer&gt;
248 |
249 |     &lt;script src="js/rules_data.js"&gt;&lt;/script&gt;
250 |     &lt;script src="js/sample_data.js"&gt;&lt;/script&gt;
251 |     &lt;script src="js/client_analyzer.js"&gt;&lt;/script&gt;
252 |     &lt;script src="js/app.js"&gt;&lt;/script&gt;
253 |     &lt;script src="js/chatbot.js" defer&gt;&lt;/script&gt;
254 | &lt;/body&gt;
255 | &lt;/html&gt;
```

File: frontend/js/chatbot.js

Size: 17162 bytes

```

1 | /**
2 |  * CorruptX - T&C AI Privacy Assistant (Chatbot)
3 |  * Intelligent NLP & Privacy Knowledge Engine (Offline + Backend Hybrid)
4 |  */
5 |
6 | (function () {
7 |     // Knowledge Base Patterns
8 |     const PRIVACY_KNOWLEDGE = [
9 |         {
10 |             keywords: ["arbitration", "class action", "court", "sue", "dispute", "lawsuit", "jury"],
11 |             title: "■ Mandatory Arbitration & Class Action Waivers",
12 |             reply: `<b>Mandatory Arbitration Clauses</b> strip away your constitutional right to take a company to
13 | <b>Why it's a Red Flag:</b>
14 | • Disputes are settled behind closed doors by private arbitrators often hired by the corporation.
15 | • You cannot band together with other affected users (Class Action Waiver).
16 | • It drastically lowers the company's liability for privacy breaches.<br><br>
17 | <i>Tip: Some companies allow you to "opt-out" of arbitration by sending a written notice within 30 days of openi
18 |     },
19 |     {
20 |         keywords: ["sell", "sold", "monetiz", "third party", "third-party", "partners", "advertis", "brokers", "sharing"],
21 |         title: "■ Data Selling & Third-Party Sharing",
22 |         reply: `<b>Data Selling & Sharing Clauses</b> permit companies to package your personal informatio
23 | <b>What to look out for:</b>
24 | • Phrases like <i>"We may share aggregated or anonymized data with third parties for marketing purposes."</i>
25 | • Phrases like <i>"Partners may receive device identifiers and demographic data."</i><br><br>
26 | <i>CorruptX Advice: Under CCPA (California) and GDPR (Europe), you have the legal right to click "Do Not Sell
27 |     },
28 |     {
29 |         keywords: ["track", "location", "gps", "biometric", "face", "fingerprint", "microphone", "camera", "listen", "recor
30 |         title: "■ Continuous Surveillance & Tracking",
31 |         reply: `<b>Surveillance & Tracking Clauses</b> allow apps to log your physical location, device se
32 | <b>Red Flag Triggers:</b>
33 | • Background GPS / Geofencing tracking.
34 | • Access to contact lists, audio/video data, or device fingerprinting.<br><br>
35 | <i>Defense: Always restrict background location permissions in your iOS/Android settings!</i>`
36 |     },
37 |     {
38 |         keywords: ["change", "modify", "update", "unilateral", "without notice", "discretion"],
39 |         title: "■ Unilateral Contract Modifications",
40 |         reply: `<b>Unilateral Modification Clauses</b> state that the company can change the terms at any time
41 | <b>Why it's Dangerous:</b>
42 | • A company could introduce data-selling or fee increases tomorrow, and your continued use of the app constitutes "agree
43 | <i>Standard Best Practice: Reputable services notify users via email 30 days prior to major policy changes.</i>
44 |     },
45 |     {
46 |         keywords: ["delete", "deletion", "retain", "retention", "forever", "indefinitely", "store", "erase"],
47 |         title: "■ Account Deletion & Data Retention",
48 |         reply: `<b>Data Retention Clauses</b> specify how long a platform keeps your data after you delete you
49 | <b>Watch out for:</b>
50 | • <i>"We retain backup copies of user content indefinitely."</i>
51 | • <i>"Deleted accounts may take up to 90 days or longer to erase from servers."</i><br><br>
52 | <i>GDPR Right: Under GDPR Article 17 ("Right to Erasure"), platforms are legally mandated to erase personal da
53 |     },
54 |     {
55 |         keywords: ["copyright", "intellectual property", "license", "royalty-free", "own", "ownership", "photos", "content
56 |         title: "■ Content Rights & IP Ownership",
57 |         reply: `<b>IP Ownership Clauses</b> determine who owns the photos, text, or files you upload.<br>
58 | <b>Common Trap:</b> While policies claim "you retain ownership", hidden text grants them a <i>"worldwi
59 | <i>Risk: This allows companies to use your uploaded photos to train AI models or use them in ads without payin
60 |     },
61 |     {
62 |         keywords: ["gdpr", "ccpa", "right", "rights", "privacy law", "opt out", "opt-out", "erase", "access"],
63 |         title: "■ Digital Privacy Laws (GDPR & CCPA)",
64 |         reply: `<b>Your Legal Privacy Rights:</b><br><br>
65 | 1. <b>Right to Access:</b> Request a complete zip file of all data collected on you.<br>
66 | 2. <b>Right to Erasure / Forgotten:</b> Request full deletion of your profile.<br>
67 | 3. <b>Right to Opt-Out:</b> Require companies to cease selling/sharing data.<br>

```

```

68 | 4. <b>Right to Portability:</b> Transfer your data to another platform.`
69 | },
70 | {
71 |     keywords: ["score", "calculate", "how it works", "corruptx", "algorithm", "weight", "negation"],
72 |     title: "■ How CorruptX Risk Scoring Works",
73 |     reply: `<b>CorruptX Dual Engine Scoring:</b><br><br>
74 | • <b>Clause Segmentation:</b> Splits policies into individual semantic clauses.<br>
75 | • <b>Category Pattern Matching:</b> Scans against 14 cybersecurity threat categories (Arbitration, Data Sell
76 | • <b>Negation Protection:</b> Recognizes privacy promises like <i>"We DO NOT sell data"</i>, so y
77 | • <b>Weighted Risk Score (0-100):</b> 0 = Maximum Privacy, 100 = Severe Privacy & Legal Violations.`
78 | },
79 | {
80 |     keywords: ["scan", "scanned", "my document", "my policy", "results", "red flag", "flagged"],
81 |     isScannedDocQuery: true
82 | }
83 | ];
84 |
85 | // DOM Elements setup
86 | document.addEventListener("DOMContentLoaded", () => {
87 |     injectChatbotUI();
88 |     initChatbotEvents();
89 | });
90 |
91 | function injectChatbotUI() {
92 |     if (document.getElementById("tc-chatbot-widget")) return;
93 |
94 |     const widgetHtml = `
95 |         <div id="tc-chatbot-widget">
96 |             <!-- Floating Trigger Button -->
97 |             <button id="tc-chatbot-toggle" title="Open T&C AI Privacy Assistant">
98 |                 <div class="chatbot-toggle-icon">
99 |                     <svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="cu
100 |                 </div>
101 |                 <span class="chatbot-toggle-text">AI Privacy Advisor</span>
102 |                 <span class="chatbot-pulse-dot"></span>
103 |             </button>
104 |
105 |             <!-- Chat Panel Window -->
106 |             <div id="tc-chatbot-modal" class="chatbot-modal hidden">
107 |                 <!-- Header -->
108 |                 <div class="chatbot-header">
109 |                     <div class="chatbot-header-info">
110 |                         <div class="chatbot-avatar">■</div>
111 |                     </div>
112 |                     <div class="chatbot-title">CorruptX Privacy Assistant</div>
113 |                     <div class="chatbot-status"><span class="status-indicator-dot"></span> Online • NLP Cy
114 |                     </div>
115 |                 </div>
116 |                 <div class="chatbot-header-actions">
117 |                     <button id="chatbot-clear-btn" title="Clear Chat History">■</button>
118 |                     <button id="chatbot-close-btn" title="Close Panel">X</button>
119 |                 </div>
120 |             </div>
121 |
122 |             <!-- Body / Messages -->
123 |             <div id="chatbot-messages" class="chatbot-messages">
124 |                 <div class="chat-message assistant">
125 |                     <div class="message-content">
126 |                         ■ Hello! I am your <b>CorruptX AI Privacy Assistant</b>.<br><br>
127 |                         Ask me anything about Terms & Conditions, privacy rights, data selling, forced arbitration, or your
128 |                     </div>
129 |                 </div>
130 |             </div>
131 |
132 |             <!-- Suggestion Chips -->
133 |             <div id="chatbot-chips" class="chatbot-chips">
134 |                 <button class="chip-btn" data-query="Does my scanned policy sell data?">■ Analyze my scanned policy<
135 |                 <button class="chip-btn" data-query="What is forced arbitration?">■ Forced Arbitration</button>
136 |                 <button class="chip-btn" data-query="Is my data sold to third parties?">■ Data Selling</button>

```

```

137 |         <button class="chip-btn" data-query="What are my GDPR rights?">GDPR Rights</button>
138 |     </div>
139 |
140 |     <!-- Input Footer -->
141 |     <div class="chatbot-footer">
142 |         <input type="text" id="chatbot-input" placeholder="Ask a question about T&C clauses..." autocomplete=
143 |         <button id="chatbot-send-btn" title="Send Message">
144 |             <svg xmlns="http://www.w3.org/2000/svg" width="18" height="18" viewBox="0 0 24 24" fill="none" stroke=
145 |         </button>
146 |     </div>
147 | </div>
148 | </div>
149 | `;
150 |
151 | document.body.insertAdjacentHTML("beforeend", widgetHtml);
152 | }
153 |
154 | function initChatbotEvents() {
155 |     const toggleBtn = document.getElementById("tc-chatbot-toggle");
156 |     const closeBtn = document.getElementById("chatbot-close-btn");
157 |     const clearBtn = document.getElementById("chatbot-clear-btn");
158 |     const modal = document.getElementById("tc-chatbot-modal");
159 |     const sendBtn = document.getElementById("chatbot-send-btn");
160 |     const inputEl = document.getElementById("chatbot-input");
161 |     const chipsContainer = document.getElementById("chatbot-chips");
162 |
163 |     if (!toggleBtn || !modal) return;
164 |
165 |     toggleBtn.addEventListener("click", () => {
166 |         modal.classList.toggle("hidden");
167 |         if (!modal.classList.contains("hidden")) {
168 |             inputEl.focus();
169 |             scrollToBottom();
170 |         }
171 |     });
172 |
173 |     closeBtn.addEventListener("click", () => {
174 |         modal.classList.add("hidden");
175 |     });
176 |
177 |     clearBtn.addEventListener("click", () => {
178 |         const messagesEl = document.getElementById("chatbot-messages");
179 |         messagesEl.innerHTML = `
180 |             <div class="chat-message assistant">
181 |                 <div class="message-content">
182 |                     Conversation cleared. What privacy or T&C topic would you like to explore?
183 |                 </div>
184 |             </div>
185 |         `;
186 |     });
187 |
188 |     sendBtn.addEventListener("click", () => handleUserSend());
189 |
190 |     inputEl.addEventListener("keydown", (e) => {
191 |         if (e.key === "Enter") {
192 |             e.preventDefault();
193 |             handleUserSend();
194 |         }
195 |     });
196 |
197 |     if (chipsContainer) {
198 |         chipsContainer.addEventListener("click", (e) => {
199 |             const chip = e.target.closest(".chip-btn");
200 |             if (chip) {
201 |                 const query = chip.getAttribute("data-query");
202 |                 if (query) {
203 |                     inputEl.value = query;
204 |                     handleUserSend();
205 |                 }

```

```

206 |     }
207 |   });
208 | }
209 | }
210 |
211 | async function handleUserSend() {
212 |   const inputEl = document.getElementById("chatbot-input");
213 |   const text = inputEl.value ? inputEl.value.trim() : "";
214 |   if (!text) return;
215 |
216 |   // Append user message
217 |   appendMessage("user", text);
218 |   inputEl.value = "";
219 |
220 |   // Show typing indicator
221 |   const typingId = appendTypingIndicator();
222 |   scrollToBottom();
223 |
224 |   // Small natural delay
225 |   await sleep(400);
226 |
227 |   // Try backend AI endpoint first if online, else client engine
228 |   let responseHtml = "";
229 |   if (window.AppState.isBackendAvailable) {
230 |     try {
231 |       const activeScan = window.getActiveScan ? window.getActiveScan() : null;
232 |       const res = await fetch((window.AppState.apiBase || "") + "/api/chat", {
233 |         method: "POST",
234 |         headers: { "Content-Type": "application/json" },
235 |         body: JSON.stringify({ message: text, active_scan: activeScan })
236 |       });
237 |       if (res.ok) {
238 |         const data = await res.json();
239 |         responseHtml = data.reply;
240 |       }
241 |     } catch (e) {
242 |       console.warn("Backend chat endpoint unavailable, falling back to offline assistant:", e);
243 |     }
244 |   }
245 |
246 |   if (!responseHtml) {
247 |     responseHtml = generateOfflineResponse(text);
248 |   }
249 |
250 |   removeTypingIndicator(typingId);
251 |   appendMessage("assistant", responseHtml);
252 |   scrollToBottom();
253 | }
254 |
255 | function generateOfflineResponse(query) {
256 |   const qLower = query.toLowerCase();
257 |   const activeScan = window.getActiveScan ? window.getActiveScan() : null;
258 |
259 |   // Check if query is about the scanned document
260 |   const isDocQuery = qLower.includes("my policy") || qLower.includes("scanned") || qLower.includes("my document") || qLower.includes("my scan");
261 |
262 |   if (isDocQuery) {
263 |     if (!activeScan) {
264 |       return `<b>No active document scan found.</b><br>Please paste a policy into the <a href="https://www.corruptx.com/policy">policy</a>.`;
265 |     }
266 |
267 |     const score = activeScan.risk_score || 0;
268 |     const level = activeScan.risk_level || "SAFE";
269 |     const docName = activeScan.document_name || "Uploaded Policy";
270 |     const redFlagsCount = activeScan.red_flags || 0;
271 |     const flagged = activeScan.flagged_clauses || [];
272 |
273 |     if (qLower.includes("sell") || qLower.includes("sharing") || qLower.includes("third party")) {
274 |       const dataSellingFlag = flagged.find(c => (c.matches || []).some(m => m.category_id === "DATA_SELLING") || c.category_id === "DATA_SELLING");

```

```

275 |         if (dataSellingFlag) {
276 |             return `■■ <b>Data Selling Detected in "${docName}":</b><br><br><br>
277 | Line snippet: <i>${(dataSellingFlag.text || "").substring(0, 150)}...</i><br><br>
278 | This policy explicitly permits sharing or monetizing user data with third parties.`;
279 |         } else {
280 |             return `■ <b>No explicit Data Selling clauses</b> were flagged in your scan of <b>"${docName}
281 |         }
282 |     }
283 |
284 |     if (qLower.includes("arbitration") || qLower.includes("court") || qLower.includes("sue")) {
285 |         const arbFlag = flagged.find(c => (c.matches || []).some(m => m.category_id === "ARBITRATION_DISPUTES"));
286 |         if (arbFlag) {
287 |             return `■■ <b>Forced Arbitration Clause Found in "${docName}":</b><br><br>
288 | Snippet: <i>${(arbFlag.text || "").substring(0, 150)}...</i><br><br>
289 | You waive your right to sue in court or join class actions under this contract.`;
290 |         } else {
291 |             return `■ <b>No Mandatory Arbitration clauses</b> were detected in <b>"${docName}"</b>
292 |         }
293 |     }
294 |
295 |     // Default scan summary response
296 |     let flagsList = flagged.slice(0, 3).map(f => `• <b>${f.matches[0]?.category_name || "Risk"}:</b> <
297 |     return `■ <b>Summary for "${docName}":</b><br><br>
298 | • <b>Risk Score:</b> ${score}/100 (<span style="color:${level==='HIGH RISK'?`var(--accent-crimson)`:`
299 | • <b>Red Flags Detected:</b> ${redFlagsCount}<br><br>
300 | ${flagsList ? "<b>Top Highlighted Flags:</b>" + flagsList : "No critical red flags detected."}<br>
301 | <a href="results.html" style="color:var(--accent-cyan); font-weight:bold;">View full interactive report </a>
302 | }
303 |
304 | // Knowledge base matching
305 | for (const item of PRIVACY_KNOWLEDGE) {
306 |     if (item.isScannedDocQuery) continue;
307 |     if (item.keywords.some(kw => qLower.includes(kw))) {
308 |         return item.reply;
309 |     }
310 | }
311 |
312 | // Default Fallback
313 | return `I evaluated your query about <b>"${query}"</b>.<br><br>
314 | Here are key privacy principles to keep in mind:<br>
315 | 1. <b>Always check for Mandatory Arbitration</b> clauses before accepting.<br>
316 | 2. <b>Look out for "Data Sharing"</b> with ambiguous "partners".<br>
317 | 3. <b>Check if terms change without notice.</b><br><br>
318 | You can paste any policy into the <a href="scanner.html" style="color:var(--accent-cyan);">CorruptX Scanner</a>
319 | }
320 |
321 | function appendMessage(sender, htmlContent) {
322 |     const messagesEl = document.getElementById("chatbot-messages");
323 |     if (!messagesEl) return;
324 |
325 |     const msgDiv = document.createElement("div");
326 |     msgDiv.className = `chat-message ${sender}`;
327 |     msgDiv.innerHTML = `<div class="message-content">${htmlContent}</div>`;
328 |     messagesEl.appendChild(msgDiv);
329 | }
330 |
331 | function appendTypingIndicator() {
332 |     const messagesEl = document.getElementById("chatbot-messages");
333 |     if (!messagesEl) return null;
334 |
335 |     const id = "typing_" + Date.now();
336 |     const typingDiv = document.createElement("div");
337 |     typingDiv.id = id;
338 |     typingDiv.className = "chat-message assistant typing";
339 |     typingDiv.innerHTML = `
340 | <div class="message-content">
341 | <span class="typing-dot"></span>
342 | <span class="typing-dot"></span>
343 | <span class="typing-dot"></span>

```

```
344 |     &lt;/div>;
345 |     `;
346 |     messagesEl.appendChild(typingDiv);
347 |     return id;
348 | }
349 |
350 | function removeTypingIndicator(id) {
351 |     if (!id) return;
352 |     const el = document.getElementById(id);
353 |     if (el) el.remove();
354 | }
355 |
356 | function scrollToBottom() {
357 |     const messagesEl = document.getElementById("chatbot-messages");
358 |     if (messagesEl) {
359 |         messagesEl.scrollTop = messagesEl.scrollHeight;
360 |     }
361 | }
362 |
363 | function sleep(ms) {
364 |     return new Promise(resolve => setTimeout(resolve, ms));
365 | }
366 | })();
```

File: frontend/js/client_analyzer.js

Size: 12789 bytes

```

1 | /**
2 |  * CorruptX T&C Red Flag Scanner - Client-Side Analysis Engine
3 |  * 100% Offline Standalone Architecture
4 |  * Provides identical NLP rule processing and scoring when running in pure browser mode.
5 |  */
6 |
7 | class ClientClauseAnalyzer {
8 |   constructor(rulesData) {
9 |     this.rulesData = rulesData || window.TC_RISK_RULES || { categories: [] };
10 |    this.categories = this.rulesData.categories || [];
11 |    this.compiledRules = [];
12 |    this.initRules();
13 |  }
14 |
15 |   initRules() {
16 |     this.compiledRules = this.categories.map(cat => {
17 |       const compiledPatterns = [];
18 |       (cat.patterns || []).forEach(patternStr => {
19 |         try {
20 |           compiledPatterns.push(new RegExp(patternStr, "i"));
21 |         } catch (e) {
22 |           console.warn("Regex compile failed:", patternStr, e);
23 |         }
24 |       });
25 |
26 |       return {
27 |         id: cat.id,
28 |         name: cat.name,
29 |         severity: cat.severity || "LOW",
30 |         weight: cat.weight || 5,
31 |         color: cat.color || "#38bdf8",
32 |         explanation: cat.explanation || "",
33 |         why_it_matters: cat.why_it_matters || "",
34 |         recommendation: cat.recommendation || "",
35 |         keywords: (cat.keywords || []).map(k => k.toLowerCase()),
36 |         patterns: compiledPatterns
37 |       };
38 |     });
39 |   }
40 |
41 |   cleanText(rawText) {
42 |     if (!rawText) return "";
43 |     let cleaned = rawText
44 |       .replace(/["]/g, "'")
45 |       .replace(/[']/g, '"')
46 |       .replace(/[--]/g, " - ")
47 |       .replace(/\r\n/g, "\n")
48 |       .replace(/\r/g, "\n");
49 |     return cleaned.trim();
50 |   }
51 |
52 |   segmentClauses(text) {
53 |     const cleaned = this.cleanText(text);
54 |     if (!cleaned) return [];
55 |
56 |     // Mask common abbreviations
57 |     const abbrMap = [
58 |       { regex: /\be\.g\.b/gi, rep: "EG_TEMP_MARKER" },
59 |       { regex: /\bi\.e\.b/gi, rep: "IE_TEMP_MARKER" },
60 |       { regex: /\bU\.S\.b/gi, rep: "US_TEMP_MARKER" },
61 |       { regex: /\betc\.b/gi, rep: "ETC_TEMP_MARKER" },
62 |       { regex: /\bInc\.b/gi, rep: "INC_TEMP_MARKER" },
63 |       { regex: /\bLtd\.b/gi, rep: "LTD_TEMP_MARKER" },
64 |       { regex: /\bLLC\.b/gi, rep: "LLC_TEMP_MARKER" },
65 |       { regex: /\bSec\.b/gi, rep: "SEC_TEMP_MARKER" },
66 |       { regex: /\bNo\.b/gi, rep: "NO_TEMP_MARKER" },
67 |       { regex: /\bv\.b/gi, rep: "V_TEMP_MARKER" },

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68 |     { regex: /\bvs\b/gi, rep: "VS_TEMP_MARKER" }
69 | ];
70 |
71 | let masked = cleaned;
72 | abbrMap.forEach(item => {
73 |     masked = masked.replace(item.regex, item.rep);
74 | });
75 |
76 | // Protect section numbering
77 | masked = masked.replace(/(\d)\.(\d)/g, "$1DECIMAL_DOT_MARKER$2");
78 |
79 | // Split on paragraphs or sentence punctuation followed by capital/digit or list headers
80 | const chunks = masked.split(/(?:\n\s*\n|(?<=[.!?])\s+(?=[A-Z0-9"])|(?<=[.!?])\s*(?=[0-9]+\.\s|[A-Z]\.\s|\([a-z0-9
81 |
82 | const clauses = [];
83 | chunks.forEach(chunk => {
84 |     let restored = chunk;
85 |     restored = restored.replace(/EG_TEMP_MARKER/g, "e.g.");
86 |     restored = restored.replace(/IE_TEMP_MARKER/g, "i.e.");
87 |     restored = restored.replace(/US_TEMP_MARKER/g, "U.S.");
88 |     restored = restored.replace(/ETC_TEMP_MARKER/g, "etc.");
89 |     restored = restored.replace(/INC_TEMP_MARKER/g, "Inc.");
90 |     restored = restored.replace(/LTD_TEMP_MARKER/g, "Ltd.");
91 |     restored = restored.replace(/LLC_TEMP_MARKER/g, "LLC.");
92 |     restored = restored.replace(/SEC_TEMP_MARKER/g, "Sec.");
93 |     restored = restored.replace(/NO_TEMP_MARKER/g, "No.");
94 |     restored = restored.replace(/V_TEMP_MARKER/g, "v.");
95 |     restored = restored.replace(/VS_TEMP_MARKER/g, "vs.");
96 |     restored = restored.replace(/DECIMAL_DOT_MARKER/g, ".");
97 |     restored = restored.replace(/\s+/g, " ").trim();
98 |
99 |     if (restored.length >= 12 || /[a-zA-Z]{3,}/.test(restored)) {
100 |         clauses.push(restored);
101 |     }
102 | });
103 |
104 | return clauses.length > 0 ? clauses : (cleaned ? [cleaned] : []);
105 | }
106 |
107 | isNegatedProtection(clauseText, categoryId) {
108 |     const clauseLower = clauseText.toLowerCase();
109 |
110 |     const negationRules = {
111 |         data_sharing: [
112 |             /(?:we|company|our\s+service)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:sell|share|disclose|rent|monetize|
113 |             /no\s+(?:third-party\s+sale|selling|sharing\s+with\s+third\s+parties)/,
114 |             /(?:we\s+do\s+not|never)\s+disclose\s+personal\s+data/
115 |         ],
116 |         tracking: [
117 |             /(?:we|company|our\s+service)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:track|use\s+tracking|record\s+keys
118 |             /(?:no|zero)\s+(?:tracking|fingerprinting|telemetry|session\s+replay)/,
119 |             /do\s+not\s+use\s+(?:tracking\s+cookies|device\s+fingerprinting|analytics\s+pixels)/
120 |         ],
121 |         data_collection: [
122 |             /(?:we|company)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+collect\s+(?:biometric|location|sensitive|contacts|
123 |             /do\s+not\s+collect\s+location\s+data/
124 |         ],
125 |         location_tracking: [
126 |             /(?:we|company)\s+(?:do\s+not|will\s+not|does\s+not|never)\s+(?:collect|track)\s+(?:your\s+)?location/,
127 |             /do\s+not\s+collect\s+location\s+data/
128 |         ],
129 |         advertising_profiling: [
130 |             /no\s+advertising\s+networks\s+are\s+embedded/,
131 |             /(?:we|company)\s+(?:do\s+not|will\s+not|never)\s+(?:serve\s+targeted|profile|sell\s+ads)/
132 |         ],
133 |         arbitration: [
134 |             /(?:we|company)\s+(?:do\s+not|will\s+not)\s+enforce\s+(?:mandatory\s+)?(?:binding\s+)?arbitration/,
135 |             /do\s+not\s+enforce\s+mandatory\s+binding\s+arbitration/
136 |         ]

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137 |     };
138 |
139 |     const rules = negationRules[categoryId] || [];
140 |     for (let r of rules) {
141 |         if (r.test(clauseLower)) return true;
142 |     }
143 |     return false;
144 | }
145 |
146 | analyzeClause(clauseText, index) {
147 |     const clauseLower = clauseText.toLowerCase();
148 |     const matchedCategories = [];
149 |
150 |     for (const rule of this.compiledRules) {
151 |         if (this.isNegatedProtection(clauseText, rule.id)) {
152 |             continue;
153 |         }
154 |
155 |         let matchedFlag = false;
156 |         let matchedSnippet = null;
157 |
158 |         for (const pattern of rule.patterns) {
159 |             const m = clauseText.match(pattern);
160 |             if (m) {
161 |                 matchedFlag = true;
162 |                 matchedSnippet = m[0];
163 |                 break;
164 |             }
165 |         }
166 |
167 |         if (!matchedFlag) {
168 |             for (const kw of rule.keywords) {
169 |                 if (clauseLower.includes(kw)) {
170 |                     matchedFlag = true;
171 |                     matchedSnippet = kw;
172 |                     break;
173 |                 }
174 |             }
175 |         }
176 |
177 |         if (matchedFlag) {
178 |             matchedCategories.push({
179 |                 category_id: rule.id,
180 |                 category_name: rule.name,
181 |                 severity: rule.severity,
182 |                 weight: rule.weight,
183 |                 color: rule.color,
184 |                 explanation: rule.explanation,
185 |                 why_it_matters: rule.why_it_matters,
186 |                 recommendation: rule.recommendation,
187 |                 snippet: matchedSnippet
188 |             });
189 |         }
190 |     }
191 |
192 |     const rank = { HIGH: 3, MEDIUM: 2, LOW: 1 };
193 |
194 |     if (matchedCategories.length > 0) {
195 |         matchedCategories.sort((a, b) => {
196 |             const rDiff = (rank[b.severity] || 0) - (rank[a.severity] || 0);
197 |             return rDiff !== 0 ? rDiff : b.weight - a.weight;
198 |         });
199 |
200 |         const top = matchedCategories[0];
201 |         return {
202 |             index: index,
203 |             text: clauseText,
204 |             is_flagged: true,
205 |             category_id: top.category_id,

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206 |         category_name: top.category_name,
207 |         severity: top.severity,
208 |         weight: top.weight,
209 |         color: top.color,
210 |         explanation: top.explanation,
211 |         why_it_matters: top.why_it_matters,
212 |         recommendation: top.recommendation,
213 |         all_matched_categories: matchedCategories.map(m => m.category_name)
214 |     };
215 | } else {
216 |     return {
217 |         index: index,
218 |         text: clauseText,
219 |         is_flagged: false,
220 |         category_id: null,
221 |         category_name: "Standard Clause",
222 |         severity: "SAFE",
223 |         weight: 0,
224 |         color: "#00ff88",
225 |         explanation: "No elevated risk indicators detected in this clause.",
226 |         why_it_matters: "Standard service provision or routine operational statement.",
227 |         recommendation: "Clause appears customary; verify context if unsure.",
228 |         all_matched_categories: []
229 |     };
230 | }
231 | }
232 |
233 | calculateRiskScore(totalClauses, flaggedClauses) {
234 |     if (totalClauses <= 0 || flaggedClauses.length === 0) {
235 |         return 5;
236 |     }
237 |
238 |     const highCount = flaggedClauses.filter(c => c.severity === "HIGH").length;
239 |     const medCount = flaggedClauses.filter(c => c.severity === "MEDIUM").length;
240 |     const lowCount = flaggedClauses.filter(c => c.severity === "LOW").length;
241 |
242 |     const weightedSum = (highCount * 16) + (medCount * 8) + (lowCount * 4);
243 |     const density = flaggedClauses.length / Math.max(totalClauses, 1);
244 |     const densityBonus = Math.min(20, Math.floor(density * 35));
245 |
246 |     let rawScore = Math.floor((weightedSum * 0.82) + densityBonus);
247 |
248 |     if (highCount >= 3) {
249 |         rawScore = Math.max(rawScore, 68);
250 |     } else if (highCount >= 1) {
251 |         rawScore = Math.max(rawScore, 42);
252 |     } else if (medCount >= 2) {
253 |         rawScore = Math.max(rawScore, 35);
254 |     }
255 |
256 |     return Math.max(0, Math.min(100, rawScore));
257 | }
258 |
259 | determineRiskLevel(score) {
260 |     if (score <= 30) return "SAFE";
261 |     if (score <= 60) return "CAUTION";
262 |     return "HIGH RISK";
263 | }
264 |
265 | scan(text, docName = "Submitted Policy") {
266 |     if (!text || text.trim().length === 0) {
267 |         return {
268 |             error: "Input text is empty. Please paste or upload a policy to scan.",
269 |             risk_score: 0,
270 |             risk_level: "SAFE",
271 |             clauses_analyzed: 0,
272 |             red_flags: 0
273 |         };
274 |     }

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```

275 |
276 |     const clauses = this.segmentClauses(text);
277 |     const totalClauses = clauses.length;
278 |     const analyzedClauses = [];
279 |     const flaggedClauses = [];
280 |     const categoryCounts = {};
281 |
282 |     clauses.forEach((cl, idx) => {
283 |         const res = this.analyzeClause(cl, idx);
284 |         analyzedClauses.append ? analyzedClauses.append(res) : analyzedClauses.push(res);
285 |         if (res.is_flagged) {
286 |             flaggedClauses.push(res);
287 |             const catId = res.category_id;
288 |             if (!categoryCounts[catId]) {
289 |                 categoryCounts[catId] = {
290 |                     category_id: catId,
291 |                     category_name: res.category_name,
292 |                     severity: res.severity,
293 |                     color: res.color,
294 |                     count: 0,
295 |                     clauses: []
296 |                 };
297 |             }
298 |             categoryCounts[catId].count++;
299 |             categoryCounts[catId].clauses.push(res.index);
300 |         }
301 |     });
302 |
303 |     const score = this.calculateRiskScore(totalClauses, flaggedClauses);
304 |     const level = this.determineRiskLevel(score);
305 |
306 |     const severityCounts = {
307 |         HIGH: flaggedClauses.filter(c => c.severity === "HIGH").length,
308 |         MEDIUM: flaggedClauses.filter(c => c.severity === "MEDIUM").length,
309 |         LOW: flaggedClauses.filter(c => c.severity === "LOW").length
310 |     };
311 |
312 |     const sortedCategories = Object.values(categoryCounts).sort((a, b) => {
313 |         const diff = b.count - a.count;
314 |         if (diff !== 0) return diff;
315 |         const rank = { HIGH: 3, MEDIUM: 2, LOW: 1 };
316 |         return (rank[b.severity] || 0) - (rank[a.severity] || 0);
317 |     });
318 |
319 |     const summary = this.generateSummary(score, level, severityCounts, totalClauses);
320 |
321 |     return {
322 |         document_name: docName,
323 |         analyzed_at: new Date().toISOString(),
324 |         risk_score: score,
325 |         risk_level: level,
326 |         summary: summary,
327 |         clauses_analyzed: totalClauses,
328 |         red_flags: flaggedClauses.length,
329 |         severity_counts: severityCounts,
330 |         categories: sortedCategories,
331 |         flagged_clauses: flaggedClauses,
332 |         all_clauses: analyzedClauses,
333 |         engine_mode: "client-offline",
334 |         disclaimer: "This tool provides automated cybersecurity risk analysis for informational purposes and is not a subs
335 |     };
336 | }
337 |
338 | generateSummary(score, level, severities, totalClauses) {
339 |     const high = severities.HIGH || 0;
340 |     const med = severities.MEDIUM || 0;
341 |     const low = severities.LOW || 0;
342 |     const totalFlags = high + med + low;
343 |

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```
344 |     if (level === "HIGH RISK") {
345 |         return `CRITICAL PRIVACY & LEGAL EXPOSURE DETECTED. Analysis identified ${totalFlags} red flags (${high} High-
346 |     } else if (level === "CAUTION") {
347 |         return `MODERATE USER EXPOSURE DETECTED. Identified ${totalFlags} flagged clauses (${high} High, ${med} Medium) ou
348 |     } else {
349 |         return `ACCEPTABLE RISK PROFILE. Minimal legal or data exposure detected across ${totalClauses} clauses. Only ${to
350 |     }
351 | }
352 | }
353 |
354 | // Global client-side scanner instance
355 | window.clientClauseAnalyzer = new ClientClauseAnalyzer(window.TC_RISK_RULES);
```

File: frontend/js/rules_data.js

Size: 20648 bytes

```

1 | /**
2 |  * CorruptX T&C Risk Taxonomy - Client-Side Rule Definitions
3 |  * Bundled for 100% true offline standalone capability.
4 |  */
5 | window.TC_RISK_RULES = {
6 |   "version": "1.0.0",
7 |   "taxonomy_name": "CorruptX CyberSecurity T&C Risk Taxonomy",
8 |   "categories": [
9 |     {
10 |      "id": "data_sharing",
11 |      "name": "Data Sharing",
12 |      "severity": "HIGH",
13 |      "weight": 15,
14 |      "color": "#ff3366",
15 |      "description": "Sharing, selling, transferring, or disclosing personal data to external third parties, business pa
16 |      "patterns": [
17 |        "share(?:s|d|ing)?\\s+(?:your|user|personal|sensitive)\\s+(?:information|data|details)\\s+with\\s+third\\s+parti
18 |        "disclose(?:s|d|ing)?\\s+(?:your|personal|user)\\s+(?:information|data)\\s+to\\s+(?:third\\s+parties|affiliates|
19 |        "sell(?:s|ing)?\\s+(?:or\\s+rent(?:s|ing)?\\s+)?(?:your|personal|user)\\s+(?:data|information)",
20 |        "transfer(?:s|red|ring)?\\s+(?:your|personal)\\s+data\\s+to\\s+(?:other\\s+entities|third\\s+parties|external)",
21 |        "commercial\\s+partners\\s+may\\s+(?:receive|access|use)\\s+your\\s+data",
22 |        "we\\s+reserve\\s+the\\s+right\\s+to\\s+share\\s+(?:aggregated|anonymized|de-identified|any)?\\s*(?:information|
23 |        "monetize(?:s|d|ing)?\\s+(?:user|personal)\\s+information",
24 |        "in\\s+the\\s+event\\s+of\\s+a\\s+merger(?:,|\\s+or)\\s+(?:acquisition|sale|bankruptcy).*data.*transferred"
25 |      ],
26 |      "keywords": [
27 |        "sell your data",
28 |        "share your information with third parties",
29 |        "disclose personal data to partners",
30 |        "monetize user information",
31 |        "share with advertisers"
32 |      ],
33 |      "explanation": "This clause allows the service to share, distribute, or sell your personal details with third-part
34 |      "why_it_matters": "Once your information is handed over to outside parties, you lose direct control over how it is
35 |      "recommendation": "Check if an opt-out mechanism is available or avoid submitting sensitive confidential data."
36 |    },
37 |    {
38 |      "id": "tracking",
39 |      "name": "Tracking & Telemetry",
40 |      "severity": "HIGH",
41 |      "weight": 14,
42 |      "color": "#ff3366",
43 |      "description": "Continuous device fingerprinting, behavioral tracking, cross-site telemetry, or keystroke monitori
44 |      "patterns": [
45 |        "track(?:s|ed|ing)?\\s+your\\s+(?:browsing|activity|behavior|location|online\\s+habits)",
46 |        "fingerprint(?:ing)?\\s+(?:your\\s+device|hardware|browser)",
47 |        "cross-site\\s+tracking",
48 |        "canvas\\s+fingerprinting",
49 |        "beacon(?:s)?|pixel\\s+tags?|web\\s+bugs?",
50 |        "record(?:s|ing)?\\s+(?:your\\s+screen|keystrokes|mouse\\s+movements|session\\s+replay)",
51 |        "monitor(?:s|ing)?\\s+(?:all|your)?\\s*(?:communications|browsing\\s+history|actions)",
52 |        "collect(?:s|ed|ing)?\\s+information\\s+about\\s+your\\s+activity\\s+across\\s+(?:third-party|other)\\s+websites
53 |      ],
54 |      "keywords": [
55 |        "cross-site tracking",
56 |        "device fingerprinting",
57 |        "session replay",
58 |        "keystroke recording",
59 |        "track your browsing activity"
60 |      ],
61 |      "explanation": "This clause enables active monitoring, device fingerprinting, or tracking of your online behavior
62 |      "why_it_matters": "Cross-site and hardware telemetry can build a persistent digital shadow profile of your habits,
63 |      "recommendation": "Use privacy-focused browsers, anti-fingerprinting shields, and strict tracker-blocking extensio
64 |    },
65 |    {
66 |      "id": "data_collection",
67 |      "name": "Data Collection",

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68 |         "severity": "HIGH",
69 |         "weight": 12,
70 |         "color": "#ff3366",
71 |         "description": "Excessive collection of sensitive personal data, biometrics, contacts, communications, or device i
72 |         "patterns": [
73 |             "collect(?:s|ed|ing)?\\s+(?:biometric|health|financial|genetic|sensitive)\\s+(?:information|data)",
74 |             "access(?:es)?\\s+(?:to\\s+)?your\\s+(?:contacts|address\\s+book|microphone|camera|stored\\s+files|photos)",
75 |             "scrape(?:s|d|ing)?\\s+(?:data|information|content)",
76 |             "collect(?:s|ed|ing)?\\s+(?:all|any)\\s+information\\s+you\\s+(?:transmit|submit|upload|post|create)",
77 |             "gather(?:s|ed|ing)?\\s+(?:details|information)\\s+from\\s+external\\s+public\\s+and\\s+private\\s+records",
78 |             "including\\s+but\\s+not\\s+limited\\s+to\\s+social\\s+security|credit\\s+card|bank\\s+account"
79 |         ],
80 |         "keywords": [
81 |             "biometric data",
82 |             "access to your contacts",
83 |             "microphone and camera access",
84 |             "sensitive personal data",
85 |             "financial information"
86 |         ],
87 |         "explanation": "The provider reserves extensive rights to gather invasive categories of personal data, including c
88 |         "why_it_matters": "Over-collection expands your attack surface; if the provider suffers a data breach, your sensit
89 |         "recommendation": "Review permissions requested on your device and restrict sensor/contact access."
90 |     },
91 |     {
92 |         "id": "arbitration",
93 |         "name": "Mandatory Arbitration",
94 |         "severity": "HIGH",
95 |         "weight": 14,
96 |         "color": "#ff3366",
97 |         "description": "Forced binding arbitration and waiver of court trials and collective class action lawsuits.",
98 |         "patterns": [
99 |             "binding\\s+arbitration",
100 |             "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+)?(?:a\\s+)?jury\\s+trial",
101 |             "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+participate\\s+in\\s+a\\s+)?class\\s+action",
102 |             "class\\s+action\\s+waiver",
103 |             "disputes\\s+shall\\s+be\\s+resolved\\s+(?:exclusively\\s+)?(?:by|through)\\s+(?:confidential|individual|binding
104 |             "no\\s+class\\s+or\\s+representative\\s+actions?",
105 |             "agree\\s+not\\s+to\\s+sue\\s+in\\s+court"
106 |         ],
107 |         "keywords": [
108 |             "binding arbitration",
109 |             "class action waiver",
110 |             "waive jury trial",
111 |             "resolve disputes through arbitration",
112 |             "no class action"
113 |         ],
114 |         "explanation": "You surrender your constitutional right to take this company to court or participate in a joint co
115 |         "why_it_matters": "Arbitration proceedings are usually private, costly for individuals, and systematically favor c
116 |         "recommendation": "Check if an arbitration opt-out letter can be submitted within 30 days of registration."
117 |     },
118 |     {
119 |         "id": "unilateral_changes",
120 |         "name": "Unilateral Changes",
121 |         "severity": "HIGH",
122 |         "weight": 13,
123 |         "color": "#ff3366",
124 |         "description": "Changing legal terms or privacy policies at any time without notifying users or requiring fresh co
125 |         "patterns": [
126 |             "reserve\\s+the\\s+right\\s+to\\s+(?:modify|change|alter|update)\\s+(?:these\\s+terms|this\\s+agreement|policy)\\s+",
127 |             "(?:modify|change|update)\\s+without\\s+(?:prior\\s+)?notice",
128 |             "continued\\s+use\\s+(?:of\\s+the\\s+service\\s+)?(?:constitutes|shall\\s+constitute|signifies)\\s+(?:your\\s+)?",
129 |             "without\\s+obligation\\s+to\\s+notify\\s+you",
130 |             "effective\\s+immediately\\s+upon\\s+posting"
131 |         ],
132 |         "keywords": [
133 |             "modify without prior notice",
134 |             "at our sole discretion without notice",
135 |             "continued use constitutes acceptance",
136 |             "change these terms at any time"

```

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137 |     ],
138 |     "explanation": "The company can alter policies, fees, or data handling rules at any time, and your continued usage",
139 |     "why_it_matters": "You could wake up to new invasive terms or higher subscription fees without ever receiving an e",
140 |     "recommendation": "Periodically review terms updates or monitor policy changes using change-detection tools."
141 | },
142 | {
143 |     "id": "liability_waiver",
144 |     "name": "Liability Waiver",
145 |     "severity": "MEDIUM",
146 |     "weight": 9,
147 |     "color": "#ffb700",
148 |     "description": "Total disclaimer of liability for data loss, service outages, bugs, or security vulnerabilities.",
149 |     "patterns": [
150 |         "provided\\s+[\\']?as\\s+is[\\']?\\s+and\\s+[\\']?as\\s+available[\\']?",
151 |         "disclaim(?:s)?\\s+(?:all|any)\\s+warranties",
152 |         "(?:under\\s+no\\s+circumstances\\s+shall|in\\s+no\\s+event\\s+shall)\\s+(?:the\\s+company|we|us|our\\s+affiliat",
153 |         "limit(?:ation)?\\s+of\\s+liability\\s+shall\\s+not\\s+exceed\\s+(?:the\\s+amount\\s+paid|\\$\\d+|one\\s+hundred",
154 |         "not\\s+liable\\s+for\\s+(?:any\\s+)?(?:data\\s+loss|security\\s+breaches|unauthorized\\s+access)"
155 |     ],
156 |     "keywords": [
157 |         "provided as is",
158 |         "in no event shall we be liable",
159 |         "disclaim all warranties",
160 |         "not liable for data loss or breach",
161 |         "limitation of liability"
162 |     ],
163 |     "explanation": "The company provides the product 'as-is' and completely disclaims legal responsibility if their so",
164 |     "why_it_matters": "If a critical security bug causes data corruption or identity theft, this clause shields the ve",
165 |     "recommendation": "Never rely solely on this vendor for sole data backups; maintain external independent copies."
166 | },
167 | {
168 |     "id": "auto_renewal",
169 |     "name": "Auto-Renewal & Recurring Billing",
170 |     "severity": "MEDIUM",
171 |     "weight": 8,
172 |     "color": "#ffb700",
173 |     "description": "Automatic recurring subscription renewals, often accompanied by strict cancellation timelines.",
174 |     "patterns": [
175 |         "auto(?:matically)?-?renew(?:s|ed|ing)?",
176 |         "renew(?:s|ed|ing)?\\s+automatically\\s+unless\\s+cancelled",
177 |         "recurring\\s+(?:billing|charge|subscription|fee)",
178 |         "charge(?:s|d)?\\s+your\\s+(?:payment\\s+method|credit\\s+card)\\s+without\\s+(?:additional|further)\\s+authoriz",
179 |         "cancel(?:ation)?\\s+at\\s+least\\s+\\d+\\s+days\\s+prior\\s+to\\s+(?:the\\s+end|renewal)",
180 |         "non-refundable\\s+(?:fees|charges|payments)"
181 |     ],
182 |     "keywords": [
183 |         "automatically renew",
184 |         "recurring subscription",
185 |         "cancel at least 24 hours prior",
186 |         "non-refundable fees",
187 |         "charge without further authorization"
188 |     ],
189 |     "explanation": "Your subscription will silently renew and charge your card automatically unless manually terminate",
190 |     "why_it_matters": "Users frequently get trapped in zombie billing cycles for unused services due to obscure cancel",
191 |     "recommendation": "Set calendar alerts 3-5 days ahead of billing cycles or use virtual credit cards with spend lim",
192 | },
193 | {
194 |     "id": "account_termination",
195 |     "name": "Account Termination",
196 |     "severity": "MEDIUM",
197 |     "weight": 8,
198 |     "color": "#ffb700",
199 |     "description": "Suspension or deletion of user accounts and content without prior notice, justification, or refund",
200 |     "patterns": [
201 |         "terminate\\s+(?:or\\s+suspend\\s+)?(?:your\\s+account|access)\\s+at\\s+(?:any\\s+time|our\\s+sole\\s+discretion",
202 |         "(?:terminate|suspend)\\s+without\\s+(?:prior\\s+)?notice\\s+(?:or\\s+liability)?",
203 |         "for\\s+any\\s+reason\\s+or\\s+(?:for\\s+)?no\\s+reason",
204 |         "forfeit(?:ure)?\\s+(?:of\\s+)?(?:all|any)\\s+(?:content|credits|funds|data)\\s+upon\\s+termination",
205 |         "no\\s+obligation\\s+to\\s+refund\\s+(?:unused\\s+)?fees"

```

```

206 | ],
207 | "keywords": [
208 |     "terminate at our sole discretion",
209 |     "suspend without notice",
210 |     "for any reason or no reason",
211 |     "forfeit content upon termination"
212 | ],
213 | "explanation": "The provider can ban your account, delete your assets, and block your access at will without explanation.",
214 | "why_it_matters": "You risk sudden loss of work, purchases, access to historical records, or communication channels.",
215 | "recommendation": "Regularly export and download backups of your cloud files, projects, and transactions."
216 | },
217 | {
218 |     "id": "third_party_access",
219 |     "name": "Third-Party Access",
220 |     "severity": "HIGH",
221 |     "weight": 11,
222 |     "color": "#ff3366",
223 |     "description": "Permitting outside integrations, ad networks, or vendors to access user accounts and content.",
224 |     "patterns": [
225 |         "third-party\\s+(?:service\\s+providers|vendors|contractors|integrations)\\s+may\\s+access",
226 |         "not\\s+responsible\\s+for\\s+(?:the\\s+privacy\\s+practices|actions|content)\\s+of\\s+third\\s+parties",
227 |         "allow(?:s)?\\s+third\\s+parties\\s+to\\s+(?:serve\\s+ads|collect\\s+data|place\\s+cookies)",
228 |         "integrated\\s+third-party\\s+services\\s+governed\\s+by\\s+their\\s+own\\s+terms",
229 |         "share\\s+with\\s+sub-processors\\s+without\\s+notification"
230 |     ],
231 |     "keywords": [
232 |         "third-party service providers may access",
233 |         "not responsible for third-party privacy",
234 |         "third parties to serve ads and collect",
235 |         "governed by third-party policies"
236 |     ],
237 |     "explanation": "External software vendors or third-party plug-ins can interface directly with your account or read your data.",
238 |     "why_it_matters": "Third-party supply chain vulnerabilities are a leading vector for corporate data leaks and identity theft.",
239 |     "recommendation": "Audit and disconnect unused connected apps and third-party account authorizations."
240 | },
241 | {
242 |     "id": "data_retention",
243 |     "name": "Indefinite Data Retention",
244 |     "severity": "MEDIUM",
245 |     "weight": 7,
246 |     "color": "#ffb700",
247 |     "description": "Retaining personal data, messages, or backups indefinitely even after account deletion.",
248 |     "patterns": [
249 |         "retain(?:s|ed|ing)?\\s+(?:your\\s+)?(?:data|information|content)\\s+(?:indefinitely|for\\s+an\\s+unlimited\\s+period)",
250 |         "may\\s+retain\\s+(?:archived|backup|residual)\\s+copies\\s+even\\s+after\\s+(?:you\\s+delete|account\\s+closure)",
251 |         "no\\s+obligation\\s+to\\s+delete\\s+(?:your\\s+)?(?:information|data)",
252 |         "retain\\s+and\\s+use\\s+your\\s+information\\s+to\\s+comply\\s+with\\s+legal\\s+obligations.*resolve\\s+disputes"
253 |     ],
254 |     "keywords": [
255 |         "retain data indefinitely",
256 |         "retain backup copies even after deletion",
257 |         "as long as we deem necessary",
258 |         "no obligation to delete"
259 |     ],
260 |     "explanation": "Deleting your account does not guarantee deletion of your personal records, which may be archived or retained.",
261 |     "why_it_matters": "Zombie data retained forever creates a permanent target for future database breaches or subpoenas.",
262 |     "recommendation": "Exercise formal GDPR/CCPA 'Right to Erasure' requests where legally applicable."
263 | },
264 | {
265 |     "id": "location_tracking",
266 |     "name": "Location Tracking",
267 |     "severity": "HIGH",
268 |     "weight": 12,
269 |     "color": "#ff3366",
270 |     "description": "Harvesting precise GPS coordinates, Wi-Fi triangulation, cell tower IDs, or continuous background location data.",
271 |     "patterns": [
272 |         "collect(?:s|ed|ing)?\\s+(?:precise|exact|real-time|continuous)?\\s+(?:location|geo|location|gps)\\s+data",
273 |         "location\\s+(?:even\\s+when\\s+the\\s+app\\s+is\\s+closed|in\\s+the\\s+background)",
274 |         "track(?:s|ed|ing)?\\s+(?:your\\s+physical\\s+location|whereabouts|movements)",

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```

275 |         "derive\\s+location\\s+from\\s+(?:ip\\s+address|wi-fi\\s+access\\s+points|cell\\s+towers|bluetooth)"
276 |     ],
277 |     "keywords": [
278 |         "precise geolocation data",
279 |         "location in the background",
280 |         "track your physical location",
281 |         "real-time location tracking"
282 |     ],
283 |     "explanation": "The app tracks or approximates your real-world geographic coordinates, potentially even when running in the background.",
284 |     "why_it_matters": "Real-time location records expose your residence, workplace, daily routines, and intimate personal information.",
285 |     "recommendation": "Set system location permissions to 'Only While Using App' or disable location entirely if not needed.",
286 | },
287 | {
288 |     "id": "advertising_profiling",
289 |     "name": "Advertising & Profiling",
290 |     "severity": "MEDIUM",
291 |     "weight": 8,
292 |     "color": "#ffb700",
293 |     "description": "Targeted behavioral ads, algorithmic psychological profiling, and interest-based segmentation.",
294 |     "patterns": [
295 |         "personalized\\s+(?:advertisements|ads|marketing|promotions)",
296 |         "interest-based\\s+advertising",
297 |         "create\\s+(?:a\\s+)?profile\\s+about\\s+you",
298 |         "target(?:ed)?\\s+advertising\\s+based\\s+on\\s+your\\s+(?:preferences|activity|browsing)",
299 |         "deliver\\s+relevant\\s+ads\\s+by\\s+analyzing\\s+your\\s+(?:interests|profile|demographics)"
300 |     ],
301 |     "keywords": [
302 |         "personalized advertisements",
303 |         "interest-based advertising",
304 |         "create a profile about you",
305 |         "targeted advertising based on"
306 |     ],
307 |     "explanation": "Your activity, searches, and demographic attributes are used to assemble an advertising profile for targeted marketing.",
308 |     "why_it_matters": "Targeted profiling can exploit behavioral vulnerabilities, shape opinions, and expose your privacy.",
309 |     "recommendation": "Opt out of personalized advertising within service settings and reset your advertising ID."
310 | },
311 | {
312 |     "id": "user_rights",
313 |     "name": "User Rights Surrender",
314 |     "severity": "HIGH",
315 |     "weight": 11,
316 |     "color": "#ff3366",
317 |     "description": "Surrendering intellectual property rights, granting irrevocable global content licenses, or moral rights waivers.",
318 |     "patterns": [
319 |         "grant(?:s)?\\s+(?:us\\s+)?(?:a\\s+)?(?:worldwide|perpetual|irrevocable|royalty-free|unrestricted)\\s+(?:license|agreement)",
320 |         "waive(?:s)?\\s+(?:all\\s+)?moral\\s+rights",
321 |         "you\\s+assign\\s+(?:all\\s+)?(?:ownership|intellectual\\s+property|rights)\\s+to\\s+us",
322 |         "we\\s+own\\s+all\\s+(?:feedback|submissions|content|ideas)\\s+provided\\s+by\\s+you",
323 |         "right\\s+to\\s+sublicense\\s+and\\s+commercialize\\s+your\\s+content"
324 |     ],
325 |     "keywords": [
326 |         "perpetual irrevocable royalty-free license",
327 |         "worldwide license to use and modify",
328 |         "waive moral rights",
329 |         "assign all intellectual property",
330 |         "sublicense and commercialize your content"
331 |     ],
332 |     "explanation": "You grant the company a permanent, global, royalty-free license to use, publish, monetize, or adapt your content without restriction.",
333 |     "why_it_matters": "You may lose commercial ownership or credit for creative works, files, or original designs uploaded to the platform.",
334 |     "recommendation": "Avoid storing or creating proprietary work or copyrighted media on platforms demanding broad IP licenses.",
335 | },
336 | {
337 |     "id": "cancellation_restrictions",
338 |     "name": "Cancellation Restrictions",
339 |     "severity": "LOW",
340 |     "weight": 5,
341 |     "color": "#38bdf8",
342 |     "description": "Complex hurdle-filled cancellation procedures, early termination penalties, or restricted contact during disputes.",
343 |     "patterns": [

```

```
344 |         "cancel(?:lation)?\\s+(?:must\\s+be\\s+made\\s+by|only\\s+by)\\s+(?:phone|written\\s+notice|certified\\s+mail)",
345 |         "early\\s+termination\\s+fee(?:s)?",
346 |         "penalt(?:y|ies)\\s+for\\s+(?:early\\s+)?cancellation",
347 |         "must\\s+provide\\s+(?:at\\s+least\\s+)?(?:30|60|90)\\s+days\\s+(?:prior\\s+)?notice\\s+to\\s+cancel",
348 |         "cannot\\s+be\\s+cancelled\\s+online"
349 |     ],
350 |     "keywords": [
351 |         "early termination fee",
352 |         "cancel only by phone",
353 |         "penalty for cancellation",
354 |         "certified mail to cancel"
355 |     ],
356 |     "explanation": "The provider imposes restrictive hurdles or fees to make terminating your account or contract deli
357 |     "why_it_matters": "Dark pattern cancellation flows prevent easy departure, forcing unneeded payments or unnecessar
358 |     "recommendation": "Check the exact cancellation procedures prior to providing payment details or signing long agre
359 | }
360 | ]
361 | };
```

File: frontend/js/app.js

Size: 10522 bytes

```

1 | /**
2 |  * CorruptX - Shared Application Logic
3 |  * Offline-first Architecture & Client Utilities
4 |  */
5 |
6 | const AppState = {
7 |   apiBase: "",
8 |   isBackendAvailable: false,
9 |   offlineMode: false
10 | };
11 |
12 | // SVG Icon Collection (Zero CDN dependencies)
13 | const ICONS = {
14 |   shield: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="cur
15 |   alertTriangle: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stro
16 |   checkCircle: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke
17 |   info: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="curre
18 |   externalLink: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" strok
19 |   copy: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="curre
20 |   trash: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="curre
21 |   search: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="cur
22 |   fileText: `<svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="c
23 | };
24 |
25 | document.addEventListener("DOMContentLoaded", () => {
26 |   initMobileMenu();
27 |   detectBackendStatus();
28 |   highlightActiveNav();
29 | });
30 |
31 | /**
32 |  * Mobile navigation drawer
33 |  */
34 | function initMobileMenu() {
35 |   const toggleBtn = document.querySelector(".mobile-menu-btn");
36 |
37 |   const nav = document.querySelector(".main-nav");
38 |   if (toggleBtn && nav) {
39 |     toggleBtn.addEventListener("click", () => {
40 |       nav.classList.toggle("mobile-open");
41 |     });
42 |   }
43 | }
44 |
45 | /**
46 |  * Active navigation link highlighter
47 |  */
48 | function highlightActiveNav() {
49 |   const path = window.location.pathname;
50 |   const navLinks = document.querySelectorAll(".nav-link");
51 |   navLinks.forEach(link => {
52 |     const href = link.getAttribute("href");
53 |     if (href && (path.endsWith(href) || (href === "/" && (path === "/" || path.endsWith("index.html")))))
54 |       link.classList.add("active");
55 |   });
56 | }
57 |
58 | /**
59 |  * Detect whether the local Flask backend is reachable.
60 |  * If not, gracefully switch to the built-in offline engine.
61 |  */
62 | async function detectBackendStatus() {
63 |   const statusEl = document.getElementById("system-status-indicator");
64 |   const statusText = document.getElementById("system-status-text");
65 |
66 |   // Determine potential API roots
67 |   let candidates = ["/api/health"];

```

```

68 |   if (window.location.protocol === "file:" || window.location.port !== "5000") {
69 |       candidates.unshift("http://127.0.0.1:5000/api/health");
70 |   }
71 |
72 |   let healthy = false;
73 |   for (const url of candidates) {
74 |       try {
75 |           const res = await fetch(url, { method: "GET", mode: "cors", cache: "no-cache" });
76 |           if (res.ok) {
77 |               const data = await res.json();
78 |               if (data.status === "healthy") {
79 |                   AppState.isBackendAvailable = true;
80 |                   AppState.apiBase = url.replace("/api/health", "");
81 |                   healthy = true;
82 |                   break;
83 |               }
84 |           }
85 |       } catch (e) {
86 |           // Backend not running on this candidate
87 |       }
88 |   }
89 |
90 |   if (healthy) {
91 |       if (statusText) statusText.textContent = "ENGINE ONLINE (FLASK)";
92 |   } else {
93 |       AppState.isBackendAvailable = false;
94 |       AppState.offlineMode = true;
95 |       if (statusText) statusText.textContent = "OFFLINE ENGINE ACTIVE";
96 |       if (statusEl) {
97 |           statusEl.style.borderColor = "rgba(0, 242, 255, 0.4)";
98 |           statusEl.style.color = "var(--accent-cyan)";
99 |           const dot = statusEl.querySelector(".status-dot");
100 |           if (dot) {
101 |               dot.style.background = "var(--accent-cyan)";
102 |               dot.style.boxShadow = "0 0 8px var(--accent-cyan)";
103 |           }
104 |       }
105 |   }
106 | }
107 |
108 | /**
109 |  * Toast Notification System
110 |  */
111 | function showToast(message, type = "info") {
112 |     let container = document.getElementById("toast-container");
113 |     if (!container) {
114 |         container = document.createElement("div");
115 |         container.id = "toast-container";
116 |         document.body.appendChild(container);
117 |     }
118 |
119 |     const toast = document.createElement("div");
120 |     toast.className = `toast toast-${type}`;
121 |     toast.innerHTML = `
122 |         <span>${message}</span>
123 |     `;
124 |
125 |     container.appendChild(toast);
126 |     setTimeout(() => {
127 |         toast.style.opacity = "0";
128 |         toast.style.transform = "translateY(10px)";
129 |         setTimeout(() => toast.remove(), 300);
130 |     }, 3500);
131 | }
132 |
133 | /**
134 |  * LocalStorage Scan History & Active Scan Manager
135 |  * Includes memory fallback and auto-pruning to prevent QuotaExceededError
136 |  */

```

```

137 | const HISTORY_KEY = "corruptx_tc_scan_history";
138 | const CURRENT_SCAN_KEY = "corruptx_tc_active_scan";
139 |
140 | // In-memory global fallback to ensure scan is never lost during page transitions
141 | window.__CORRUPTX_ACTIVE_SCAN__ = null;
142 |
143 | function getScanHistory() {
144 |   try {
145 |     const raw = localStorage.getItem(HISTORY_KEY);
146 |     return raw ? JSON.parse(raw) : [];
147 |   } catch (e) {
148 |     console.error("Failed to read scan history:", e);
149 |     return [];
150 |   }
151 | }
152 |
153 | /**
154 |  * Creates a lightweight version of scan data for persistent history storage
155 |  */
156 | function createLightweightScanRecord(scanResult) {
157 |   // Prune large repetitive content to prevent LocalStorage quota overflow
158 |   const cleanResult = { ...scanResult };
159 |
160 |   // Keep only up to 50 clauses in all_clauses for history preview to keep size small
161 |   if (Array.isArray(cleanResult.all_clauses) && cleanResult.all_clauses.length > 50) {
162 |     cleanResult.all_clauses = cleanResult.all_clauses.slice(0, 50).map(c => ({
163 |       index: c.index,
164 |       snippet: (c.text || "").substring(0, 150) + "...",
165 |       is_flagged: c.is_flagged
166 |     }));
167 |   }
168 |
169 |   return {
170 |     id: "scan_" + Date.now() + "_" + Math.random().toString(36).substr(2, 5),
171 |     document_name: scanResult.document_name || "Submitted Policy",
172 |     analyzed_at: scanResult.analyzed_at || new Date().toISOString(),
173 |     risk_score: scanResult.risk_score,
174 |     risk_level: scanResult.risk_level,
175 |     clauses_analyzed: scanResult.clauses_analyzed,
176 |     red_flags: scanResult.red_flags,
177 |     severity_counts: scanResult.severity_counts,
178 |     full_data: cleanResult
179 |   };
180 | }
181 |
182 | function saveScanToHistory(scanResult) {
183 |   try {
184 |     const history = getScanHistory();
185 |     const record = createLightweightScanRecord(scanResult);
186 |
187 |     // Prepend new scan
188 |     history.unshift(record);
189 |     // Keep last 20 scans max
190 |     let trimmed = history.slice(0, 20);
191 |
192 |     try {
193 |       localStorage.setItem(HISTORY_KEY, JSON.stringify(trimmed));
194 |     } catch (quotaErr) {
195 |       console.warn("LocalStorage quota reached, pruning older history items...");
196 |       // Strip full_data from older items if quota is tight
197 |       trimmed = trimmed.slice(0, 5).map((item, idx) => {
198 |         if (idx > 0 && item.full_data) {
199 |           delete item.full_data.all_clauses;
200 |         }
201 |         return item;
202 |       });
203 |       localStorage.setItem(HISTORY_KEY, JSON.stringify(trimmed));
204 |     }
205 |     return record.id;

```

```
206 |     } catch (e) {
207 |         console.error("Failed to save scan history:", e);
208 |     }
209 | }
210 |
211 | function deleteScanHistoryItem(id) {
212 |     try {
213 |         const history = getScanHistory().filter(item => item.id !== id);
214 |         localStorage.setItem(HISTORY_KEY, JSON.stringify(history));
215 |     } catch (e) {
216 |         console.error("Failed to delete item:", e);
217 |     }
218 | }
219 |
220 | function clearScanHistory() {
221 |     try {
222 |         localStorage.removeItem(HISTORY_KEY);
223 |     } catch (e) {
224 |         console.error("Failed to clear history:", e);
225 |     }
226 | }
227 |
228 | function setActiveScan(scanResult) {
229 |     // Always update in-memory cache
230 |     window.__CORRUPTX_ACTIVE_SCAN__ = scanResult;
231 |     try {
232 |         sessionStorage.setItem(CURRENT_SCAN_KEY, JSON.stringify(scanResult));
233 |     } catch (e) {
234 |         console.warn("SessionStorage failed to set active scan, falling back to in-memory/localStorage:", e);
235 |         try {
236 |             // Try saving lightweight version if session storage full
237 |             const light = createLightweightScanRecord(scanResult).full_data;
238 |             sessionStorage.setItem(CURRENT_SCAN_KEY, JSON.stringify(light));
239 |         } catch (e2) {
240 |             console.warn("SessionStorage quota exceeded completely. Active scan preserved in memory.");
241 |         }
242 |     }
243 | }
244 |
245 | function getActiveScan() {
246 |     if (window.__CORRUPTX_ACTIVE_SCAN__) {
247 |         return window.__CORRUPTX_ACTIVE_SCAN__;
248 |     }
249 |     try {
250 |         const raw = sessionStorage.getItem(CURRENT_SCAN_KEY);
251 |         if (raw) {
252 |             const parsed = JSON.parse(raw);
253 |             window.__CORRUPTX_ACTIVE_SCAN__ = parsed;
254 |             return parsed;
255 |         }
256 |     } catch (e) {
257 |         console.error("Failed to load active scan from sessionStorage:", e);
258 |     }
259 |
260 |     // Fallback to latest history scan
261 |     const history = getScanHistory();
262 |     if (history.length > 0 && history[0].full_data) {
263 |         window.__CORRUPTX_ACTIVE_SCAN__ = history[0].full_data;
264 |         return history[0].full_data;
265 |     }
266 |
267 |     return null;
268 | }
```

File: frontend/js/scanner.js

Size: 9570 bytes

```

1 | /**
2 |  * CorruptX - Scanner Page Controller
3 |  * Handles document intake, file drag/drop, sample loading,
4 |  * multi-step scan animation, and offline/online API dispatch.
5 |  */
6 |
7 | document.addEventListener("DOMContentLoaded", () => {
8 |     const policyTextarea = document.getElementById("policy-text");
9 |     const docNameInput = document.getElementById("doc-name-input");
10 |    const charCountEl = document.getElementById("char-count");
11 |    const wordCountEl = document.getElementById("word-count");
12 |    const clauseEstimateEl = document.getElementById("clause-estimate");
13 |
14 |    const btnPaste = document.getElementById("btn-paste");
15 |    const btnClear = document.getElementById("btn-clear");
16 |    const btnLoadDemo = document.getElementById("btn-load-demo");
17 |    const btnScan = document.getElementById("btn-start-scan");
18 |
19 |    const fileInput = document.getElementById("file-upload-input");
20 |    const dropzone = document.getElementById("file-dropzone");
21 |
22 |    const scanningOverlay = document.getElementById("scanning-overlay");
23 |    const progressFill = document.getElementById("scan-progress-fill");
24 |
25 |    // Multi-step elements
26 |    const step1 = document.getElementById("step-1");
27 |    const step2 = document.getElementById("step-2");
28 |    const step3 = document.getElementById("step-3");
29 |    const step4 = document.getElementById("step-4");
30 |
31 |    // Check if URL has ?demo=1 or similar query parameter
32 |    const urlParams = new URLSearchParams(window.location.search);
33 |    if (urlParams.get("demo")) {
34 |        loadDemoPolicy(urlParams.get("demo") || "high_risk_social");
35 |    }
36 |
37 |    // Textarea input event
38 |    if (policyTextarea) {
39 |        policyTextarea.addEventListener("input", updateStats);
40 |        // Initial stats
41 |        updateStats();
42 |    }
43 |
44 |    function updateStats() {
45 |        const text = policyTextarea.value || "";
46 |        const chars = text.length;
47 |        const words = text.trim() ? text.trim().split(/\s+/).length : 0;
48 |        // Rough estimate of clauses (split on periods / newlines)
49 |        const clauses = text.trim() ? Math.max(1, text.split(/(?:\?|!|\.|\n)+/).filter(s => s.trim().length > 10).length) : 0;
50 |
51 |        if (charCountEl) charCountEl.textContent = chars.toLocaleString();
52 |        if (wordCountEl) wordCountEl.textContent = words.toLocaleString();
53 |        if (clauseEstimateEl) clauseEstimateEl.textContent = `~${clauses}`;
54 |    }
55 |
56 |    // Paste button
57 |    if (btnPaste) {
58 |        btnPaste.addEventListener("click", async () => {
59 |            try {
60 |                if (navigator.clipboard && navigator.clipboard.readText) {
61 |                    const clipText = await navigator.clipboard.readText();
62 |                    if (clipText) {
63 |                        policyTextarea.value = clipText;
64 |                        updateStats();
65 |                        showToast("Policy text pasted from clipboard", "success");
66 |                        return;
67 |                    }

```

```

68 |     }
69 |   } catch (e) {
70 |     // Clipboard permission denied or unsupported
71 |   }
72 |   policyTextarea.focus();
73 |   showToast("Please use Ctrl+V / Cmd+V to paste directly", "info");
74 | });
75 | }
76 |
77 | // Clear button
78 | if (btnClear) {
79 |   btnClear.addEventListener("click", () => {
80 |     policyTextarea.value = "";
81 |     docNameInput.value = "";
82 |     updateStats();
83 |     showToast("Cleared workspace", "info");
84 |   });
85 | }
86 |
87 | // Load Demo Policy
88 | if (btnLoadDemo) {
89 |   btnLoadDemo.addEventListener("click", () => {
90 |     loadDemoPolicy("high_risk_social");
91 |   });
92 | }
93 |
94 | function loadDemoPolicy(sampleId = "high_risk_social") {
95 |   let sample = null;
96 |   if (window.TC_SAMPLE_POLICIES.length > 0) {
97 |     sample = window.TC_SAMPLE_POLICIES.find(s => s.id === sampleId) || window.TC_SAMPLE_POLICIES[0];
98 |   }
99 |
100 |   if (sample) {
101 |     policyTextarea.value = sample.content;
102 |     if (docNameInput) docNameInput.value = sample.title;
103 |     updateStats();
104 |     showToast(`Loaded "${sample.title}" (${sample.expected_risk})`, "success");
105 |   } else {
106 |     showToast("Sample policy data not found", "error");
107 |   }
108 | }
109 |
110 | // File Upload Handlers
111 | if (fileInput) {
112 |   fileInput.addEventListener("change", (e) => {
113 |     const file = e.target.files[0];
114 |     if (file) handleUploadedFile(file);
115 |   });
116 | }
117 |
118 | if (dropzone) {
119 |   dropzone.addEventListener("click", () => fileInput.click());
120 |
121 |   ["dragenter", "dragover"].forEach(evt => {
122 |     dropzone.addEventListener(evt, (e) => {
123 |       e.preventDefault();
124 |       dropzone.classList.add("dragover");
125 |     });
126 |   });
127 |
128 |   ["dragleave", "drop"].forEach(evt => {
129 |     dropzone.addEventListener(evt, (e) => {
130 |       e.preventDefault();
131 |       dropzone.classList.remove("dragover");
132 |     });
133 |   });
134 |
135 |   dropzone.addEventListener("drop", (e) => {
136 |     const file = e.dataTransfer.files[0];

```

```

137 |     if (file) handleUploadedFile(file);
138 |   });
139 | }
140 |
141 | function handleUploadedFile(file) {
142 |   if (!file) return;
143 |
144 |   if (!file.name.match(/\.(txt|md|html|htm)$/i)) {
145 |     showToast("Please upload a .txt, .md, or .html file", "error");
146 |     return;
147 |   }
148 |
149 |   const reader = new FileReader();
150 |   reader.onload = (e) => {
151 |     let content = e.target.result || "";
152 |
153 |     // Strip basic HTML if html file
154 |     if (file.name.match(/\.(html|htm)$/i)) {
155 |       const parser = new DOMParser();
156 |       const doc = parser.parseFromString(content, "text/html");
157 |       // Remove scripts/styles
158 |       doc.querySelectorAll("script, style, noscript, meta").forEach(el => el.remove());
159 |       content = doc.body ? doc.body.innerText : content;
160 |     }
161 |
162 |     policyTextarea.value = content;
163 |     if (docNameInput) {
164 |       docNameInput.value = file.name.replace(/\.([^.]+)$/, "");
165 |     }
166 |     updateStats();
167 |     showToast(`File "${file.name}" loaded successfully`, "success");
168 |   };
169 |
170 |   reader.onerror = () => {
171 |     showToast("Error reading file", "error");
172 |   };
173 |
174 |   reader.readAsText(file);
175 | }
176 |
177 | // Start Scan Action
178 | if (btnScan) {
179 |   btnScan.addEventListener("click", () => {
180 |     startAnalysis();
181 |   });
182 | }
183 |
184 | async function startAnalysis() {
185 |   const text = policyTextarea.value ? policyTextarea.value.trim() : "";
186 |   const docName = (docNameInput.value && docNameInput.value.trim())
187 |     ? docNameInput.value.trim()
188 |     : "Analyzed Terms & Conditions";
189 |
190 |   if (!text) {
191 |     showToast("Please paste or upload Terms & Conditions text first.", "error");
192 |     policyTextarea.focus();
193 |     return;
194 |   }
195 |
196 |   if (text.length < 20) {
197 |     showToast("Input text is too short. Please paste full policy clauses.", "error");
198 |     return;
199 |   }
200 |
201 |   // Activate scanning overlay
202 |   if (scanningOverlay) {
203 |     scanningOverlay.classList.add("active");
204 |   }
205 |

```

```

206 | // Step 1
207 | updateStepUI(1);
208 | if (progressFill) progressFill.style.width = "25%";
209 |
210 | try {
211 |     // Step 2 after small delay for animation
212 |     await sleep(350);
213 |     updateStepUI(2);
214 |     if (progressFill) progressFill.style.width = "50%";
215 |
216 |     let result = null;
217 |
218 |     // Try Backend if active, otherwise use client analyzer
219 |     if (AppState.isBackendAvailable && window.location.protocol !== "file:") {
220 |         try {
221 |             const apiEndpoint = (AppState.apiBase || "") + "/api/scan";
222 |             const res = await fetch(apiEndpoint, {
223 |                 method: "POST",
224 |                 headers: { "Content-Type": "application/json" },
225 |                 body: JSON.stringify({ text: text, document_name: docName })
226 |             });
227 |
228 |             if (res.ok) {
229 |                 result = await res.json();
230 |             } else {
231 |                 console.warn("Backend returned error, switching to client-side engine");
232 |             }
233 |         } catch (e) {
234 |             console.warn("Backend unreachable, switching to client-side engine:", e);
235 |         }
236 |     }
237 |
238 |     // If backend was unreachable or in offline mode, run client-side analyzer
239 |     if (!result) {
240 |         if (window.clientClauseAnalyzer) {
241 |             result = window.clientClauseAnalyzer.scan(text, docName);
242 |         } else {
243 |             throw new Error("Client analysis engine not initialized.");
244 |         }
245 |     }
246 |
247 |     // Step 3
248 |     await sleep(300);
249 |     updateStepUI(3);
250 |     if (progressFill) progressFill.style.width = "75%";
251 |
252 |     // Step 4
253 |     await sleep(250);
254 |     updateStepUI(4);
255 |     if (progressFill) progressFill.style.width = "100%";
256 |
257 |     // Save scan to sessionStorage for Results page
258 |     setActiveScan(result);
259 |     // Also save to scan history in localStorage
260 |     saveScanToHistory(result);
261 |
262 |     await sleep(300);
263 |
264 |     // Redirect to results page
265 |     // Handle both server routes (/results) and file protocol (results.html)
266 |     if (window.location.protocol === "file:" || window.location.pathname.endsWith(".html")) {
267 |         window.location.href = "results.html";
268 |     } else {
269 |         window.location.href = "/results";
270 |     }
271 |
272 | } catch (err) {
273 |     console.error("Scan execution failed:", err);
274 |     if (scanningOverlay) scanningOverlay.classList.remove("active");

```

```
275 |     showToast(`Scan failed: ${err.message || err}`, "error");
276 |   }
277 | }
278 |
279 | function updateStepUI(activeStepNum) {
280 |   const steps = [step1, step2, step3, step4];
281 |   steps.forEach((st, idx) => {
282 |     if (!st) return;
283 |     const stepIdx = idx + 1;
284 |     const iconBox = st.querySelector(".step-icon-indicator");
285 |
286 |     if (stepIdx < activeStepNum) {
287 |       st.className = "scan-step-item done";
288 |       if (iconBox) iconBox.innerHTML = ICONS.checkCircle;
289 |     } else if (stepIdx === activeStepNum) {
290 |       st.className = "scan-step-item active";
291 |       if (iconBox) iconBox.innerHTML = `<div class="step-spinner"></div>`;
292 |     } else {
293 |       st.className = "scan-step-item pending";
294 |       if (iconBox) iconBox.innerHTML = `•`;
295 |     }
296 |   });
297 | }
298 |
299 | function sleep(ms) {
300 |   return new Promise(resolve => setTimeout(resolve, ms));
301 | }
302 | });
```

File: frontend/js/results.js

Size: 16093 bytes

```

1 | /**
2 |  * CorruptX - Results Dashboard Controller
3 |  * Circular Risk Gauge, Filterable Red Flag Cards,
4 |  * Interactive Policy Syntax Highlighting, and Security Inspector Drawer.
5 |  */
6 |
7 | document.addEventListener("DOMContentLoaded", () => {
8 |     let scanData = getActiveScan();
9 |
10 |    // If no scan in sessionStorage, check history or fallback to demo
11 |    if (!scanData) {
12 |        const history = getScanHistory();
13 |        if (history.length > 0) {
14 |            scanData = history[0].full_data;
15 |        } else if (window.clientClauseAnalyzer && window.TC_SAMPLE_POLICIES && window.TC_SAMPLE_POLICIES.length) {
16 |            // Auto-scan the demo policy so the results page is never blank
17 |            const demo = window.TC_SAMPLE_POLICIES[0];
18 |            scanData = window.clientClauseAnalyzer.scan(demo.content, demo.title);
19 |            saveScanToHistory(scanData);
20 |            setActiveScan(scanData);
21 |        }
22 |    }
23 |
24 |    if (!scanData) {
25 |        window.location.href = "scanner.html";
26 |        return;
27 |    }
28 |
29 |    // Initialize UI components
30 |    renderOverview(scanData);
31 |    renderRedFlagCards(scanData);
32 |    renderPolicyHighlighter(scanData);
33 |    renderCategoryBreakdown(scanData);
34 |    initTabNavigation();
35 |    initActions(scanData);
36 | });
37 |
38 | /**
39 |  * Render Overview Banner and Circular Score Gauge
40 |  */
41 | function renderOverview(data) {
42 |     const docTitleEl = document.getElementById("res-doc-title");
43 |     const docDateEl = document.getElementById("res-doc-date");
44 |     const riskBadgeEl = document.getElementById("res-risk-badge");
45 |     const summaryTextEl = document.getElementById("res-summary-text");
46 |
47 |     const metricClausesEl = document.getElementById("metric-clauses");
48 |     const metricRedflagsEl = document.getElementById("metric-redflags");
49 |     const metricHighEl = document.getElementById("metric-high");
50 |     const metricMedEl = document.getElementById("metric-med");
51 |     const metricLowEl = document.getElementById("metric-low");
52 |
53 |     if (docTitleEl) docTitleEl.textContent = data.document_name || "Analyzed Document";
54 |     if (docDateEl) {
55 |         const dateStr = data.analyzed_at ? new Date(data.analyzed_at).toLocaleString() : new Date().toLocaleString();
56 |         docDateEl.textContent = `Scanned: ${dateStr}`;
57 |     }
58 |
59 |     // Risk Badge & Colors
60 |     const score = data.risk_score || 0;
61 |     const level = data.risk_level || "SAFE";
62 |
63 |     let badgeClass = "badge-safe";
64 |     let gaugeColor = "var(--accent-green)";
65 |
66 |     if (level === "HIGH RISK") {
67 |         badgeClass = "badge-high";

```

```

68 |     gaugeColor = "var(--accent-crimson)";
69 | } else if (level === "CAUTION") {
70 |     badgeClass = "badge-medium";
71 |     gaugeColor = "var(--accent-amber)";
72 | }
73 |
74 | if (riskBadgeEl) {
75 |     riskBadgeEl.className = `badge ${badgeClass} risk-level-badge`;
76 |     riskBadgeEl.textContent = level;
77 | }
78 |
79 | if (summaryTextEl) {
80 |     summaryTextEl.textContent = data.summary || "";
81 | }
82 |
83 | // Metrics counts
84 | if (metricClausesEl) metricClausesEl.textContent = data.clauses_analyzed || 0;
85 | if (metricRedflagsEl) metricRedflagsEl.textContent = data.red_flags || 0;
86 | if (metricHighEl) metricHighEl.textContent = data.severity_counts?.HIGH || 0;
87 | if (metricMedEl) metricMedEl.textContent = data.severity_counts?.MEDIUM || 0;
88 | if (metricLowEl) metricLowEl.textContent = data.severity_counts?.LOW || 0;
89 |
90 | // Animate Circular Gauge
91 | animateCircularGauge(score, gaugeColor);
92 | }
93 |
94 | function animateCircularGauge(score, strokeColor) {
95 |     const fillPath = document.getElementById("gauge-fill-path");
96 |     const scoreText = document.getElementById("gauge-score-value");
97 |
98 |     if (!fillPath || !scoreText) return;
99 |
100 |     const radius = 80;
101 |     const circumference = 2 * Math.PI * radius;
102 |     fillPath.style.strokeDasharray = `${circumference}`;
103 |
104 |     // Start with full offset (empty circle)
105 |     fillPath.style.strokeDashoffset = `${circumference}`;
106 |     fillPath.style.stroke = strokeColor;
107 |
108 |     // Trigger animation after slight timeout
109 |     setTimeout(() => {
110 |         const targetOffset = circumference - (score / 100) * circumference;
111 |         fillPath.style.strokeDashoffset = `${targetOffset}`;
112 |
113 |         // Counter animation
114 |         let current = 0;
115 |         const stepTime = 1200 / Math.max(score, 1);
116 |         const counter = setInterval(() => {
117 |             if (current >= score) {
118 |                 scoreText.textContent = score;
119 |                 clearInterval(counter);
120 |             } else {
121 |                 current++;
122 |                 scoreText.textContent = current;
123 |             }
124 |         }, Math.max(10, stepTime));
125 |     }, 100);
126 | }
127 |
128 | /**
129 |  * Render Detected Red Flag Cards
130 |  */
131 | function renderRedFlagCards(data) {
132 |     const container = document.getElementById("red-flags-container");
133 |     if (!container) return;
134 |
135 |     const flagged = data.flagged_clauses || [];
136 |

```

```

137 | // Update tab counter badge
138 | const flagsCountBadge = document.getElementById("tab-count-flags");
139 | if (flagsCountBadge) flagsCountBadge.textContent = flagged.length;
140 |
141 | if (flagged.length === 0) {
142 |     container.innerHTML = `
143 |         <div class="card" style="text-align: center; padding: 48px 24px;">
144 |             <div style="color: var(--accent-green); margin-bottom: 12px;">${ICONS.checkCircle}</div>
145 |             <h3 style="color: #fff; margin-bottom: 8px;">No Critical Red Flags Detected</h3>
146 |             <p style="max-width: 480px; margin: 0 auto;">This document does not contain clauses triggering high-severi
147 |         </div>
148 |     `;
149 |     return;
150 | }
151 |
152 | container.innerHTML = flagged.map((flag, i) => {
153 |     const severityLower = flag.severity.toLowerCase();
154 |     return `
155 |         <div class="flag-card flag-card-${severityLower}" data-severity="${flag.severity}" id="flag-card-${flag.index}">
156 |             <div class="flag-card-header">
157 |                 <div class="flag-category-title">
158 |                     <span class="category-name-text">${escapeHtml(flag.category_name)}</span>
159 |                     <span class="badge badge-${severityLower}">${flag.severity} RISK</span>
160 |                 </div>
161 |                 <div class="flag-card-actions">
162 |                     <button class="btn btn-outline btn-sm btn-inspect-clause" data-clause-index="${flag.index}">
163 |                         Inspect in Policy
164 |                     </button>
165 |                 </div>
166 |             </div>
167 |
168 |             <div class="clause-quote-box">
169 |                 "${escapeHtml(flag.text)}"
170 |             </div>
171 |
172 |             <div class="flag-analysis-body">
173 |                 <div class="analysis-callout">
174 |                     <div class="callout-header">
175 |                         ${ICONS.info}
176 |                         <span>What this clause means</span>
177 |                     </div>
178 |                     <div class="callout-text">
179 |                         ${escapeHtml(flag.explanation)}
180 |                     </div>
181 |                 </div>
182 |
183 |                 <div class="analysis-callout">
184 |                     <div class="callout-header" style="color: var(--accent-crimson);">
185 |                         ${ICONS.alertTriangle}
186 |                         <span>Why this matters</span>
187 |                     </div>
188 |                     <div class="callout-text">
189 |                         ${escapeHtml(flag.why_it_matters)}
190 |                     </div>
191 |                 </div>
192 |             </div>
193 |
194 |             <div class="recommendation-bar">
195 |                 ${ICONS.checkCircle}
196 |                 <div><strong>Suggested Action:</strong> ${escapeHtml(flag.recommendation)}</div>
197 |             </div>
198 |         `;
199 |     }).join("");
200 |
201 |
202 | // Attach click listeners to "Inspect in Policy" buttons
203 | container.querySelectorAll(".btn-inspect-clause").forEach(btn => {
204 |     btn.addEventListener("click", () => {
205 |         const idx = parseInt(btn.getAttribute("data-clause-index"), 10);

```

```

206 |     switchToHighlighterTab(idx);
207 | });
208 | });
209 |
210 | // Filter handlers
211 | initFilters(flagged);
212 | }
213 |
214 | function initFilters(allFlags) {
215 |     const filterBtns = document.querySelectorAll(".filter-btn");
216 |     filterBtns.forEach(btn => {
217 |         btn.addEventListener("click", () => {
218 |             filterBtns.forEach(b => { b.classList.remove("active");
219 |             btn.classList.add("active");
220 |
221 |             const filter = btn.getAttribute("data-filter");
222 |             const cards = document.querySelectorAll(".flag-card");
223 |
224 |             cards.forEach(card => {
225 |                 const sev = card.getAttribute("data-severity");
226 |                 if (filter === "all" || sev === filter) {
227 |                     card.style.display = "block";
228 |                 } else {
229 |                     card.style.display = "none";
230 |                 }
231 |             });
232 |         });
233 |     });
234 | }
235 |
236 | /**
237 |  * Render Interactive Policy Analysis (Full Text with Clause Highlighting)
238 |  */
239 | function renderPolicyHighlighter(data) {
240 |     const docContainer = document.getElementById("policy-full-text-container");
241 |     if (!docContainer) return;
242 |
243 |     const allClauses = data.all_clauses || [];
244 |
245 |     if (allClauses.length === 0) {
246 |         docContainer.textContent = "No document text available.";
247 |         return;
248 |     }
249 |
250 |     docContainer.innerHTML = allClauses.map(clause => {
251 |         let highlightClass = "highlight-safe";
252 |         if (clause.is_flagged) {
253 |             if (clause.severity === "HIGH") highlightClass = "highlight-high";
254 |             else if (clause.severity === "MEDIUM") highlightClass = "highlight-medium";
255 |             else if (clause.severity === "LOW") highlightClass = "highlight-low";
256 |         }
257 |
258 |         return `<span class="clause-highlight ${highlightClass}" data-clause-index="${clause.index}" title="${clause.is_f
259 |     }).join("");
260 |
261 |     // Attach click events on highlighted spans
262 |     docContainer.querySelectorAll(".clause-highlight").forEach(span => {
263 |         span.addEventListener("click", () => {
264 |             const idx = parseInt(span.getAttribute("data-clause-index"), 10);
265 |             const clauseObj = allClauses[idx];
266 |             inspectClause(clauseObj, span);
267 |         });
268 |     });
269 | }
270 |
271 | function inspectClause(clause, targetSpan) {
272 |     // Highlight currently inspected span
273 |     document.querySelectorAll(".clause-highlight").forEach(el => { el.classList.remove("active-inspected");
274 |     if (targetSpan) targetSpan.classList.add("active-inspected");

```

```

275 |
276 | const pane = document.getElementById("inspector-content-pane");
277 | const placeholder = document.getElementById("inspector-placeholder");
278 |
279 | if (!pane || !placeholder) return;
280 |
281 | placeholder.style.display = "none";
282 | pane.style.display = "flex";
283 |
284 | const sevLower = (clause.severity || "safe").toLowerCase();
285 |
286 | pane.innerHTML = `
287 |   <div style="display: flex; align-items: center; justify-content: space-between;">
288 |     <span class="badge badge-${sevLower}">${clause.severity} RISK</span>
289 |     <span style="font-family: var(--font-mono); font-size: 0.78rem; color: var(--text-dim);">Clause #${clause.in
290 |   </div>
291 |
292 |   <div>
293 |     <h4 style="font-size: 1.05rem; color: #fff; margin-bottom: 4px;">${escapeHtml(clause.category_name || "Stand
294 |     <div style="font-size: 0.8rem; color: var(--text-dim); font-family: var(--font-mono);">
295 |       Taxonomy Risk Weight: ${clause.weight || 0} pts
296 |     </div>
297 |   </div>
298 |
299 |   <div style="background: rgba(8, 12, 20, 0.8); border: 1px solid var(--border-card); border-radius: 6px; padding:
300 |     "${escapeHtml(clause.text)}"
301 |   </div>
302 |
303 |   <div class="analysis-callout" style="padding: 10px;">
304 |     <div class="callout-header">${ICONS.info} <span>Plain English</span></div>
305 |     <div class="callout-text" style="font-size: 0.84rem;">${escapeHtml(clause.explanation)}</div>
306 |   </div>
307 |
308 |   <div class="analysis-callout" style="padding: 10px;">
309 |     <div class="callout-header" style="color: var(--accent-crimson);">${ICONS.alertTriangle} <span>Why it
310 |     <div class="callout-text" style="font-size: 0.84rem;">${escapeHtml(clause.why_it_matters)}</div>
311 |   </div>
312 |
313 |   <div class="recommendation-bar" style="padding: 8px 12px; font-size: 0.82rem;">
314 |     <div><strong>Action:</strong> ${escapeHtml(clause.recommendation)}</div>
315 |   </div>
316 | `;
317 | }
318 |
319 | function switchToHighlighterTab(clauseIndex) {
320 |   const highlighterTabBtn = document.querySelector('[data-tab="highlighter"]');
321 |   if (highlighterTabBtn) highlighterTabBtn.click();
322 |
323 |   setTimeout(() => {
324 |     const span = document.querySelector(`.clause-highlight[data-clause-index="${clauseIndex}"]`);
325 |     if (span) {
326 |       span.scrollIntoView({ behavior: "smooth", block: "center" });
327 |       span.click();
328 |     }
329 |   }, 150);
330 | }
331 |
332 | /**
333 |  * Render Category Risk Breakdown Chart
334 |  */
335 | function renderCategoryBreakdown(data) {
336 |   const container = document.getElementById("category-breakdown-container");
337 |   if (!container) return;
338 |
339 |   const categories = data.categories || [];
340 |
341 |   if (categories.length === 0) {
342 |     container.innerHTML = `<p style="text-align: center; color: var(--text-muted); padding: 32px;">No risk categor
343 |     return;

```

```

344 |   }
345 |
346 |   const maxCount = Math.max(...categories.map(c => c.count), 1);
347 |
348 |   container.innerHTML = `
349 |     <div class="breakdown-list">
350 |       ${categories.map(cat => {
351 |         const percent = Math.min(100, Math.round((cat.count / maxCount) * 100));
352 |         const sevLower = cat.severity.toLowerCase();
353 |         let color = "var(--accent-blue)";
354 |         if (cat.severity === "HIGH") color = "var(--accent-crimson)";
355 |         else if (cat.severity === "MEDIUM") color = "var(--accent-amber)";
356 |
357 |         return `
358 |           <div class="breakdown-item">
359 |             <div class="breakdown-item-header">
360 |               <div class="breakdown-category-label">
361 |                 <span>${escapeHtml(cat.category_name)}</span>
362 |                 <span class="badge badge-${sevLower}">${cat.severity}</span>
363 |               </div>
364 |               <div class="breakdown-counts">
365 |                 <span>${cat.count} clause${cat.count > 1 ? 's' : ''} flagged</span>
366 |               </div>
367 |             </div>
368 |             <div class="breakdown-track">
369 |               <div class="breakdown-fill" style="width: ${percent}%; background: ${color}; box-shadow: 0 0 10px ${color}>
370 |             </div>
371 |           </div>
372 |         `;
373 |       }).join("")}
374 |     </div>
375 | `;
376 | }
377 |
378 | /**
379 |  * Tab Navigation (Red Flags vs Highlighter vs Breakdown)
380 |  */
381 | function initTabNavigation() {
382 |   const tabs = document.querySelectorAll(".tab-btn");
383 |   const views = {
384 |     cards: document.getElementById("view-red-flags"),
385 |     highlighter: document.getElementById("view-policy-analysis"),
386 |     breakdown: document.getElementById("view-category-breakdown")
387 |   };
388 |
389 |   tabs.forEach(tab => {
390 |     tab.addEventListener("click", () => {
391 |       tabs.forEach(t => t.classList.remove("active"));
392 |       tab.classList.add("active");
393 |
394 |       const target = tab.getAttribute("data-tab");
395 |       Object.keys(views).forEach(k => {
396 |         if (views[k]) {
397 |           views[k].style.display = (k === target) ? "block" : "none";
398 |         }
399 |       });
400 |     });
401 |   });
402 | }
403 |
404 | /**
405 |  * Export & Action handlers
406 |  */
407 | function initActions(data) {
408 |   const btnExport = document.getElementById("btn-export-report");
409 |   const btnCopy = document.getElementById("btn-copy-summary");
410 |   const btnRescan = document.getElementById("btn-rescan-new");
411 |
412 |   if (btnExport) {

```

```

413 |     btnExport.addEventListener("click", () => {
414 |         const blob = new Blob([JSON.stringify(data, null, 2)], { type: "application/json" });
415 |         const url = URL.createObjectURL(blob);
416 |         const a = document.createElement("a");
417 |         a.href = url;
418 |         a.download = `corruptx_tc_report_${Date.now()}.json`;
419 |         a.click();
420 |         URL.revokeObjectURL(url);
421 |         showToast("Security report exported as JSON", "success");
422 |     });
423 | }
424 |
425 | if (btnCopy) {
426 |     btnCopy.addEventListener("click", () => {
427 |         const summaryMarkdown = `
428 | # CORRUPTX T&C SECURITY REPORT
429 | **Target:** ${data.document_name}
430 | **Risk Score:** ${data.risk_score}/100 [${data.risk_level}]
431 | **Analyzed Clauses:** ${data.clauses_analyzed}
432 | **Red Flags Identified:** ${data.red_flags}
433 |
434 | ## Critical Red Flags:
435 | ${((data.flagged_clauses || []).map(f => `- **${f.category_name} (${f.severity})**: "${f.text.substring(0, 100)}..."` -
436 |
437 | *Generated by CorruptX T&C Red Flag Scanner (Cybersecurity Engine)*
438 |     `).trim());
439 |
440 |         navigator.clipboard.writeText(summaryMarkdown).then(() => {
441 |             showToast("Executive summary copied to clipboard", "success");
442 |         }).catch(() => {
443 |             showToast("Clipboard copy failed", "error");
444 |         });
445 |     });
446 | }
447 |
448 | if (btnRescan) {
449 |     btnRescan.addEventListener("click", () => {
450 |         window.location.href = "scanner.html";
451 |     });
452 | }
453 | }
454 |
455 | function escapeHtml(text) {
456 |     if (!text) return "";
457 |     return text
458 |         .replace(/&/g, "&amp;")
459 |         .replace(/</g, "&lt;")
460 |         .replace(/>/g, "&gt;")
461 |         .replace(/"/g, "&quot;")
462 |         .replace(/'/g, "&#039;");
463 | }

```

File: frontend/js/history.js

Size: 6276 bytes

```

1 | /**
2 |  * CorruptX - Scan History Controller
3 |  * Manages persistent client-side scan logs with localStorage,
4 |  * restoration into Results dashboard, deletion, and export.
5 |  */
6 |
7 | document.addEventListener("DOMContentLoaded", () => {
8 |   renderHistoryView();
9 |   initHistoryControls();
10 | });
11 |
12 | function renderHistoryView() {
13 |   const history = getScanHistory();
14 |   const listContainer = document.getElementById("history-list-container");
15 |   const emptyState = document.getElementById("history-empty-state");
16 |
17 |   const statTotalScans = document.getElementById("stat-total-scans");
18 |   const statAvgScore = document.getElementById("stat-avg-score");
19 |   const statHighRiskScans = document.getElementById("stat-high-risk-scans");
20 |
21 |   if (!listContainer) return;
22 |
23 |   // Compute metrics
24 |   const total = history.length;
25 |   const avg = total > 0 ? Math.round(history.reduce((acc, h) => acc + (h.risk_score || 0), 0) / total) : 0;
26 |   const highRiskCount = history.filter(h => h.risk_level === "HIGH RISK").length;
27 |
28 |   if (statTotalScans) statTotalScans.textContent = total;
29 |   if (statAvgScore) statAvgScore.textContent = `${avg}/100`;
30 |   if (statHighRiskScans) statHighRiskScans.textContent = highRiskCount;
31 |
32 |   if (total === 0) {
33 |     if (emptyState) emptyState.style.display = "block";
34 |     listContainer.innerHTML = "";
35 |     return;
36 |   }
37 |
38 |   if (emptyState) emptyState.style.display = "none";
39 |
40 |   listContainer.innerHTML = history.map(item => {
41 |     const sevLower = (item.risk_level || "SAFE").toLowerCase().replace(" ", "-");
42 |     let badgeClass = "badge-safe";
43 |     if (item.risk_level === "HIGH RISK") badgeClass = "badge-high";
44 |     else if (item.risk_level === "CAUTION") badgeClass = "badge-medium";
45 |
46 |     const dateFormatted = new Date(item.analyzed_at).toLocaleString();
47 |
48 |     return `
49 |       <div class="card history-card" id="hist-item-${item.id}" style="margin-bottom: 16px; padding: 20px;">
50 |         <div style="display: flex; align-items: center; justify-content: space-between; flex-wrap: wrap; gap: 16px;">
51 |           <div style="display: flex; align-items: center; gap: 16px;">
52 |             <div style="width: 48px; height: 48px; border-radius: 10px; background: rgba(15, 23, 42, 0.9); border: 1px solid #fff;">
53 |               <span style="font-size: 1.15rem; font-weight: 800; color: #fff; line-height: 1;">${item.risk_score}</span>
54 |             <span style="font-size: 0.6rem; color: var(--text-dim); font-family: var(--font-mono);">/${item.risk_level}</span>
55 |           </div>
56 |           <div>
57 |             <h3 style="font-size: 1.05rem; font-weight: 700; color: #fff; margin-bottom: 4px;">${escapeHtml(item.title)}
58 |             <div style="display: flex; align-items: center; gap: 12px; font-size: 0.8rem; color: var(--text-dim); font-family: var(--font-mono);">
59 |               <span>${dateFormatted}</span>
60 |               <span>•</span>
61 |               <span>${item.clauses_analyzed} Clauses</span>
62 |               <span>•</span>
63 |               <span>${item.red_flags} Red Flags</span>
64 |             </div>
65 |           </div>
66 |         </div>
67 |       `

```

```

68 |         <div style="display: flex; align-items: center; gap: 12px;">
69 |             <span class="badge ${badgeClass}">${item.risk_level}</span>
70 |             <button class="btn btn-outline btn-sm btn-view-scan" data-id="${item.id}">
71 |                 View Analysis
72 |             </button>
73 |             <button class="btn btn-danger btn-sm btn-delete-scan" data-id="${item.id}" title="Delete Record">
74 |                 ${ICONS.trash}
75 |             </button>
76 |         </div>
77 |     </div>
78 | </div>
79 | `;
80 | }).join("");
81 |
82 | // Attach event handlers
83 | listContainer.querySelectorAll(".btn-view-scan").forEach(btn => {
84 |     btn.addEventListener("click", () => {
85 |         const id = btn.getAttribute("data-id");
86 |         const record = history.find(h => h.id === id);
87 |         if (record && record.full_data) {
88 |             setActiveScan(record.full_data);
89 |             if (window.location.protocol === "file:" || window.location.pathname.endsWith(".html")) {
90 |                 window.location.href = "results.html";
91 |             } else {
92 |                 window.location.href = "/results";
93 |             }
94 |         }
95 |     });
96 | });
97 |
98 | listContainer.querySelectorAll(".btn-delete-scan").forEach(btn => {
99 |     btn.addEventListener("click", () => {
100 |         const id = btn.getAttribute("data-id");
101 |         deleteScanHistoryItem(id);
102 |         showToast("Scan record removed", "info");
103 |         renderHistoryView();
104 |     });
105 | });
106 | }
107 |
108 | function initHistoryControls() {
109 |     const btnClear = document.getElementById("btn-clear-history");
110 |     const btnExportAll = document.getElementById("btn-export-all-history");
111 |     const btnLoadSampleHistory = document.getElementById("btn-load-sample-history");
112 |
113 |     if (btnClear) {
114 |         btnClear.addEventListener("click", () => {
115 |             if (confirm("Are you sure you want to clear all local scan history?")) {
116 |                 clearScanHistory();
117 |                 renderHistoryView();
118 |                 showToast("Scan history cleared", "info");
119 |             }
120 |         });
121 |     }
122 |
123 |     if (btnExportAll) {
124 |         btnExportAll.addEventListener("click", () => {
125 |             const history = getScanHistory();
126 |             if (history.length === 0) {
127 |                 showToast("No scan records to export", "error");
128 |                 return;
129 |             }
130 |             const blob = new Blob([JSON.stringify(history, null, 2)], { type: "application/json" });
131 |             const url = URL.createObjectURL(blob);
132 |             const a = document.createElement("a");
133 |             a.href = url;
134 |             a.download = `corruptx_history_${Date.now()}.json`;
135 |             a.click();
136 |             URL.revokeObjectURL(url);

```

```
137 |     showToast("Full history exported", "success");
138 |   });
139 | }
140 |
141 | if (btnLoadSampleHistory) {
142 |   btnLoadSampleHistory.addEventListener("click", () => {
143 |     if (window.clientClauseAnalyzer && window.TC_SAMPLE_POLICIES) {
144 |       window.TC_SAMPLE_POLICIES.forEach(sample => {
145 |         const scan = window.clientClauseAnalyzer.scan(sample.content, sample.title);
146 |         saveScanToHistory(scan);
147 |       });
148 |       renderHistoryView();
149 |       showToast("Loaded sample scan history", "success");
150 |     }
151 |   });
152 | }
153 | }
154 |
155 | function escapeHtml(text) {
156 |   if (!text) return "";
157 |   return text
158 |     .replace(/&/g, "&amp;")
159 |     .replace(/</g, "&lt;")
160 |     .replace(/>/g, "&gt;")
161 |     .replace(/"/g, "&quot;")
162 |     .replace(/'/g, "&#039;");
163 | }
```

File: frontend/js/sample_data.js

Size: 6986 bytes

```
1 | /**
2 |  * Preloaded Hackathon Demo Policies
3 |  * Allows zero-network instant testing of the scanner.
4 |  */
5 | window.TC_SAMPLE_POLICIES = [
6 |   {
7 |     "id": "high_risk_social",
8 |     "title": "HyperNet Social App Terms & Privacy Policy",
9 |     "expected_risk": "HIGH RISK",
10 |    "description": "Aggressive mobile app with tracking, third-party data sales, unilateral modifications, and forced an
11 |    "content": "Welcome to HyperNet. By accessing or using our services, mobile application, or websites, you agree to b
12 |   },
13 |   {
14 |     "id": "moderate_risk_saas",
15 |     "title": "NovaDesk Cloud Workspace Agreement",
16 |     "expected_risk": "CAUTION",
17 |     "description": "A business SaaS platform with commercial auto-renewal, standard liability caps, unilateral maintenar
18 |     "content": "NovaDesk Cloud Services Master Subscription Agreement.\n\n1. SCOPE AND SERVICES\nNovaDesk provides onlin
19 |   },
20 |   {
21 |     "id": "safe_open_source",
22 |     "title": "AegisGuard Privacy-First Client Terms",
23 |     "expected_risk": "SAFE",
24 |     "description": "Open-source zero-knowledge utility with end-to-end encryption, zero tracking, and no third-party sha
25 |     "content": "AegisGuard Terms of Use and Privacy Charter.\n\n1. ZERO KNOWLEDGE ARCHITECTURE\nAegisGuard operates as a
26 |   }
27 | ];
```

File: frontend/css/style.css

Size: 12574 bytes

```

1 | /* =====
2 |    CORRUPTX - T&C RED FLAG SCANNER
3 |    Cybersecurity Theme & Design System
4 |    ===== */
5 |
6 | :root {
7 |   --bg-primary: #07090e;
8 |   --bg-secondary: #0c1018;
9 |   --bg-tertiary: #111827;
10 |   --bg-card: rgba(13, 19, 31, 0.78);
11 |   --bg-card-hover: rgba(18, 27, 44, 0.9);
12 |   --bg-glass: rgba(15, 23, 42, 0.65);
13 |
14 |   --border-subtle: rgba(0, 242, 255, 0.12);
15 |   --border-focus: rgba(0, 242, 255, 0.45);
16 |   --border-card: rgba(255, 255, 255, 0.07);
17 |
18 |   --accent-cyan: #00f2ff;
19 |   --accent-cyan-glow: rgba(0, 242, 255, 0.25);
20 |   --accent-green: #00ff88;
21 |   --accent-green-glow: rgba(0, 255, 136, 0.2);
22 |   --accent-amber: #ffb700;
23 |   --accent-amber-glow: rgba(255, 183, 0, 0.2);
24 |   --accent-crimson: #ff3366;
25 |   --accent-crimson-glow: rgba(255, 51, 102, 0.25);
26 |   --accent-blue: #38bdf8;
27 |
28 |   --text-main: #f1f5f9;
29 |   --text-muted: #94a3b8;
30 |   --text-dim: #64748b;
31 |   --text-highlight: #ffffff;
32 |
33 |   --font-sans: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Inter", "Helvetica Neue", Arial, sans-serif;
34 |   --font-mono: ui-monospace, "SF Mono", "Cascadia Code", "JetBrains Mono", Consolas, monospace;
35 |
36 |   --radius-sm: 6px;
37 |   --radius-md: 10px;
38 |   --radius-lg: 16px;
39 |   --radius-full: 9999px;
40 |
41 |   --shadow-cyber: 0 8px 32px 0 rgba(0, 0, 0, 0.5), 0 0 15px rgba(0, 242, 255, 0.08);
42 |   --shadow-glow: 0 0 20px var(--accent-cyan-glow);
43 |   --shadow-red-glow: 0 0 20px var(--accent-crimson-glow);
44 | }
45 |
46 | /* Base Resets */
47 | *, *::before, *::after {
48 |   box-sizing: border-box;
49 |   margin: 0;
50 |   padding: 0;
51 | }
52 |
53 | html {
54 |   font-size: 16px;
55 |   scroll-behavior: smooth;
56 |   background-color: var(--bg-primary);
57 |   color: var(--text-main);
58 |   font-family: var(--font-sans);
59 |   line-height: 1.6;
60 |   -webkit-font-smoothing: antialiased;
61 | }
62 |
63 | body {
64 |   min-height: 100vh;
65 |   display: flex;
66 |   flex-direction: column;
67 |   background-color: var(--bg-primary);

```

```
68 | background-image:
69 |     radial-gradient(circle at 50% 0%, rgba(0, 242, 255, 0.04) 0%, transparent 60%),
70 |     radial-gradient(circle at 90% 80%, rgba(255, 51, 102, 0.03) 0%, transparent 50%),
71 |     linear-gradient(rgba(255, 255, 255, 0.015) 1px, transparent 1px),
72 |     linear-gradient(90deg, rgba(255, 255, 255, 0.015) 1px, transparent 1px);
73 | background-size: 100% 100%, 100% 100%, 48px 48px, 48px 48px;
74 | overflow-x: hidden;
75 | }
76 |
77 | /* Typography */
78 | h1, h2, h3, h4, h5, h6 {
79 |     color: var(--text-highlight);
80 |     font-weight: 700;
81 |     line-height: 1.25;
82 |     letter-spacing: -0.02em;
83 | }
84 |
85 | p {
86 |     color: var(--text-muted);
87 | }
88 |
89 | a {
90 |     color: var(--accent-cyan);
91 |     text-decoration: none;
92 |     transition: all 0.2s ease;
93 | }
94 |
95 | a:hover {
96 |     color: #fff;
97 |     text-shadow: 0 0 8px var(--accent-cyan);
98 | }
99 |
100 | code, .code-font {
101 |     font-family: var(--font-mono);
102 | }
103 |
104 | /* Layout Container */
105 | .container {
106 |     width: 100%;
107 |     max-width: 1240px;
108 |     margin: 0 auto;
109 |     padding: 0 24px;
110 | }
111 |
112 | /* Header & Navbar */
113 | .site-header {
114 |     position: sticky;
115 |     top: 0;
116 |     z-index: 100;
117 |     background: rgba(7, 9, 14, 0.85);
118 |     backdrop-filter: blur(14px);
119 |     -webkit-backdrop-filter: blur(14px);
120 |     border-bottom: 1px solid var(--border-subtle);
121 |     height: 72px;
122 |     display: flex;
123 |     align-items: center;
124 | }
125 |
126 | .header-inner {
127 |     display: flex;
128 |     align-items: center;
129 |     justify-content: space-between;
130 |     width: 100%;
131 | }
132 |
133 | .brand-wrapper {
134 |     display: flex;
135 |     align-items: center;
136 |     gap: 12px;
```

```
137 | }
138 |
139 | .brand-shield-icon {
140 |   width: 36px;
141 |   height: 36px;
142 |   display: flex;
143 |   align-items: center;
144 |   justify-content: center;
145 |   background: rgba(0, 242, 255, 0.08);
146 |   border: 1px solid var(--border-focus);
147 |   border-radius: var(--radius-sm);
148 |   color: var(--accent-cyan);
149 |   box-shadow: 0 0 14px var(--accent-cyan-glow);
150 | }
151 |
152 | .brand-shield-icon svg {
153 |   width: 22px;
154 |   height: 22px;
155 | }
156 |
157 | .brand-meta {
158 |   display: flex;
159 |   flex-direction: column;
160 | }
161 |
162 | .brand-team {
163 |   font-size: 0.68rem;
164 |   text-transform: uppercase;
165 |   letter-spacing: 0.14em;
166 |   color: var(--accent-cyan);
167 |   font-weight: 800;
168 |   line-height: 1;
169 | }
170 |
171 | .brand-name {
172 |   font-size: 1.15rem;
173 |   font-weight: 800;
174 |   letter-spacing: 0.02em;
175 |   color: var(--text-highlight);
176 |   line-height: 1.2;
177 | }
178 |
179 | .brand-name span {
180 |   color: var(--accent-crimson);
181 | }
182 |
183 | /* Navigation Links */
184 | .main-nav {
185 |   display: flex;
186 |   align-items: center;
187 |   gap: 28px;
188 | }
189 |
190 | .nav-link {
191 |   font-size: 0.92rem;
192 |   font-weight: 500;
193 |   color: var(--text-muted);
194 |   position: relative;
195 |   padding: 6px 0;
196 |   display: flex;
197 |   align-items: center;
198 |   gap: 6px;
199 |   transition: color 0.2s ease;
200 | }
201 |
202 | .nav-link:hover, .nav-link.active {
203 |   color: var(--text-highlight);
204 | }
205 |
```

```
206 | .nav-link.active::after {
207 |   content: "";
208 |   position: absolute;
209 |   bottom: 0;
210 |   left: 0;
211 |   width: 100%;
212 |   height: 2px;
213 |   background: var(--accent-cyan);
214 |   box-shadow: 0 0 8px var(--accent-cyan);
215 |   border-radius: 2px;
216 | }
217 |
218 | .header-actions {
219 |   display: flex;
220 |   align-items: center;
221 |   gap: 16px;
222 | }
223 |
224 | .system-status {
225 |   display: flex;
226 |   align-items: center;
227 |   gap: 8px;
228 |   padding: 4px 12px;
229 |   border-radius: var(--radius-full);
230 |   background: rgba(0, 255, 136, 0.08);
231 |   border: 1px solid rgba(0, 255, 136, 0.25);
232 |   font-size: 0.72rem;
233 |   font-family: var(--font-mono);
234 |   color: var(--accent-green);
235 |   letter-spacing: 0.05em;
236 |   text-transform: uppercase;
237 | }
238 |
239 | .status-dot {
240 |   width: 7px;
241 |   height: 7px;
242 |   background: var(--accent-green);
243 |   border-radius: 50%;
244 |   box-shadow: 0 0 8px var(--accent-green);
245 |   animation: pulse-green 2s infinite ease-in-out;
246 | }
247 |
248 | @keyframes pulse-green {
249 |   0%, 100% { opacity: 1; transform: scale(1); }
250 |   50% { opacity: 0.4; transform: scale(0.85); }
251 | }
252 |
253 | /* Mobile Toggle */
254 | .mobile-menu-btn {
255 |   display: none;
256 |   background: transparent;
257 |   border: 1px solid var(--border-subtle);
258 |   border-radius: var(--radius-sm);
259 |   color: var(--text-main);
260 |   padding: 8px;
261 |   cursor: pointer;
262 | }
263 |
264 | .mobile-menu-btn svg {
265 |   width: 22px;
266 |   height: 22px;
267 |   display: block;
268 | }
269 |
270 | /* Cyber Buttons */
271 | .btn {
272 |   display: inline-flex;
273 |   align-items: center;
274 |   justify-content: center;
```

```
275 | gap: 8px;
276 | padding: 10px 20px;
277 | font-size: 0.92rem;
278 | font-weight: 600;
279 | border-radius: var(--radius-md);
280 | border: 1px solid transparent;
281 | cursor: pointer;
282 | transition: all 0.22s cubic-bezier(0.16, 1, 0.3, 1);
283 | text-decoration: none;
284 | white-space: nowrap;
285 | }
286 |
287 | .btn-primary {
288 |   background: linear-gradient(135deg, #00d2df 0%, #0088cc 100%);
289 |   color: #020712;
290 |   box-shadow: 0 4px 18px var(--accent-cyan-glow);
291 |   font-weight: 700;
292 | }
293 |
294 | .btn-primary:hover {
295 |   background: linear-gradient(135deg, #24f6ff 0%, #009be6 100%);
296 |   box-shadow: 0 6px 24px rgba(0, 242, 255, 0.45);
297 |   transform: translateY(-1px);
298 |   color: #000;
299 | }
300 |
301 | .btn-secondary {
302 |   background: rgba(15, 23, 42, 0.6);
303 |   border-color: var(--border-subtle);
304 |   color: var(--text-main);
305 | }
306 |
307 | .btn-secondary:hover {
308 |   background: rgba(30, 41, 59, 0.8);
309 |   border-color: var(--border-focus);
310 |   color: #fff;
311 |   transform: translateY(-1px);
312 | }
313 |
314 | .btn-outline {
315 |   background: transparent;
316 |   border: 1px solid var(--border-subtle);
317 |   color: var(--accent-cyan);
318 | }
319 |
320 | .btn-outline:hover {
321 |   border-color: var(--accent-cyan);
322 |   background: rgba(0, 242, 255, 0.08);
323 |   color: #fff;
324 | }
325 |
326 | .btn-danger {
327 |   background: rgba(255, 51, 102, 0.15);
328 |   border: 1px solid rgba(255, 51, 102, 0.4);
329 |   color: #ff3366;
330 | }
331 |
332 | .btn-danger:hover {
333 |   background: rgba(255, 51, 102, 0.3);
334 |   box-shadow: 0 0 15px var(--accent-crimson-glow);
335 |   color: #fff;
336 | }
337 |
338 | .btn-sm {
339 |   padding: 6px 12px;
340 |   font-size: 0.8rem;
341 | }
342 |
343 | .btn-lg {
```

```
344 | padding: 14px 28px;
345 | font-size: 1.05rem;
346 | }
347 |
348 | /* Glass Cards */
349 | .card {
350 | background: var(--bg-card);
351 | backdrop-filter: blur(12px);
352 | -webkit-backdrop-filter: blur(12px);
353 | border: 1px solid var(--border-card);
354 | border-radius: var(--radius-lg);
355 | padding: 24px;
356 | position: relative;
357 | overflow: hidden;
358 | box-shadow: var(--shadow-cyber);
359 | transition: transform 0.25s ease, border-color 0.25s ease, box-shadow 0.25s ease;
360 | }
361 |
362 | .card:hover {
363 | border-color: var(--border-subtle);
364 | }
365 |
366 | .card-cyber-accent::before {
367 | content: "";
368 | position: absolute;
369 | top: 0;
370 | left: 0;
371 | right: 0;
372 | height: 2px;
373 | background: linear-gradient(90deg, transparent, var(--accent-cyan), transparent);
374 | }
375 |
376 | /* Badges */
377 | .badge {
378 | display: inline-flex;
379 | align-items: center;
380 | gap: 5px;
381 | padding: 4px 10px;
382 | border-radius: var(--radius-sm);
383 | font-size: 0.75rem;
384 | font-weight: 700;
385 | text-transform: uppercase;
386 | letter-spacing: 0.06em;
387 | font-family: var(--font-mono);
388 | }
389 |
390 | .badge-high {
391 | background: rgba(255, 51, 102, 0.15);
392 | color: #ff3366;
393 | border: 1px solid rgba(255, 51, 102, 0.4);
394 | box-shadow: 0 0 10px rgba(255, 51, 102, 0.2);
395 | }
396 |
397 | .badge-medium {
398 | background: rgba(255, 183, 0, 0.15);
399 | color: #ffb700;
400 | border: 1px solid rgba(255, 183, 0, 0.4);
401 | }
402 |
403 | .badge-low {
404 | background: rgba(56, 189, 248, 0.15);
405 | color: #38bdf8;
406 | border: 1px solid rgba(56, 189, 248, 0.4);
407 | }
408 |
409 | .badge-safe {
410 | background: rgba(0, 255, 136, 0.15);
411 | color: #00ff88;
412 | border: 1px solid rgba(0, 255, 136, 0.4);
```

```
413 | }
414 |
415 | /* Toast Notifications */
416 | #toast-container {
417 |   position: fixed;
418 |   bottom: 24px;
419 |   right: 24px;
420 |   z-index: 1000;
421 |   display: flex;
422 |   flex-direction: column;
423 |   gap: 10px;
424 |   pointer-events: none;
425 | }
426 |
427 | .toast {
428 |   pointer-events: auto;
429 |   background: rgba(13, 19, 31, 0.95);
430 |   backdrop-filter: blur(12px);
431 |   border: 1px solid var(--border-focus);
432 |   box-shadow: 0 8px 30px rgba(0, 0, 0, 0.7);
433 |   border-radius: var(--radius-md);
434 |   padding: 12px 18px;
435 |   display: flex;
436 |   align-items: center;
437 |   gap: 12px;
438 |   color: var(--text-main);
439 |   font-size: 0.88rem;
440 |   max-width: 360px;
441 |   animation: toast-in 0.3s cubic-bezier(0.16, 1, 0.3, 1) forwards;
442 | }
443 |
444 | .toast.toast-error {
445 |   border-color: var(--accent-crimson);
446 |   color: #ff859f;
447 | }
448 |
449 | .toast.toast-success {
450 |   border-color: var(--accent-green);
451 |   color: #a7f3d0;
452 | }
453 |
454 | @keyframes toast-in {
455 |   from { opacity: 0; transform: translateY(12px) scale(0.95); }
456 |   to { opacity: 1; transform: translateY(0) scale(1); }
457 | }
458 |
459 | /* Modals & Drawers */
460 | .modal-backdrop {
461 |   position: fixed;
462 |   top: 0;
463 |   left: 0;
464 |   right: 0;
465 |   bottom: 0;
466 |   background: rgba(4, 6, 10, 0.82);
467 |   backdrop-filter: blur(8px);
468 |   z-index: 500;
469 |   display: flex;
470 |   align-items: center;
471 |   justify-content: center;
472 |   padding: 20px;
473 |   opacity: 0;
474 |   visibility: hidden;
475 |   transition: opacity 0.25s ease, visibility 0.25s ease;
476 | }
477 |
478 | .modal-backdrop.active {
479 |   opacity: 1;
480 |   visibility: visible;
481 | }
```

```
482 |  
483 | .modal-content {  
484 |   background: var(--bg-secondary);  
485 |   border: 1px solid var(--border-focus);  
486 |   border-radius: var(--radius-lg);  
487 |   box-shadow: 0 20px 60px rgba(0, 0, 0, 0.8), 0 0 30px var(--accent-cyan-glow);  
488 |   width: 100%;  
489 |   max-width: 680px;  
490 |   max-height: 85vh;  
  
491 |   display: flex;  
492 |   flex-direction: column;  
493 |   overflow: hidden;  
494 |   transform: translateY(20px) scale(0.97);  
495 |   transition: transform 0.25s cubic-bezier(0.16, 1, 0.3, 1);  
496 | }  
497 |  
498 | .modal-backdrop.active .modal-content {  
499 |   transform: translateY(0) scale(1);  
500 | }  
501 |  
502 | .modal-header {  
503 |   padding: 18px 24px;  
504 |   border-bottom: 1px solid var(--border-card);  
505 |   display: flex;  
506 |   align-items: center;  
507 |   justify-content: space-between;  
508 | }  
509 |  
510 | .modal-title {  
511 |   font-size: 1.15rem;  
512 |   font-weight: 700;  
513 |   color: #fff;  
514 | }  
515 |  
516 | .modal-close-btn {  
517 |   background: transparent;  
518 |   border: none;  
519 |   color: var(--text-muted);  
520 |   cursor: pointer;  
521 |   padding: 4px;  
522 | }  
523 |  
524 | .modal-close-btn:hover {  
525 |   color: #fff;  
526 | }  
527 |  
528 | .modal-body {  
529 |   padding: 24px;  
530 |   overflow-y: auto;  
531 |   display: flex;  
532 |   flex-direction: column;  
533 |   gap: 20px;  
534 | }  
535 |  
536 | /* Footer */  
537 | .site-footer {  
538 |   margin-top: auto;  
539 |   border-top: 1px solid var(--border-card);  
540 |   background: rgba(6, 8, 12, 0.9);  
541 |   padding: 40px 0 24px;  
542 |   font-size: 0.88rem;  
543 | }  
544 |  
545 | .footer-top {  
546 |   display: flex;  
547 |   justify-content: space-between;  
548 |   align-items: flex-start;  
549 |   flex-wrap: wrap;  
550 |   gap: 32px;
```

```
551 |     margin-bottom: 30px;
552 | }
553 |
554 | .footer-brand {
555 |     max-width: 380px;
556 | }
557 |
558 | .footer-desc {
559 |     margin-top: 12px;
560 |     font-size: 0.85rem;
561 |     line-height: 1.5;
562 | }
563 |
564 | .footer-links {
565 |     display: flex;
566 |     gap: 40px;
567 | }
568 |
569 | .footer-group-title {
570 |     font-size: 0.78rem;
571 |     text-transform: uppercase;
572 |     letter-spacing: 0.1em;
573 |     color: var(--accent-cyan);
574 |     font-family: var(--font-mono);
575 |     margin-bottom: 12px;
576 | }
577 |
578 | .footer-link-list {
579 |     list-style: none;
580 |     display: flex;
581 |     flex-direction: column;
582 |     gap: 8px;
583 | }
584 |
585 | .footer-link-list a {
586 |     color: var(--text-muted);
587 |     font-size: 0.85rem;
588 | }
589 |
590 | .footer-link-list a:hover {
591 |     color: #fff;
592 | }
593 |
594 | .footer-disclaimer {
595 |     padding: 16px 20px;
596 |     background: rgba(17, 24, 39, 0.5);
597 |     border: 1px solid rgba(255, 255, 255, 0.05);
598 |     border-radius: var(--radius-md);
599 |     margin-bottom: 24px;
600 |     font-size: 0.8rem;
601 |     color: var(--text-dim);
602 |     line-height: 1.5;
603 | }
604 |
605 | .footer-bottom {
606 |     display: flex;
607 |     justify-content: space-between;
608 |     align-items: center;
609 |     flex-wrap: wrap;
610 |     gap: 16px;
611 |     border-top: 1px solid rgba(255, 255, 255, 0.05);
612 |     padding-top: 20px;
613 |     color: var(--text-dim);
614 |     font-size: 0.8rem;
615 | }
```

File: frontend/css/chatbot.css

Size: 6382 bytes

```

1 | /**
2 |  * CorruptX - T&C AI Assistant (Chatbot) Styles
3 |  */
4 |
5 | #tc-chatbot-widget {
6 |   position: fixed;
7 |   bottom: 24px;
8 |   right: 24px;
9 |   z-index: 99999;
10 |   font-family: var(--font-main, 'Inter', -apple-system, sans-serif);
11 | }
12 |
13 | /* Floating Trigger Button */
14 | #tc-chatbot-toggle {
15 |   display: flex;
16 |   align-items: center;
17 |   gap: 10px;
18 |   background: rgba(15, 23, 42, 0.85);
19 |   border: 1px solid var(--accent-cyan, #00f2ff);
20 |   color: #ffffff;
21 |   padding: 12px 18px;
22 |   border-radius: 50px;
23 |   cursor: pointer;
24 |   box-shadow: 0 8px 24px rgba(0, 0, 0, 0.4), 0 0 15px rgba(0, 242, 255, 0.25);
25 |   backdrop-filter: blur(12px);
26 |   transition: all 0.25s ease;
27 |   user-select: none;
28 | }
29 |
30 | #tc-chatbot-toggle:hover {
31 |   transform: translateY(-3px) scale(1.03);
32 |   box-shadow: 0 12px 28px rgba(0, 0, 0, 0.5), 0 0 22px rgba(0, 242, 255, 0.45);
33 |   background: rgba(15, 23, 42, 0.95);
34 | }
35 |
36 | .chatbot-toggle-icon {
37 |   display: flex;
38 |   align-items: center;
39 |   justify-content: center;
40 |   color: var(--accent-cyan, #00f2ff);
41 | }
42 |
43 | .chatbot-toggle-text {
44 |   font-weight: 600;
45 |   font-size: 0.9rem;
46 |   letter-spacing: 0.3px;
47 | }
48 |
49 | .chatbot-pulse-dot {
50 |   width: 8px;
51 |   height: 8px;
52 |   background-color: var(--accent-green, #10b981);
53 |   border-radius: 50%;
54 |   box-shadow: 0 0 8px var(--accent-green, #10b981);
55 |   animation: pulse-dot-anim 1.8s infinite;
56 | }
57 |
58 | @keyframes pulse-dot-anim {
59 |   0% { transform: scale(0.95); opacity: 0.8; }
60 |   50% { transform: scale(1.3); opacity: 1; }
61 |   100% { transform: scale(0.95); opacity: 0.8; }
62 | }
63 |
64 | /* Chat Panel Window */
65 | .chatbot-modal {
66 |   position: absolute;
67 |   bottom: 60px;

```

```
68 | right: 0;
69 | width: 380px;
70 | max-width: calc(100vw - 32px);
71 | height: 520px;
72 | max-height: calc(100vh - 100px);
73 | background: rgba(15, 23, 42, 0.94);
74 | border: 1px solid rgba(0, 242, 255, 0.3);
75 | border-radius: 16px;
76 | box-shadow: 0 20px 40px rgba(0, 0, 0, 0.6), 0 0 30px rgba(0, 242, 255, 0.15);
77 | backdrop-filter: blur(16px);
78 | display: flex;
79 | flex-direction: column;
80 | overflow: hidden;
81 | transition: all 0.3s cubic-bezier(0.16, 1, 0.3, 1);
82 | opacity: 1;
83 | transform: translateY(0) scale(1);
84 | transform-origin: bottom right;
85 | }
86 |
87 | .chatbot-modal.hidden {
88 |   opacity: 0;
89 |   transform: translateY(20px) scale(0.9);
90 |   pointer-events: none;
91 | }
92 |
93 | /* Header */
94 | .chatbot-header {
95 |   padding: 14px 18px;
96 |   background: rgba(30, 41, 59, 0.7);
97 |   border-bottom: 1px solid rgba(255, 255, 255, 0.08);
98 |   display: flex;
99 |   align-items: center;
100 |   justify-content: space-between;
101 | }
102 |
103 | .chatbot-header-info {
104 |   display: flex;
105 |   align-items: center;
106 |   gap: 12px;
107 | }
108 |
109 | .chatbot-avatar {
110 |   width: 34px;
111 |   height: 34px;
112 |   border-radius: 50%;
113 |   background: rgba(0, 242, 255, 0.15);
114 |   border: 1px solid var(--accent-cyan, #00f2ff);
115 |   display: flex;
116 |   align-items: center;
117 |   justify-content: center;
118 |   font-size: 1.1rem;
119 | }
120 |
121 | .chatbot-title {
122 |   font-weight: 700;
123 |   font-size: 0.95rem;
124 |   color: #ffffff;
125 | }
126 |
127 | .chatbot-status {
128 |   font-size: 0.75rem;
129 |   color: #94a3b8;
130 |   display: flex;
131 |   align-items: center;
132 |   gap: 5px;
133 | }
134 |
135 | .status-indicator-dot {
136 |   width: 6px;
```

```
137 | height: 6px;
138 | border-radius: 50%;
139 | background: var(--accent-green, #10b981);
140 | }
141 |
142 | .chatbot-header-actions button {
143 |   background: transparent;
144 |   border: none;
145 |   color: #94a3b8;
146 |   font-size: 1.1rem;
147 |   cursor: pointer;
148 |   padding: 4px 8px;
149 |   border-radius: 6px;
150 |   transition: all 0.2s ease;
151 | }
152 |
153 | .chatbot-header-actions button:hover {
154 |   color: #ffffff;
155 |   background: rgba(255, 255, 255, 0.1);
156 | }
157 |
158 | /* Messages Body */
159 | .chatbot-messages {
160 |   flex: 1;
161 |   padding: 16px;
162 |   overflow-y: auto;
163 |   display: flex;
164 |   flex-direction: column;
165 |   gap: 14px;
166 |   scroll-behavior: smooth;
167 | }
168 |
169 | .chat-message {
170 |   display: flex;
171 |   max-width: 86%;
172 | }
173 |
174 | .chat-message.user {
175 |   align-self: flex-end;
176 | }
177 |
178 | .chat-message.assistant {
179 |   align-self: flex-start;
180 | }
181 |
182 | .message-content {
183 |   padding: 10px 14px;
184 |   border-radius: 12px;
185 |   font-size: 0.88rem;
186 |   line-height: 1.45;
187 |   color: #e2e8f0;
188 | }
189 |
190 | .chat-message.user .message-content {
191 |   background: linear-gradient(135deg, rgba(0, 242, 255, 0.2), rgba(16, 185, 129, 0.2));
192 |   border: 1px solid rgba(0, 242, 255, 0.3);
193 |   border-bottom-right-radius: 2px;
194 |   color: #ffffff;
195 | }
196 |
197 | .chat-message.assistant .message-content {
198 |   background: rgba(30, 41, 59, 0.8);
199 |   border: 1px solid rgba(255, 255, 255, 0.08);
200 |   border-bottom-left-radius: 2px;
201 | }
202 |
203 | /* Typing indicator */
204 | .chat-message.typing .message-content {
205 |   display: flex;
```

```
206 | gap: 6px;
207 | align-items: center;
208 | padding: 12px 18px;
209 | }
210 |
211 | .typing-dot {
212 |   width: 6px;
213 |   height: 6px;
214 |   background: var(--accent-cyan, #00f2ff);
215 |   border-radius: 50%;
216 |   animation: typing-bounce 1.4s infinite ease-in-out both;
217 | }
218 |
219 | .typing-dot:nth-child(1) { animation-delay: -0.32s; }
220 | .typing-dot:nth-child(2) { animation-delay: -0.16s; }
221 |
222 | @keyframes typing-bounce {
223 |   0%, 80%, 100% { transform: scale(0); }
224 |   40% { transform: scale(1); }
225 | }
226 |
227 | /* Suggestion Chips */
228 | .chatbot-chips {
229 |   padding: 8px 12px;
230 |   display: flex;
231 |   gap: 6px;
232 |   overflow-x: auto;
233 |   border-top: 1px solid rgba(255, 255, 255, 0.05);
234 |   scrollbar-width: none;
235 | }
236 |
237 | .chatbot-chips::-webkit-scrollbar {
238 |   display: none;
239 | }
240 |
241 | .chip-btn {
242 |   white-space: nowrap;
243 |   background: rgba(30, 41, 59, 0.6);
244 |   border: 1px solid rgba(0, 242, 255, 0.2);
245 |   color: #cbd5e1;
246 |   font-size: 0.78rem;
247 |   padding: 5px 10px;
248 |   border-radius: 20px;
249 |   cursor: pointer;
250 |   transition: all 0.2s ease;
251 | }
252 |
253 | .chip-btn:hover {
254 |   background: rgba(0, 242, 255, 0.15);
255 |   border-color: var(--accent-cyan, #00f2ff);
256 |   color: #ffffff;
257 | }
258 |
259 | /* Footer Input */
260 | .chatbot-footer {
261 |   padding: 12px;
262 |   background: rgba(15, 23, 42, 0.9);
263 |   border-top: 1px solid rgba(255, 255, 255, 0.08);
264 |   display: flex;
265 |   gap: 8px;
266 | }
267 |
268 | .chatbot-footer input {
269 |   flex: 1;
270 |   background: rgba(30, 41, 59, 0.8);
271 |   border: 1px solid rgba(255, 255, 255, 0.1);
272 |   border-radius: 8px;
273 |   padding: 10px 12px;
274 |   color: #ffffff;
```

```
275 |     font-size: 0.88rem;
276 |     outline: none;
277 |     transition: border 0.2s ease;
278 | }
279 |
280 | .chatbot-footer input:focus {
281 |     border-color: var(--accent-cyan, #00f2ff);
282 | }
283 |
284 | .chatbot-footer button {
285 |     background: var(--accent-cyan, #00f2ff);
286 |     border: none;
287 |     color: #0f172a;
288 |     padding: 0 14px;
289 |     border-radius: 8px;
290 |     cursor: pointer;
291 |     display: flex;
292 |     align-items: center;
293 |     justify-content: center;
294 |     transition: all 0.2s ease;
295 | }
296 |
297 | .chatbot-footer button:hover {
298 |     background: #38bdf8;
299 |     transform: scale(1.05);
300 | }
301 |
302 | /* Responsive adjust */
303 | @media (max-width: 480px) {
304 |     #tc-chatbot-widget {
305 |         bottom: 16px;
306 |         right: 16px;
307 |     }
308 |     .chatbot-modal {
309 |         width: calc(100vw - 32px);
310 |         height: 480px;
311 |     }
312 | }
```

File: frontend/css/scanner.css

Size: 8214 bytes

```

1 | /* =====
2 |    CORRUPTX - Scanner Page Styling
3 |    ===== */
4 |
5 | .scanner-hero {
6 |     padding: 48px 0 28px;
7 |     text-align: center;
8 | }
9 |
10 | .scanner-badge {
11 |     display: inline-flex;
12 |     align-items: center;
13 |     gap: 8px;
14 |     padding: 6px 14px;
15 |     background: rgba(0, 242, 255, 0.08);
16 |     border: 1px solid var(--border-focus);
17 |     border-radius: var(--radius-full);
18 |     color: var(--accent-cyan);
19 |     font-family: var(--font-mono);
20 |     font-size: 0.78rem;
21 |     letter-spacing: 0.08em;
22 |     margin-bottom: 16px;
23 |     text-transform: uppercase;
24 | }
25 |
26 | .scanner-title {
27 |     font-size: 2.4rem;
28 |     font-weight: 800;
29 |     margin-bottom: 12px;
30 |     background: linear-gradient(135deg, #ffffff 0%, #cbd5e1 60%, #38bdf8 100%);
31 |     -webkit-background-clip: text;
32 |     -webkit-text-fill-color: transparent;
33 | }
34 |
35 | .scanner-subtitle {
36 |     font-size: 1.05rem;
37 |     max-width: 640px;
38 |     margin: 0 auto;
39 | }
40 |
41 | /* Scanner Workspace Box */
42 | .scanner-workspace {
43 |     margin-bottom: 60px;
44 | }
45 |
46 | .scanner-card {
47 |     background: var(--bg-card);
48 |     backdrop-filter: blur(16px);
49 |     -webkit-backdrop-filter: blur(16px);
50 |     border: 1px solid var(--border-subtle);
51 |     border-radius: var(--radius-lg);
52 |     padding: 24px;
53 |     box-shadow: 0 10px 40px rgba(0, 0, 0, 0.6), 0 0 20px rgba(0, 242, 255, 0.05);
54 | }
55 |
56 | /* Document Meta Bar */
57 | .doc-meta-bar {
58 |     display: flex;
59 |     align-items: center;
60 |     justify-content: space-between;
61 |     flex-wrap: wrap;
62 |     gap: 16px;
63 |     margin-bottom: 18px;
64 |     padding-bottom: 16px;
65 |     border-bottom: 1px solid var(--border-card);
66 | }
67 |

```

```
68 | .doc-name-input-group {
69 |   display: flex;
70 |   align-items: center;
71 |   gap: 10px;
72 |   flex: 1;
73 |   min-width: 260px;
74 | }
75 |
76 | .doc-name-label {
77 |   font-size: 0.82rem;
78 |   font-family: var(--font-mono);
79 |   color: var(--accent-cyan);
80 |   text-transform: uppercase;
81 | }
82 |
83 | .doc-name-input {
84 |   background: rgba(11, 17, 28, 0.8);
85 |   border: 1px solid var(--border-subtle);
86 |   border-radius: var(--radius-sm);
87 |   padding: 8px 12px;
88 |   color: #fff;
89 |   font-size: 0.9rem;
90 |   font-family: var(--font-sans);
91 |   flex: 1;
92 |   outline: none;
93 |   transition: border-color 0.2s ease;
94 | }
95 |
96 | .doc-name-input:focus {
97 |   border-color: var(--accent-cyan);
98 |   box-shadow: 0 0 10px var(--accent-cyan-glow);
99 | }
100 |
101 | /* Action Toolbar */
102 | .scanner-actions-bar {
103 |   display: flex;
104 |   align-items: center;
105 |   gap: 10px;
106 |   flex-wrap: wrap;
107 | }
108 |
109 | /* Text Input Area */
110 | .textarea-wrapper {
111 |   position: relative;
112 |   width: 100%;
113 |   margin-bottom: 14px;
114 | }
115 |
116 | .policy-textarea {
117 |   width: 100%;
118 |   min-height: 380px;
119 |   max-height: 600px;
120 |   background: rgba(7, 11, 19, 0.85);
121 |   border: 1px solid var(--border-card);
122 |   border-radius: var(--radius-md);
123 |   padding: 20px;
124 |   color: #e2e8f0;
125 |   font-family: var(--font-sans);
126 |   font-size: 0.94rem;
127 |   line-height: 1.7;
128 |   resize: vertical;
129 |   outline: none;
130 |   transition: all 0.25s ease;
131 | }
132 |
133 | .policy-textarea:focus {
134 |   border-color: var(--accent-cyan);
135 |   box-shadow: 0 0 20px var(--accent-cyan-glow);
136 |   background: rgba(10, 15, 26, 0.95);
```

```
137 | }
138 |
139 | .policy-textarea::placeholder {
140 |   color: #475569;
141 | }
142 |
143 | /* Textarea Stats Bar */
144 | .textarea-stats-bar {
145 |   display: flex;
146 |   align-items: center;
147 |   justify-content: space-between;
148 |   flex-wrap: wrap;
149 |   gap: 12px;
150 |   margin-bottom: 24px;
151 |   font-size: 0.82rem;
152 |   font-family: var(--font-mono);
153 |   color: var(--text-dim);
154 | }
155 |
156 | .stats-group {
157 |   display: flex;
158 |   gap: 16px;
159 | }
160 |
161 | .stat-item span {
162 |   color: var(--accent-cyan);
163 |   font-weight: 700;
164 | }
165 |
166 | /* Dropzone */
167 | .file-dropzone {
168 |   border: 2px dashed rgba(0, 242, 255, 0.2);
169 |   background: rgba(0, 242, 255, 0.02);
170 |   border-radius: var(--radius-md);
171 |   padding: 20px;
172 |   text-align: center;
173 |   cursor: pointer;
174 |   transition: all 0.25s ease;
175 |   margin-bottom: 24px;
176 | }
177 |
178 | .file-dropzone:hover, .file-dropzone.dragover {
179 |   border-color: var(--accent-cyan);
180 |   background: rgba(0, 242, 255, 0.06);
181 |   box-shadow: 0 0 20px var(--accent-cyan-glow);
182 | }
183 |
184 | .dropzone-inner {
185 |   display: flex;
186 |   flex-direction: column;
187 |   align-items: center;
188 |   gap: 8px;
189 | }
190 |
191 | .dropzone-icon {
192 |   color: var(--accent-cyan);
193 |   width: 28px;
194 |   height: 28px;
195 | }
196 |
197 | .dropzone-text {
198 |   font-size: 0.9rem;
199 |   color: var(--text-muted);
200 | }
201 |
202 | .dropzone-subtext {
203 |   font-size: 0.78rem;
204 |   color: var(--text-dim);
205 |   font-family: var(--font-mono);
```

```
206 | }
207 |
208 | /* Scanner Footer Launch Bar */
209 | .scanner-launch-bar {
210 |   display: flex;
211 |   align-items: center;
212 |   justify-content: space-between;
213 |   flex-wrap: wrap;
214 |   gap: 16px;
215 |   padding-top: 16px;
216 |   border-top: 1px solid var(--border-card);
217 | }
218 |
219 | .launch-tips {
220 |   display: flex;
221 |   align-items: center;
222 |   gap: 8px;
223 |   font-size: 0.8rem;
224 |   color: var(--text-dim);
225 | }
226 |
227 | .launch-tips svg {
228 |   color: var(--accent-cyan);
229 |   width: 16px;
230 |   height: 16px;
231 | }
232 |
233 | .btn-scan-main {
234 |   padding: 14px 36px;
235 |   font-size: 1.05rem;
236 |   letter-spacing: 0.02em;
237 | }
238 |
239 | /* Scanning Overlay Animation */
240 | .scanning-overlay {
241 |   position: fixed;
242 |   top: 0;
243 |   left: 0;
244 |   right: 0;
245 |   bottom: 0;
246 |   background: rgba(4, 7, 12, 0.92);
247 |   backdrop-filter: blur(12px);
248 |   -webkit-backdrop-filter: blur(12px);
249 |   z-index: 1000;
250 |   display: flex;
251 |   align-items: center;
252 |   justify-content: center;
253 |   padding: 20px;
254 |   opacity: 0;
255 |   visibility: hidden;
256 |   transition: opacity 0.3s ease, visibility 0.3s ease;
257 | }
258 |
259 | .scanning-overlay.active {
260 |   opacity: 1;
261 |   visibility: visible;
262 | }
263 |
264 | .scanning-modal {
265 |   background: var(--bg-secondary);
266 |   border: 1px solid var(--border-focus);
267 |   border-radius: var(--radius-lg);
268 |   padding: 36px;
269 |   width: 100%;
270 |   max-width: 520px;
271 |   box-shadow: 0 0 50px rgba(0, 242, 255, 0.25);
272 |   text-align: center;
273 | }
274 |
```

```
275 | /* Cyber Radar Visual */
276 | .radar-box {
277 |     width: 100px;
278 |     height: 100px;
279 |     margin: 0 auto 24px;
280 |     position: relative;
281 |
282 |     display: flex;
283 |     align-items: center;
284 |     justify-content: center;
285 | }
286 | .radar-ring {
287 |     position: absolute;
288 |     border: 1px solid rgba(0, 242, 255, 0.3);
289 |     border-radius: 50%;
290 |     box-sizing: border-box;
291 | }
292 |
293 | .radar-ring.ring-1 { width: 100%; height: 100%; animation: pulse-ring 2s infinite ease-out; }
294 | .radar-ring.ring-2 { width: 70%; height: 70%; border-color: rgba(0, 242, 255, 0.5); }
295 | .radar-ring.ring-3 { width: 40%; height: 40%; border-color: rgba(0, 242, 255, 0.7); }
296 |
297 | .radar-sweep {
298 |     position: absolute;
299 |     width: 100%;
300 |     height: 100%;
301 |     border-radius: 50%;
302 |     background: conic-gradient(from 0deg, rgba(0, 242, 255, 0.4), transparent 60%);
303 |     animation: sweep 1.8s linear infinite;
304 | }
305 |
306 | .radar-center-icon {
307 |     color: #fff;
308 |     z-index: 2;
309 |     filter: drop-shadow(0 0 8px var(--accent-cyan));
310 | }
311 |
312 | @keyframes sweep {
313 |     from { transform: rotate(0deg); }
314 |     to { transform: rotate(360deg); }
315 | }
316 |
317 | @keyframes pulse-ring {
318 |     0% { transform: scale(0.95); opacity: 0.8; }
319 |     50% { transform: scale(1.05); opacity: 0.3; }
320 |     100% { transform: scale(0.95); opacity: 0.8; }
321 | }
322 |
323 | .scan-progress-title {
324 |     font-size: 1.35rem;
325 |     font-weight: 700;
326 |     margin-bottom: 8px;
327 |     color: #fff;
328 | }
329 |
330 | .scan-progress-sub {
331 |     font-size: 0.86rem;
332 |     color: var(--text-muted);
333 |     margin-bottom: 24px;
334 | }
335 |
336 | /* Scanning Step Checklist */
337 | .scan-steps-list {
338 |     display: flex;
339 |     flex-direction: column;
340 |     gap: 12px;
341 |     text-align: left;
342 |     margin-bottom: 24px;
343 |     background: rgba(9, 14, 24, 0.8);
```

```
344 |     border: 1px solid var(--border-card);
345 |     border-radius: var(--radius-md);
346 |     padding: 16px;
347 | }
348 |
349 | .scan-step-item {
350 |     display: flex;
351 |     align-items: center;
352 |     gap: 12px;
353 |     font-size: 0.85rem;
354 |     font-family: var(--font-mono);
355 |     color: var(--text-dim);
356 |     transition: color 0.25s ease;
357 | }
358 |
359 | .scan-step-item.pending {
360 |     color: var(--text-dim);
361 | }
362 |
363 | .scan-step-item.active {
364 |     color: var(--accent-cyan);
365 |     font-weight: 700;
366 | }
367 |
368 | .scan-step-item.done {
369 |     color: var(--accent-green);
370 | }
371 |
372 | .step-icon-indicator {
373 |     width: 18px;
374 |     height: 18px;
375 |     display: flex;
376 |     align-items: center;
377 |     justify-content: center;
378 | }
379 |
380 | .step-spinner {
381 |     width: 14px;
382 |     height: 14px;
383 |     border: 2px solid rgba(0, 242, 255, 0.3);
384 |     border-top-color: var(--accent-cyan);
385 |     border-radius: 50%;
386 |     animation: spin 0.8s linear infinite;
387 | }
388 |
389 | @keyframes spin {
390 |     to { transform: rotate(360deg); }
391 | }
392 |
393 | /* Progress Bar */
394 | .progress-bar-track {
395 |     width: 100%;
396 |     height: 6px;
397 |     background: rgba(255, 255, 255, 0.08);
398 |     border-radius: var(--radius-full);
399 |     overflow: hidden;
400 |     position: relative;
401 | }
402 |
403 | .progress-bar-fill {
404 |     height: 100%;
405 |     width: 0%;
406 |     background: linear-gradient(90deg, #00d2df, #00ff88);
407 |     box-shadow: 0 0 10px var(--accent-cyan);
408 |     transition: width 0.3s ease;
409 | }
```

File: frontend/css/results.css

Size: 12343 bytes

```

1 | /* =====
2 |    CORRUPTX - Results Dashboard Styling
3 |    ===== */
4 |
5 | .results-header-section {
6 |     padding: 36px 0 24px;
7 | }
8 |
9 | .results-doc-title-row {
10 |     display: flex;
11 |     align-items: center;
12 |     justify-content: space-between;
13 |     flex-wrap: wrap;
14 |     gap: 16px;
15 |     margin-bottom: 24px;
16 | }
17 |
18 | .results-title-group h1 {
19 |     font-size: 1.8rem;
20 |     font-weight: 800;
21 |     display: flex;
22 |     align-items: center;
23 |     gap: 12px;
24 | }
25 |
26 | .results-doc-subtitle {
27 |     font-size: 0.88rem;
28 |     font-family: var(--font-mono);
29 |     color: var(--text-dim);
30 |     margin-top: 4px;
31 | }
32 |
33 | .results-actions-group {
34 |     display: flex;
35 |     align-items: center;
36 |     gap: 10px;
37 |     flex-wrap: wrap;
38 | }
39 |
40 | /* =====
41 |    Top Score Overview Banner
42 |    ===== */
43 | .overview-card {
44 |     background: var(--bg-card);
45 |     backdrop-filter: blur(16px);
46 |     border: 1px solid var(--border-card);
47 |     border-radius: var(--radius-lg);
48 |     padding: 32px;
49 |     box-shadow: var(--shadow-cyber);
50 |     display: grid;
51 |     grid-template-columns: 260px 1fr;
52 |     gap: 36px;
53 |     align-items: center;
54 |     margin-bottom: 32px;
55 |     position: relative;
56 | }
57 |
58 | .overview-card::after {
59 |     content: "";
60 |     position: absolute;
61 |     top: 0;
62 |     left: 0;
63 |     right: 0;
64 |     height: 2px;
65 |     background: linear-gradient(90deg, transparent, var(--accent-cyan), transparent);
66 | }
67 |

```

```
68 | /* Circular Meter */
69 | .meter-wrapper {
70 |     display: flex;
71 |     flex-direction: column;
72 |     align-items: center;
73 |     justify-content: center;
74 | }
75 |
76 | .circular-gauge {
77 |     position: relative;
78 |     width: 190px;
79 |     height: 190px;
80 | }
81 |
82 | .gauge-svg {
83 |     width: 100%;
84 |     height: 100%;
85 |     transform: rotate(-90deg);
86 | }
87 |
88 | .gauge-track {
89 |     fill: none;
90 |     stroke: rgba(255, 255, 255, 0.06);
91 |     stroke-width: 10;
92 | }
93 |
94 | .gauge-fill {
95 |     fill: none;
96 |     stroke-width: 10;
97 |     stroke-linecap: round;
98 |     transition: stroke-dashoffset 1.2s cubic-bezier(0.16, 1, 0.3, 1), stroke 0.4s ease;
99 | }
100 |
101 | .gauge-content {
102 |     position: absolute;
103 |     top: 0;
104 |     left: 0;
105 |     width: 100%;
106 |     height: 100%;
107 |     display: flex;
108 |     flex-direction: column;
109 |     align-items: center;
110 |     justify-content: center;
111 |     text-align: center;
112 | }
113 |
114 | .gauge-score {
115 |     font-size: 3.2rem;
116 |     font-weight: 800;
117 |     line-height: 1;
118 |     font-family: var(--font-sans);
119 |     color: #fff;
120 |     letter-spacing: -0.04em;
121 | }
122 |
123 | .gauge-max {
124 |     font-size: 0.85rem;
125 |     color: var(--text-dim);
126 |     font-family: var(--font-mono);
127 |     font-weight: 600;
128 | }
129 |
130 | .gauge-label {
131 |     margin-top: 14px;
132 | }
133 |
134 | /* Overview Info Panel */
135 | .overview-info-panel {
136 |     display: flex;
```

```
137 | flex-direction: column;
138 | gap: 20px;
139 | }
140 |
141 | .overview-status-line {
142 |   display: flex;
143 |   align-items: center;
144 |   gap: 12px;
145 | }
146 |
147 | .risk-level-badge {
148 |   font-size: 0.88rem;
149 |   padding: 6px 14px;
150 |   letter-spacing: 0.08em;
151 | }
152 |
153 | .scan-timestamp {
154 |   font-size: 0.82rem;
155 |   font-family: var(--font-mono);
156 |   color: var(--text-dim);
157 | }
158 |
159 | .overview-summary-text {
160 |   font-size: 1rem;
161 |   color: #cbd5e1;
162 |   line-height: 1.6;
163 | }
164 |
165 | /* Metrics Grid */
166 | .metrics-grid {
167 |   display: grid;
168 |   grid-template-columns: repeat(auto-fit, minmax(130px, 1fr));
169 |   gap: 14px;
170 |   padding-top: 16px;
171 |   border-top: 1px solid var(--border-card);
172 | }
173 |
174 | .metric-box {
175 |   background: rgba(8, 12, 20, 0.7);
176 |   border: 1px solid var(--border-card);
177 |   border-radius: var(--radius-md);
178 |   padding: 12px 16px;
179 |   display: flex;
180 |   flex-direction: column;
181 | }
182 |
183 | .metric-label {
184 |   font-size: 0.72rem;
185 |   font-family: var(--font-mono);
186 |   color: var(--text-dim);
187 |   text-transform: uppercase;
188 |   letter-spacing: 0.05em;
189 |   margin-bottom: 4px;
190 | }
191 |
192 | .metric-value {
193 |   font-size: 1.45rem;
194 |   font-weight: 800;
195 |   color: #fff;
196 |   line-height: 1.2;
197 | }
198 |
199 | .metric-value.high-val { color: var(--accent-crimson); }
200 | .metric-value.med-val { color: var(--accent-amber); }
201 | .metric-value.low-val { color: var(--accent-blue); }
202 | .metric-value.safe-val { color: var(--accent-green); }
203 |
204 | /* =====
205 |   Tab Navigation & View Switcher
```

```
206 | ===== */
207 | .results-tabs-bar {
208 |   display: flex;
209 |   align-items: center;
210 |   justify-content: space-between;
211 |   flex-wrap: wrap;
212 |   gap: 16px;
213 |   margin-bottom: 24px;
214 |   border-bottom: 1px solid var(--border-card);
215 |   padding-bottom: 12px;
216 | }
217 |
218 | .tab-nav {
219 |   display: flex;
220 |   gap: 8px;
221 | }
222 |
223 | .tab-btn {
224 |   background: transparent;
225 |   border: 1px solid transparent;
226 |   border-radius: var(--radius-md);
227 |   color: var(--text-muted);
228 |   font-size: 0.92rem;
229 |   font-weight: 600;
230 |   padding: 8px 16px;
231 |   cursor: pointer;
232 |   display: flex;
233 |   align-items: center;
234 |   gap: 8px;
235 |   transition: all 0.2s ease;
236 | }
237 |
238 | .tab-btn:hover {
239 |   color: #fff;
240 |   background: rgba(255, 255, 255, 0.03);
241 | }
242 |
243 | .tab-btn.active {
244 |   color: var(--accent-cyan);
245 |   background: rgba(0, 242, 255, 0.08);
246 |   border-color: var(--border-focus);
247 |   box-shadow: 0 0 12px var(--accent-cyan-glow);
248 | }
249 |
250 | .tab-badge {
251 |   padding: 2px 7px;
252 |   background: rgba(255, 255, 255, 0.08);
253 |   border-radius: var(--radius-full);
254 |   font-size: 0.72rem;
255 |   font-family: var(--font-mono);
256 | }
257 |
258 | .tab-btn.active .tab-badge {
259 |   background: var(--accent-cyan);
260 |   color: #000;
261 | }
262 |
263 | /* Filter buttons */
264 | .filter-group {
265 |   display: flex;
266 |   align-items: center;
267 |   gap: 6px;
268 |   flex-wrap: wrap;
269 | }
270 |
271 | .filter-btn {
272 |   background: transparent;
273 |   border: 1px solid var(--border-card);
274 |   border-radius: var(--radius-sm);
```

```

275 |   color: var(--text-dim);
276 |   font-size: 0.78rem;
277 |   font-family: var(--font-mono);
278 |   padding: 5px 10px;
279 |   cursor: pointer;
280 |   transition: all 0.2s ease;
281 | }
282 |
283 | .filter-btn:hover {
284 |   color: #fff;
285 |   border-color: var(--border-subtle);
286 | }
287 |
288 | .filter-btn.active {
289 |   background: rgba(0, 242, 255, 0.12);
290 |   color: var(--accent-cyan);
291 |   border-color: var(--accent-cyan);
292 | }
293 |
294 | /* =====
295 |   View 1: Red Flag Cards
296 |   ===== */
297 | .red-flags-view {
298 |   display: flex;
299 |   flex-direction: column;
300 |   gap: 18px;
301 |   margin-bottom: 48px;
302 | }
303 |
304 | .flag-card {
305 |   background: var(--bg-card);
306 |   backdrop-filter: blur(12px);
307 |   border: 1px solid var(--border-card);
308 |   border-radius: var(--radius-lg);
309 |   padding: 22px 24px;
310 |   transition: all 0.22s ease;
311 |   position: relative;
312 |   overflow: hidden;
313 | }
314 |
315 | .flag-card::before {
316 |   content: "";
317 |   position: absolute;
318 |   top: 0;
319 |   left: 0;
320 |   bottom: 0;
321 |   width: 4px;
322 | }
323 |
324 | .flag-card.flag-card-high::before { background: var(--accent-crimson); box-shadow: 0 0 10px var(--accent-crimson); }
325 | .flag-card.flag-card-medium::before { background: var(--accent-amber); box-shadow: 0 0 10px var(--accent-amber); }
326 | .flag-card.flag-card-low::before { background: var(--accent-blue); box-shadow: 0 0 8px var(--accent-blue); }
327 |
328 | .flag-card:hover {
329 |   border-color: var(--border-subtle);
330 |   transform: translateY(-2px);
331 |   box-shadow: var(--shadow-cyber);
332 | }
333 |
334 | .flag-card-header {
335 |   display: flex;
336 |   align-items: center;
337 |   justify-content: space-between;
338 |   flex-wrap: wrap;
339 |   gap: 12px;
340 |   margin-bottom: 14px;
341 | }
342 |
343 | .flag-category-title {

```

```
344 |     display: flex;
345 |     align-items: center;
346 |     gap: 10px;
347 | }
348 |
349 | .category-name-text {
350 |     font-size: 1.05rem;
351 |     font-weight: 700;
352 |     color: #fff;
353 | }
354 |
355 | .flag-card-actions {
356 |     display: flex;
357 |     align-items: center;
358 |     gap: 8px;
359 | }
360 |
361 | /* Original clause snippet block */
362 | .clause-quote-box {
363 |     background: rgba(7, 10, 16, 0.85);
364 |     border: 1px solid rgba(255, 255, 255, 0.05);
365 |     border-left: 3px solid var(--accent-cyan);
366 |     border-radius: var(--radius-sm);
367 |     padding: 14px 18px;
368 |     margin-bottom: 16px;
369 |     font-size: 0.92rem;
370 |     color: #e2e8f0;
371 |     line-height: 1.6;
372 |     font-style: italic;
373 | }
374 |
375 | /* Explanation & Why this matters */
376 | .flag-analysis-body {
377 |     display: grid;
378 |     grid-template-columns: 1fr 1fr;
379 |     gap: 16px;
380 |     margin-bottom: 14px;
381 | }
382 |
383 | .analysis-callout {
384 |     background: rgba(15, 23, 42, 0.45);
385 |     border: 1px solid rgba(255, 255, 255, 0.04);
386 |     border-radius: var(--radius-md);
387 |     padding: 14px;
388 | }
389 |
390 | .callout-header {
391 |     display: flex;
392 |     align-items: center;
393 |     gap: 8px;
394 |     font-size: 0.78rem;
395 |     font-family: var(--font-mono);
396 |     text-transform: uppercase;
397 |     letter-spacing: 0.05em;
398 |     margin-bottom: 6px;
399 |     color: var(--accent-cyan);
400 | }
401 |
402 | .callout-header svg {
403 |     width: 14px;
404 |     height: 14px;
405 | }
406 |
407 | .callout-text {
408 |     font-size: 0.88rem;
409 |     color: #cbd5e1;
410 |     line-height: 1.5;
411 | }
412 |
```

```

413 | /* Action recommendation footer */
414 | .recommendation-bar {
415 |     background: rgba(0, 255, 136, 0.04);
416 |     border: 1px solid rgba(0, 255, 136, 0.2);
417 |     border-radius: var(--radius-md);
418 |     padding: 10px 16px;
419 |     display: flex;
420 |     align-items: center;
421 |     gap: 12px;
422 |     font-size: 0.86rem;
423 |     color: #a7f3d0;
424 | }
425 |
426 | .recommendation-bar svg {
427 |     color: var(--accent-green);
428 |     flex-shrink: 0;
429 | }
430 |
431 | /* =====
432 |     View 2: Interactive Policy Highlighter View
433 |     ===== */
434 | .policy-analysis-view {
435 |     display: grid;
436 |     grid-template-columns: 1fr 340px;
437 |     gap: 24px;
438 |     align-items: start;
439 |     margin-bottom: 48px;
440 | }
441 |
442 | .policy-doc-container {
443 |     background: var(--bg-card);
444 |     backdrop-filter: blur(14px);
445 |     border: 1px solid var(--border-card);
446 |     border-radius: var(--radius-lg);
447 |     padding: 28px;
448 |     box-shadow: var(--shadow-cyber);
449 |     font-size: 0.96rem;
450 |     line-height: 1.85;
451 |     color: #cbd5e1;
452 | }
453 |
454 | .policy-doc-container p {
455 |     margin-bottom: 16px;
456 |     color: #cbd5e1;
457 | }
458 |
459 | /* Clause highlight tags */
460 | .clause-highlight {
461 |     cursor: pointer;
462 |     padding: 2px 5px;
463 |     border-radius: 4px;
464 |     transition: all 0.2s ease;
465 |     position: relative;
466 | }
467 |
468 | .clause-highlight:hover {
469 |     filter: brightness(1.2);
470 |     box-shadow: 0 0 10px rgba(0, 242, 255, 0.3);
471 | }
472 |
473 | .clause-highlight.highlight-high {
474 |     background: rgba(255, 51, 102, 0.18);
475 |     border-bottom: 2px solid var(--accent-crimson);
476 |     color: #fff;
477 | }
478 |
479 | .clause-highlight.highlight-medium {
480 |     background: rgba(255, 183, 0, 0.16);
481 |     border-bottom: 2px solid var(--accent-amber);

```

```
482 |   color: #fff;
483 | }
484 |
485 | .clause-highlight.highlight-low {
486 |   background: rgba(56, 189, 248, 0.14);
487 |   border-bottom: 2px solid var(--accent-blue);
488 |   color: #fff;
489 | }
490 |
491 | .clause-highlight.highlight-safe {
492 |   background: transparent;
493 |   color: #cbd5e1;
494 | }
495 |
496 | .clause-highlight.active-inspected {
497 |   outline: 2px solid var(--accent-cyan);
498 |   box-shadow: 0 0 15px var(--accent-cyan);
499 | }
500 |
501 | /* Sticky Inspector Panel in Highlighter View */
502 | .sticky-inspector-panel {
503 |   position: sticky;
504 |   top: 96px;
505 |   background: var(--bg-card);
506 |   border: 1px solid var(--border-subtle);
507 |   border-radius: var(--radius-lg);
508 |   padding: 20px;
509 |   box-shadow: var(--shadow-cyber);
510 | }
511 |
512 | .inspector-title {
513 |   font-size: 1.05rem;
514 |   font-weight: 700;
515 |   display: flex;
516 |   align-items: center;
517 |   gap: 8px;
518 |   margin-bottom: 14px;
519 |   padding-bottom: 12px;
520 |   border-bottom: 1px solid var(--border-card);
521 |   color: #fff;
522 | }
523 |
524 | .inspector-placeholder {
525 |   padding: 32px 16px;
526 |   text-align: center;
527 |   color: var(--text-dim);
528 |   font-size: 0.88rem;
529 | }
530 |
531 | .inspector-placeholder svg {
532 |   margin-bottom: 12px;
533 |   color: var(--accent-cyan);
534 | }
535 |
536 | .inspector-content-pane {
537 |   display: flex;
538 |   flex-direction: column;
539 |   gap: 14px;
540 | }
541 |
542 | /* =====
543 |   View 3: Category Risk Breakdown
544 |   ===== */
545 | .category-breakdown-view {
546 |   background: var(--bg-card);
547 |   backdrop-filter: blur(14px);
548 |   border: 1px solid var(--border-card);
549 |   border-radius: var(--radius-lg);
550 |   padding: 28px;
```

```
551 |     box-shadow: var(--shadow-cyber);
552 |     margin-bottom: 48px;
553 | }
554 |
555 | .breakdown-title-row {
556 |     display: flex;
557 |     align-items: center;
558 |     justify-content: space-between;
559 |     margin-bottom: 24px;
560 | }
561 |
562 | .breakdown-list {
563 |     display: flex;
564 |     flex-direction: column;
565 |     gap: 20px;
566 | }
567 |
568 | .breakdown-item {
569 |     display: flex;
570 |     flex-direction: column;
571 |     gap: 8px;
572 | }
573 |
574 | .breakdown-item-header {
575 |     display: flex;
576 |     justify-content: space-between;
577 |     align-items: center;
578 | }
579 |
580 | .breakdown-category-label {
581 |     font-weight: 600;
582 |     font-size: 0.95rem;
583 |     color: #fff;
584 |     display: flex;
585 |     align-items: center;
586 |     gap: 10px;
587 | }
588 |
589 | .breakdown-counts {
590 |     font-family: var(--font-mono);
591 |     font-size: 0.82rem;
592 |     color: var(--text-muted);
593 | }
594 |
595 | .breakdown-track {
596 |     height: 8px;
597 |     background: rgba(255, 255, 255, 0.06);
598 |     border-radius: var(--radius-full);
599 |     overflow: hidden;
600 |     position: relative;
601 | }
602 |
603 | .breakdown-fill {
604 |     height: 100%;
605 |     border-radius: var(--radius-full);
606 |     transition: width 0.8s cubic-bezier(0.16, 1, 0.3, 1);
607 | }
```

File: frontend/css/responsive.css

Size: 2296 bytes

```

1 | /* =====
2 |    CORRUPTX - Responsive Breakpoints & Mobile Optimization
3 |    ===== */
4 |
5 | /* Tablets & Small Laptops (<= 1024px) */
6 | @media (max-width: 1024px) {
7 |     .overview-card {
8 |         grid-template-columns: 1fr;
9 |         text-align: center;
10 |         gap: 28px;
11 |     }
12 |
13 |     .overview-status-line {
14 |         justify-content: center;
15 |     }
16 |
17 |     .policy-analysis-view {
18 |         grid-template-columns: 1fr;
19 |     }
20 |
21 |     .sticky-inspector-panel {
22 |         position: static;
23 |         margin-top: 24px;
24 |     }
25 | }
26 |
27 | /* Tablets (<= 768px) */
28 | @media (max-width: 768px) {
29 |     .site-header {
30 |         height: 64px;
31 |     }
32 |
33 |     .main-nav {
34 |         display: none;
35 |         position: absolute;
36 |         top: 64px;
37 |         left: 0;
38 |         right: 0;
39 |         background: rgba(7, 9, 14, 0.98);
40 |         backdrop-filter: blur(20px);
41 |         border-bottom: 1px solid var(--border-subtle);
42 |         padding: 20px 24px;
43 |         flex-direction: column;
44 |         align-items: flex-start;
45 |         gap: 16px;
46 |         box-shadow: 0 10px 30px rgba(0, 0, 0, 0.8);
47 |     }
48 |
49 |     .main-nav.mobile-open {
50 |         display: flex;
51 |     }
52 |
53 |     .mobile-menu-btn {
54 |         display: block;
55 |     }
56 |
57 |     .system-status {
58 |         display: none;
59 |     }
60 |
61 |     .flag-analysis-body {
62 |         grid-template-columns: 1fr;
63 |     }
64 |
65 |     .scanner-title {
66 |         font-size: 1.8rem;
67 |     }

```

```
68 |  
69 | .policy-textarea {  
70 |     min-height: 280px;  
71 | }  
72 |  
73 | .results-tabs-bar {  
74 |     flex-direction: column;  
75 |     align-items: stretch;  
76 | }  
77 |  
78 | .tab-nav {  
79 |     overflow-x: auto;  
80 |     padding-bottom: 4px;  
81 | }  
82 |  
83 | .filter-group {  
84 |     justify-content: flex-start;  
85 | }  
86 |  
87 | .footer-top {  
88 |     flex-direction: column;  
89 |     gap: 24px;  
90 | }  
91 |  
92 | .footer-links {  
93 |     flex-wrap: wrap;  
94 |     gap: 24px;  
95 | }  
96 |  
97 | .hero-mockup-wrapper {  
98 |     margin-top: 32px;  
99 | }  
100 | }  
101 |  
102 | /* Mobile Devices (<= 480px) */  
103 | @media (max-width: 480px) {  
104 |     .container {  
105 |         padding: 0 16px;  
106 |     }  
107 |  
108 |     .scanner-card, .overview-card, .flag-card, .policy-doc-container {  
109 |         padding: 18px 16px;  
110 |     }  
111 |  
112 |     .metrics-grid {  
113 |         grid-template-columns: 1fr 1fr;  
114 |     }  
115 |  
116 |     .scanner-actions-bar {  
117 |         width: 100%;  
118 |         justify-content: space-between;  
119 |     }  
120 |  
121 |     .scanner-actions-bar .btn {  
122 |         padding: 8px 12px;  
123 |         font-size: 0.8rem;  
124 |     }  
125 |  
126 |     .btn-scan-main {  
127 |         width: 100%;  
128 |     }  
129 |  
130 |     .circular-gauge {  
131 |         width: 160px;  
132 |         height: 160px;  
133 |     }  
134 |  
135 |     .gauge-score {  
136 |         font-size: 2.7rem;
```

```
137 |   }  
138 | }
```

File: backend/risk_rules.json

Size: 20499 bytes

```

1 | {
2 |   "version": "1.0.0",
3 |   "taxonomy_name": "CorruptX CyberSecurity T&C Risk Taxonomy",
4 |   "categories": [
5 |     {
6 |       "id": "data_sharing",
7 |       "name": "Data Sharing",
8 |       "severity": "HIGH",
9 |       "weight": 15,
10 |      "color": "#ff3366",
11 |      "description": "Sharing, selling, transferring, or disclosing personal data to external third parties, business pa
12 |      "patterns": [
13 |        "share(?:s|d|ing)?\\s+(?:your|user|personal|sensitive)\\s+(?:information|data|details)\\s+with\\s+third\\s+parti
14 |        "disclose(?:s|d|ing)?\\s+(?:your|personal|user)\\s+(?:information|data)\\s+to\\s+(?:third\\s+parties|affiliates|
15 |        "sell(?:s|ing)?\\s+(?:or\\s+rent(?:s|ing)?\\s+)?(?:your|personal|user)\\s+(?:data|information)",
16 |        "transfer(?:s|red|ring)?\\s+(?:your|personal)\\s+data\\s+to\\s+(?:other\\s+entities|third\\s+parties|external)",
17 |        "commercial\\s+partners\\s+may\\s+(?:receive|access|use)\\s+your\\s+data",
18 |        "we\\s+reserve\\s+the\\s+right\\s+to\\s+share\\s+(?:aggregated|anonymized|de-identified|any)?\\s*(?:information|
19 |        "monetize(?:s|d|ing)?\\s+(?:user|personal)\\s+information",
20 |        "in\\s+the\\s+event\\s+of\\s+a\\s+merger(?:,|\\s+or)\\s+(?:acquisition|sale|bankruptcy).*data.*transferred"
21 |      ],
22 |      "keywords": [
23 |        "sell your data",
24 |        "share your information with third parties",
25 |        "disclose personal data to partners",
26 |        "monetize user information",
27 |        "share with advertisers"
28 |      ],
29 |      "explanation": "This clause allows the service to share, distribute, or sell your personal details with third-part
30 |      "why_it_matters": "Once your information is handed over to outside parties, you lose direct control over how it is
31 |      "recommendation": "Check if an opt-out mechanism is available or avoid submitting sensitive confidential data."
32 |    },
33 |    {
34 |      "id": "tracking",
35 |      "name": "Tracking & Telemetry",
36 |      "severity": "HIGH",
37 |      "weight": 14,
38 |      "color": "#ff3366",
39 |      "description": "Continuous device fingerprinting, behavioral tracking, cross-site telemetry, or keystroke monitori
40 |      "patterns": [
41 |        "track(?:s|ed|ing)?\\s+your\\s+(?:browsing|activity|behavior|location|online\\s+habits)",
42 |        "fingerprint(?:ing)?\\s+(?:your\\s+device|hardware|browser)",
43 |        "cross-site\\s+tracking",
44 |        "canvas\\s+fingerprinting",
45 |        "beacon(?:s)?|pixel\\s+tags?|web\\s+bugs?",
46 |        "record(?:s|ing)?\\s+(?:your\\s+screen|keystrokes|mouse\\s+movements|session\\s+replay)",
47 |        "monitor(?:s|ing)?\\s+(?:all|your)?\\s*(?:communications|browsing\\s+history|actions)",
48 |        "collect(?:s|ed|ing)?\\s+information\\s+about\\s+your\\s+activity\\s+across\\s+(?:third-party|other)\\s+websites
49 |      ],
50 |      "keywords": [
51 |        "cross-site tracking",
52 |        "device fingerprinting",
53 |        "session replay",
54 |        "keystroke recording",
55 |        "track your browsing activity"
56 |      ],
57 |      "explanation": "This clause enables active monitoring, device fingerprinting, or tracking of your online behavior
58 |      "why_it_matters": "Cross-site and hardware telemetry can build a persistent digital shadow profile of your habits,
59 |      "recommendation": "Use privacy-focused browsers, anti-fingerprinting shields, and strict tracker-blocking extensio
60 |    },
61 |    {
62 |      "id": "data_collection",
63 |      "name": "Data Collection",
64 |      "severity": "HIGH",
65 |      "weight": 12,
66 |      "color": "#ff3366",
67 |      "description": "Excessive collection of sensitive personal data, biometrics, contacts, communications, or device i

```

```

68 |         "patterns": [
69 |             "collect(?:s|ed|ing)?\\s+(?:biometric|health|financial|genetic|sensitive)\\s+(?:information|data)",
70 |             "access(?:es)?\\s+(?:to\\s+)?your\\s+(?:contacts|address\\s+book|microphone|camera|stored\\s+files|photos)",
71 |             "scrape(?:s|d|ing)?\\s+(?:data|information|content)",
72 |             "collect(?:s|ed|ing)?\\s+(?:all|any)\\s+information\\s+you\\s+(?:transmit|submit|upload|post|create)",
73 |             "gather(?:s|ed|ing)?\\s+(?:details|information)\\s+from\\s+external\\s+public\\s+and\\s+private\\s+records",
74 |             "including\\s+but\\s+not\\s+limited\\s+to\\s+social\\s+security|credit\\s+card|bank\\s+account"
75 |         ],
76 |         "keywords": [
77 |             "biometric data",
78 |             "access to your contacts",
79 |             "microphone and camera access",
80 |             "sensitive personal data",
81 |             "financial information"
82 |         ],
83 |         "explanation": "The provider reserves extensive rights to gather invasive categories of personal data, including c
84 |         "why_it_matters": "Over-collection expands your attack surface; if the provider suffers a data breach, your sensit
85 |         "recommendation": "Review permissions requested on your device and restrict sensor/contact access."
86 |     },
87 |     {
88 |         "id": "arbitration",
89 |         "name": "Mandatory Arbitration",
90 |         "severity": "HIGH",
91 |         "weight": 14,
92 |         "color": "#ff3366",
93 |         "description": "Forced binding arbitration and waiver of court trials and collective class action lawsuits.",
94 |         "patterns": [
95 |             "binding\\s+arbitration",
96 |             "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+)?(?:a\\s+)?jury\\s+trial",
97 |             "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+participate\\s+in\\s+a\\s+)?class\\s+action",
98 |             "class\\s+action\\s+waiver",
99 |             "disputes?\\s+shall\\s+be\\s+resolved\\s+(?:exclusively\\s+)?(?:by|through)\\s+(?:confidential|individual|bindin
100 |             "no\\s+class\\s+or\\s+representative\\s+actions?",
101 |             "agree\\s+not\\s+to\\s+sue\\s+in\\s+court"
102 |         ],
103 |         "keywords": [
104 |             "binding arbitration",
105 |             "class action waiver",
106 |             "waive jury trial",
107 |             "resolve disputes through arbitration",
108 |             "no class action"
109 |         ],
110 |         "explanation": "You surrender your constitutional right to take this company to court or participate in a joint co
111 |         "why_it_matters": "Arbitration proceedings are usually private, costly for individuals, and systematically favor c
112 |         "recommendation": "Check if an arbitration opt-out letter can be submitted within 30 days of registration."
113 |     },
114 |     {
115 |         "id": "unilateral_changes",
116 |         "name": "Unilateral Changes",
117 |         "severity": "HIGH",
118 |         "weight": 13,
119 |         "color": "#ff3366",
120 |         "description": "Changing legal terms or privacy policies at any time without notifying users or requiring fresh co
121 |         "patterns": [
122 |             "reserve\\s+the\\s+right\\s+to\\s+(?:modify|change|alter|update)\\s+(?:these\\s+terms|this\\s+agreement|policy)\\s+
123 |             "(?:modify|change|update)\\s+without\\s+(?:prior\\s+)?notice",
124 |             "continued\\s+use\\s+(?:of\\s+the\\s+service\\s+)?(?:constitutes|shall\\s+constitute|signifies)\\s+(?:your\\s+)?
125 |             "without\\s+obligation\\s+to\\s+notify\\s+you",
126 |             "effective\\s+immediately\\s+upon\\s+posting"
127 |         ],
128 |         "keywords": [
129 |             "modify without prior notice",
130 |             "at our sole discretion without notice",
131 |             "continued use constitutes acceptance",
132 |             "change these terms at any time"
133 |         ],
134 |         "explanation": "The company can alter policies, fees, or data handling rules at any time, and your continued usage
135 |         "why_it_matters": "You could wake up to new invasive terms or higher subscription fees without ever receiving an e
136 |         "recommendation": "Periodically review terms updates or monitor policy changes using change-detection tools."

```

```

137 | },
138 | {
139 |     "id": "liability_waiver",
140 |     "name": "Liability Waiver",
141 |
142 |     "severity": "MEDIUM",
143 |     "weight": 9,
144 |     "color": "#ffb700",
145 |     "description": "Total disclaimer of liability for data loss, service outages, bugs, or security vulnerabilities.",
146 |     "patterns": [
147 |         "provided\\s+[\\']?as\\s+is[\\']?\\s+and\\s+[\\']?as\\s+available[\\']?",
148 |         "disclaim(?:s)?\\s+(?:all|any)\\s+warranties",
149 |         "(?:under\\s+no\\s+circumstances\\s+shall|in\\s+no\\s+event\\s+shall)\\s+(?:the\\s+company|we|us|our\\s+affiliat",
150 |         "limit(?:ation)?\\s+of\\s+liability\\s+shall\\s+not\\s+exceed\\s+(?:the\\s+amount\\s+paid|\\$\\d+|one\\s+hundred",
151 |         "not\\s+liable\\s+for\\s+(?:any\\s+)?(?:data\\s+loss|security\\s+breaches|unauthorized\\s+access)"
152 |     ],
153 |     "keywords": [
154 |         "provided as is",
155 |         "in no event shall we be liable",
156 |         "disclaim all warranties",
157 |         "not liable for data loss or breach",
158 |         "limitation of liability"
159 |     ],
160 |     "explanation": "The company provides the product 'as-is' and completely disclaims legal responsibility if their so",
161 |     "why_it_matters": "If a critical security bug causes data corruption or identity theft, this clause shields the ve",
162 |     "recommendation": "Never rely solely on this vendor for sole data backups; maintain external independent copies."
163 | },
164 | {
165 |     "id": "auto_renewal",
166 |     "name": "Auto-Renewal & Recurring Billing",
167 |     "severity": "MEDIUM",
168 |     "weight": 8,
169 |     "color": "#ffb700",
170 |     "description": "Automatic recurring subscription renewals, often accompanied by strict cancellation timelines.",
171 |     "patterns": [
172 |         "auto(?:matically)?-?renew(?:s|ed|ing)?",
173 |         "renew(?:s|ed|ing)?\\s+automatically\\s+unless\\s+cancelled",
174 |         "recurring\\s+(?:billing|charge|subscription|fee)",
175 |         "charge(?:s|d)?\\s+your\\s+(?:payment\\s+method|credit\\s+card)\\s+without\\s+(?:additional|further)\\s+authoriz",
176 |         "cancel(?:ation)?\\s+at\\s+least\\s+\\d+\\s+days\\s+prior\\s+to\\s+(?:the\\s+end|renewal)",
177 |         "non-refundable\\s+(?:fees|charges|payments)"
178 |     ],
179 |     "keywords": [
180 |         "automatically renew",
181 |         "recurring subscription",
182 |         "cancel at least 24 hours prior",
183 |         "non-refundable fees",
184 |         "charge without further authorization"
185 |     ],
186 |     "explanation": "Your subscription will silently renew and charge your card automatically unless manually terminate",
187 |     "why_it_matters": "Users frequently get trapped in zombie billing cycles for unused services due to obscure cancel",
188 |     "recommendation": "Set calendar alerts 3-5 days ahead of billing cycles or use virtual credit cards with spend lim",
189 | },
190 | {
191 |     "id": "account_termination",
192 |     "name": "Account Termination",
193 |     "severity": "MEDIUM",
194 |     "weight": 8,
195 |     "color": "#ffb700",
196 |     "description": "Suspension or deletion of user accounts and content without prior notice, justification, or refund",
197 |     "patterns": [
198 |         "terminate\\s+(?:or\\s+suspend\\s+)?(?:your\\s+account|access)\\s+at\\s+(?:any\\s+time|our\\s+sole\\s+discretion",
199 |         "(?:terminate|suspend)\\s+without\\s+(?:prior\\s+)?notice\\s+(?:or\\s+liability)?",
200 |         "for\\s+any\\s+reason\\s+or\\s+(?:for\\s+)?no\\s+reason",
201 |         "forfeit(?:ure)?\\s+(?:of\\s+)?(?:all|any)\\s+(?:content|credits|funds|data)\\s+upon\\s+termination",
202 |         "no\\s+obligation\\s+to\\s+refund\\s+(?:unused\\s+)?fees"
203 |     ],
204 |     "keywords": [
205 |         "terminate at our sole discretion",
206 |         "suspend without notice",

```

```

206 |         "for any reason or no reason",
207 |         "forfeit content upon termination"
208 |     ],
209 |     "explanation": "The provider can ban your account, delete your assets, and block your access at will without explanation.",
210 |     "why_it_matters": "You risk sudden loss of work, purchases, access to historical records, or communication channels.",
211 |     "recommendation": "Regularly export and download backups of your cloud files, projects, and transactions."
212 | },
213 | {
214 |     "id": "third_party_access",
215 |     "name": "Third-Party Access",
216 |     "severity": "HIGH",
217 |     "weight": 11,
218 |     "color": "#ff3366",
219 |     "description": "Permitting outside integrations, ad networks, or vendors to access user accounts and content.",
220 |     "patterns": [
221 |         "third-party\\s+(?:service\\s+providers|vendors|contractors|integrations)\\s+may\\s+access",
222 |         "not\\s+responsible\\s+for\\s+(?:the\\s+privacy\\s+practices|actions|content)\\s+of\\s+third\\s+parties",
223 |         "allow(?:s)?\\s+third\\s+parties\\s+to\\s+(?:serve\\s+ads|collect\\s+data|place\\s+cookies)",
224 |         "integrated\\s+third-party\\s+services\\s+governed\\s+by\\s+their\\s+own\\s+terms",
225 |         "share\\s+with\\s+sub-processors\\s+without\\s+notification"
226 |     ],
227 |     "keywords": [
228 |         "third-party service providers may access",
229 |         "not responsible for third-party privacy",
230 |         "third parties to serve ads and collect",
231 |         "governed by third-party policies"
232 |     ],
233 |     "explanation": "External software vendors or third-party plug-ins can interface directly with your account or reach your data.",
234 |     "why_it_matters": "Third-party supply chain vulnerabilities are a leading vector for corporate data leaks and identity theft.",
235 |     "recommendation": "Audit and disconnect unused connected apps and third-party account authorizations."
236 | },
237 | {
238 |     "id": "data_retention",
239 |     "name": "Indefinite Data Retention",
240 |     "severity": "MEDIUM",
241 |     "weight": 7,
242 |     "color": "#ffb700",
243 |     "description": "Retaining personal data, messages, or backups indefinitely even after account deletion.",
244 |     "patterns": [
245 |         "retain(?:s|ed|ing)?\\s+(?:your\\s+)?(?:data|information|content)\\s+(?:indefinitely|for\\s+an\\s+unlimited\\s+period)",
246 |         "may\\s+retain\\s+(?:archived|backup|residual)\\s+copies\\s+even\\s+after\\s+(?:you\\s+delete|account\\s+closure)",
247 |         "no\\s+obligation\\s+to\\s+delete\\s+(?:your\\s+)?(?:information|data)",
248 |         "retain\\s+and\\s+use\\s+your\\s+information\\s+to\\s+comply\\s+with\\s+legal\\s+obligations.*resolve\\s+disputes"
249 |     ],
250 |     "keywords": [
251 |         "retain data indefinitely",
252 |         "retain backup copies even after deletion",
253 |         "as long as we deem necessary",
254 |         "no obligation to delete"
255 |     ],
256 |     "explanation": "Deleting your account does not guarantee deletion of your personal records, which may be archived or retained.",
257 |     "why_it_matters": "Zombie data retained forever creates a permanent target for future database breaches or subpoenas.",
258 |     "recommendation": "Exercise formal GDPR/CCPA 'Right to Erasure' requests where legally applicable."
259 | },
260 | {
261 |     "id": "location_tracking",
262 |     "name": "Location Tracking",
263 |     "severity": "HIGH",
264 |     "weight": 12,
265 |     "color": "#ff3366",
266 |     "description": "Harvesting precise GPS coordinates, Wi-Fi triangulation, cell tower IDs, or continuous background location data.",
267 |     "patterns": [
268 |         "collect(?:s|ed|ing)?\\s+(?:precise|exact|real-time|continuous)?\\s*(?:location|geolocation|gps)\\s+data",
269 |         "location\\s+(?:even\\s+when\\s+the\\s+app\\s+is\\s+closed|in\\s+the\\s+background)",
270 |         "track(?:s|ed|ing)?\\s+(?:your\\s+physical\\s+location|whereabouts|movements)",
271 |         "derive\\s+location\\s+from\\s+(?:ip\\s+address|wi-fi\\s+access\\s+points|cell\\s+towers|bluetooth)"
272 |     ],
273 |     "keywords": [
274 |         "precise geolocation data",

```

```

275 |         "location in the background",
276 |         "track your physical location",
277 |         "real-time location tracking"
278 |     ],
279 |     "explanation": "The app tracks or approximates your real-world geographic coordinates, potentially even when running in the background.",
280 |     "why_it_matters": "Real-time location records expose your residence, workplace, daily routines, and intimate personal information.",
281 |     "recommendation": "Set system location permissions to 'Only While Using App' or disable location entirely if not needed.",
282 | },
283 | {
284 |     "id": "advertising_profiling",
285 |     "name": "Advertising & Profiling",
286 |     "severity": "MEDIUM",
287 |     "weight": 8,
288 |     "color": "#ffb700",
289 |     "description": "Targeted behavioral ads, algorithmic psychological profiling, and interest-based segmentation.",
290 |     "patterns": [
291 |         "personalized\\s+(?:advertisements|ads|marketing|promotions)",
292 |         "interest-based\\s+advertising",
293 |         "create\\s+(?:a\\s+)?profile\\s+about\\s+you",
294 |         "target(?:ed)?\\s+advertising\\s+based\\s+on\\s+your\\s+(?:preferences|activity|browsing)",
295 |         "deliver\\s+relevant\\s+ads\\s+by\\s+analyzing\\s+your\\s+(?:interests|profile|demographics)"
296 |     ],
297 |     "keywords": [
298 |         "personalized advertisements",
299 |         "interest-based advertising",
300 |         "create a profile about you",
301 |         "targeted advertising based on"
302 |     ],
303 |     "explanation": "Your activity, searches, and demographic attributes are used to assemble an advertising profile for targeted marketing.",
304 |     "why_it_matters": "Targeted profiling can exploit behavioral vulnerabilities, shape opinions, and expose your private information.",
305 |     "recommendation": "Opt out of personalized advertising within service settings and reset your advertising ID."
306 | },
307 | {
308 |     "id": "user_rights",
309 |     "name": "User Rights Surrender",
310 |     "severity": "HIGH",
311 |     "weight": 11,
312 |     "color": "#ff3366",
313 |     "description": "Surrendering intellectual property rights, granting irrevocable global content licenses, or moral rights waivers.",
314 |     "patterns": [
315 |         "grant(?:s)?\\s+(?:us\\s+)?(?:a\\s+)?(?:worldwide|perpetual|irrevocable|royalty-free|unrestricted)\\s+(?:license|agreement)",
316 |         "waive(?:s|d)?\\s+(?:all\\s+)?moral\\s+rights",
317 |         "you\\s+assign\\s+(?:all\\s+)?(?:ownership|intellectual\\s+property|rights)\\s+to\\s+us",
318 |         "we\\s+own\\s+all\\s+(?:feedback|submissions|content|ideas)\\s+provided\\s+by\\s+you",
319 |         "right\\s+to\\s+sublicense\\s+and\\s+commercialize\\s+your\\s+content"
320 |     ],
321 |     "keywords": [
322 |         "perpetual irrevocable royalty-free license",
323 |         "worldwide license to use and modify",
324 |         "waive moral rights",
325 |         "assign all intellectual property",
326 |         "sublicense and commercialize your content"
327 |     ],
328 |     "explanation": "You grant the company a permanent, global, royalty-free license to use, publish, monetize, or adapt your content.",
329 |     "why_it_matters": "You may lose commercial ownership or credit for creative works, files, or original designs uploaded to the app.",
330 |     "recommendation": "Avoid storing or creating proprietary work or copyrighted media on platforms demanding broad IP rights.",
331 | },
332 | {
333 |     "id": "cancellation_restrictions",
334 |     "name": "Cancellation Restrictions",
335 |     "severity": "LOW",
336 |     "weight": 5,
337 |     "color": "#38bdf8",
338 |     "description": "Complex hurdle-filled cancellation procedures, early termination penalties, or restricted contact during cancellation.",
339 |     "patterns": [
340 |         "cancel(?:ation)?\\s+(?:must\\s+be\\s+made\\s+by|only\\s+by)\\s+(?:phone|written\\s+notice|certified\\s+mail)",
341 |         "early\\s+termination\\s+fee(?:s)?",
342 |         "penalt(?:y|ies)\\s+for\\s+(?:early\\s+)?cancellation",
343 |         "must\\s+provide\\s+(?:at\\s+least\\s+)?(?:30|60|90)\\s+days\\s+(?:prior\\s+)?notice\\s+to\\s+cancel",

```

```
344 |         "cannot\\s+be\\s+cancelled\\s+online"
345 |     ],
346 |     "keywords": [
347 |         "early termination fee",
348 |         "cancel only by phone",
349 |         "penalty for cancellation",
350 |         "certified mail to cancel"
351 |     ],
352 |     "explanation": "The provider imposes restrictive hurdles or fees to make terminating your account or contract deli
353 |     "why_it_matters": "Dark pattern cancellation flows prevent easy departure, forcing unneeded payments or unnecessar
354 |     "recommendation": "Check the exact cancellation procedures prior to providing payment details or signing long agre
355 | }
356 | ]
357 | }
```

File: backend/sample_policies.json

Size: 7003 bytes

```
1 | {
2 |   "samples": [
3 |     {
4 |       "id": "high_risk_social",
5 |       "title": "HyperNet Social App Terms & Privacy Policy",
6 |       "expected_risk": "HIGH RISK",
7 |       "description": "A consumer mobile application packed with invasive tracking, cross-site telemetry, binding arbitra
8 |       "content": "Welcome to HyperNet. By accessing or using our services, mobile application, or websites, you agree to
9 |     },
10 |    {
11 |      "id": "moderate_risk_saas",
12 |      "title": "NovaDesk Cloud Workspace Agreement",
13 |      "expected_risk": "CAUTION",
14 |      "description": "A business SaaS platform with commercial auto-renewal, standard liability caps, unilateral mainten
15 |      "content": "NovaDesk Cloud Services Master Subscription Agreement.\n\n1. SCOPE AND SERVICES\nNovaDesk provides onl
16 |    },
17 |    {
18 |      "id": "safe_open_source",
19 |      "title": "AegisGuard Privacy-First Client Terms",
20 |      "expected_risk": "SAFE",
21 |      "description": "An open-source, local-first utility with end-to-end encryption, zero tracking, transparent cancell
22 |      "content": "AegisGuard Terms of Use and Privacy Charter.\n\n1. ZERO KNOWLEDGE ARCHITECTURE\nAegisGuard operates as
23 |    }
24 |  ]
25 | }
```

File: backend/requirements.txt

Size: 54 bytes

```
1 | Flask>=3.0.0
2 | flask-cors>=4.0.0
3 | beautifulsoup4>=4.12.0
```

File: README.md

Size: 11291 bytes

```

1 | # T&C RED FLAG SCANNER
2 | &gt; ***"Before you click Agree, know what you're agreeing to."**
3 | &gt; **Team: CorruptX** | Cybersecurity Hackathon Project
4 |
5 | ---
6 |
7 | ## 1. Project Overview
8 |
9 | **T&C Red Flag Scanner** is an AI-powered cybersecurity application designed to analyze Terms & Conditions, Privacy Policies, and End User License Agreements (EULAs).
10 |
11 | Built with a **100% offline-first architecture**, the application guarantees zero third-party cloud data leakage—meaning all analysis is performed locally on your device.
12 |
13 | ---
14 |
15 | ## 2. Key Features
16 |
17 | - **Automated Clause Segmentation**: Splits complex legal prose into discrete clauses while preserving decimal sections.
18 | - **14-Category Risk Taxonomy**: Detects patterns across critical vectors including:
19 |   1. Data Sharing & Monetization
20 |   2. Tracking & Hardware Telemetry (Fingerprinting, Session Replay)
21 |   3. Excessive Data Collection (Sensors, Contacts, Biometrics)
22 |   4. Mandatory Binding Arbitration & Class Action Waivers
23 |   5. Unilateral Policy Modifications without Notice
24 |   6. Broad Liability Waivers & Warranty Disclaimers
25 |   7. Auto-Renewal & Recurring Billing Traps
26 |   8. Unilateral Account Termination & Content Forfeiture
27 |   9. Third-Party Vendor Access
28 |   10. Indefinite Data Retention
29 |   11. Real-Time Location & Background Tracking
30 |   12. Behavioral Advertising & Demographic Profiling
31 |   13. Content Licensing & User Rights Surrender
32 |   14. Restrictive Cancellation Penalties
33 | - **Negation & Privacy Protection Detection**: Intelligently distinguishes between risky clauses and positive privacy protections.
34 | - **Deterministic 0-100 Risk Score**: Computes a mathematically grounded exposure score and assigns clear risk tiers:
35 |   - `SAFE` (0-30)
36 |   - `CAUTION` (31-60)
37 |   - `HIGH RISK` (61-100)
38 | - **Interactive Policy Highlighting & Inspector**: Renders full document text with inline color-coded risk spans (`.highlight` classes).
39 | - **Client-Side Offline Engine**: Bundled with a dual-execution architecture. If the Python backend is running, it uses the Python engine; otherwise, it falls back to a lightweight JavaScript engine.
40 | - **Offline History & Reporting**: Stores previous scans locally using browser `localStorage` with view, delete, and export capabilities.
41 |
42 | ---
43 |
44 | ## 3. Technology Stack
45 |
46 | - **Backend**: Python 3 (Flask, Flask-CORS, BeautifulSoup4)
47 | - **Analysis Engine**: Rule-based NLP pattern matcher and weighted scoring matrix (`analyzer.py` + `risk_rules.json`)
48 | - **Frontend**: Modern Vanilla HTML5, CSS3 (Cyberpunk/Dark Cybersecurity Theme, Glassmorphism, CSS Grid), Vanilla ES6+ JavaScript
49 | - **Icons**: Local inline SVGs (zero CDN dependency)
50 | - **Storage**: Client-side `localStorage` and `sessionStorage`
51 |
52 | ---
53 |
54 | ## 4. Folder Structure
55 |
56 | ```
57 | tc-red-flag-scanner/
58 | ├── backend/
59 | │   ├── app.py           # Flask API server & static router
60 | │   ├── analyzer.py      # Core NLP rule engine & risk scorer
61 | │   ├── risk_rules.json  # Cybersecurity risk taxonomy (14 categories)
62 | │   ├── sample_policies.json # Preloaded hackathon demo policies
63 | │   ├── test_analyzer.py # Automated test verification script
64 | │   └── requirements.txt  # Python dependencies
65 | ├── frontend/
66 | └── ...
67 | ```

```

```

68 | ■ ■■■ index.html          # Landing page with hero, mockup & team info
69 | ■ ■■■ scanner.html       # Main analysis console & scanning modal
70 | ■ ■■■ results.html       # Results dashboard with circular risk meter
71 | ■ ■■■ history.html       # Scan history management
72 | ■ ■■■ about.html        # Mission, cybersecurity context & team details
73 | ■
74 | ■ ■■■ css/
75 | ■ ■ ■■■ style.css        # Global theme, design system & typography
76 | ■ ■ ■■■ scanner.css     # Textarea, dropzone, and radar scanning overlay
77 | ■ ■ ■■■ results.css     # Circular gauge, metrics, cards & highlighter
78 | ■ ■ ■■■ responsive.css  # Mobile, tablet, and desktop breakpoints
79 | ■
80 | ■ ■■■ js/
81 | ■ ■■■ rules_data.js     # Offline bundled taxonomy
82 | ■ ■■■ sample_data.js    # Offline bundled demo policies
83 | ■ ■■■ client_analyzer.js # Standalone client-side NLP engine
84 | ■ ■■■ app.js            # Shared state, notifications & localStorage
85 | ■ ■■■ scanner.js        # Scanner page controller & animation
86 | ■ ■■■ results.js        # Circular meter, highlighter & inspector
87 | ■ ■■■ history.js        # History manager & report exporter
88 | ■
89 | ■■■ README.md           # Complete technical documentation
90 | ■■■ .gitignore          # Git ignore rules
91 | ---
92 |
93 | ---
94 |
95 | ## 5. Installation & Setup
96 |
97 | ### Prerequisites
98 | - Python 3.9+ (Python 3.10, 3.11, 3.12, 3.14 supported)
99 | - Any modern web browser (Chrome, Edge, Firefox, Brave, Safari)
100 |
101 | ### Option A: Running with Local Flask Server (Recommended)
102 |
103 | 1. Open PowerShell / Terminal in the `tc-red-flag-scanner/backend` directory:
104 |    ```bash
105 |    cd "backend"
106 |    ```
107 |
108 | 2. Install dependencies:
109 |    ```bash
110 |    pip install -r requirements.txt
111 |    ```
112 |
113 | 3. Start the Flask application:
114 |    ```bash
115 |    python app.py
116 |    ```
117 |
118 | 4. Open your browser and navigate to:
119 |    ```
120 |    http://127.0.0.1:5000
121 |    ```
122 |
123 | ### Option B: Running 100% Offline Standalone (Zero Server Required)
124 |
125 | You can run the entire application completely without a web server or Python:
126 | 1. Navigate to the `frontend/` folder.
127 | 2. Double-click `index.html` or open it directly in any browser (`file:///.../frontend/index.html`).
128 | 3. The built-in client-side analyzer (`client_analyzer.js`) automatically activates and executes scans locally in your browser.
129 |
130 | ---
131 |
132 | ## 6. REST API Endpoints
133 |
134 | | Method | Endpoint | Description |
135 | |---|---|---|
136 | | `GET` | `/api/health` | Service health status and active rule categories count |

```

```

137 | | `GET` | `/api/rules` | Complete active risk rules taxonomy (`risk_rules.json`) |
138 | | `GET` | `/api/demo` | Curated sample policies for instant hackathon evaluation |
139 | | `POST` | `/api/scan` | Main analysis endpoint. Accepts JSON or multipart file uploads |
140 |
141 | ### Example Scan Request:
142 | ```bash
143 | curl -X POST http://127.0.0.1:5000/api/scan \
144 |   -H "Content-Type: application/json" \
145 |   -d '{"text": "We reserve the right to share your personal information with third parties. All disputes shall be resolved through arbitration."}'
146 | ```
147 |
148 | ### Example Response:
149 | ```json
150 | {
151 |   "document_name": "Test Contract",
152 |   "analyzed_at": "2026-09-22T23:15:30.123456",
153 |   "risk_score": 68,
154 |   "risk_level": "HIGH RISK",
155 |   "clauses_analyzed": 2,
156 |   "red_flags": 2,
157 |   "severity_counts": {
158 |     "HIGH": 2,
159 |     "MEDIUM": 0,
160 |     "LOW": 0
161 |   },
162 |   "categories": [
163 |     { "category_id": "data_sharing", "category_name": "Data Sharing", "severity": "HIGH", "count": 1 },
164 |     { "category_id": "arbitration", "category_name": "Mandatory Arbitration", "severity": "HIGH", "count": 1 }
165 |   ],
166 |   "flagged_clauses": [
167 |     {
168 |       "index": 0,
169 |       "text": "We reserve the right to share your personal information with third parties.",
170 |       "category_name": "Data Sharing",
171 |       "severity": "HIGH",
172 |       "explanation": "This clause allows the service to share, distribute, or sell your personal details with third-parties.",
173 |       "why_it_matters": "Once your information is handed over to outside parties, you lose direct control...",
174 |       "recommendation": "Check if an opt-out mechanism is available or avoid submitting sensitive confidential data."
175 |     }
176 |   ]
177 | }
178 | ```
179 |
180 | ---
181 |
182 | ## 7. Risk Scoring Methodology
183 |
184 | The risk score is calculated using a deterministic algorithm:
185 | 1. Clause Weighting:
186 |   - `HIGH` severity clauses: 16 points each
187 |   - `MEDIUM` severity clauses: 8 points each
188 |   - `LOW` severity clauses: 4 points each
189 | 2. Exposure Density Factor:
190 |   - Compares the ratio of flagged clauses to total document clauses:
191 |     
$$\text{Density Bonus} = \min\left(20, \left\lfloor \frac{\text{Flagged Clauses}}{\text{Total Clauses}} \right\rfloor \times 35\right)$$

192 | 3. Composite Normalization:
193 |     
$$\text{Score} = \min\left(100, \left(\text{Weighted Sum} \times 0.82\right) + \text{Density Bonus}\right)$$

194 | 4. Safety Threshold Guardrails:
195 |   - If  $\geq 3$  `HIGH` severity red flags are detected, the minimum score floor is bounded to $68$ (`HIGH RISK`).
196 |   - If  $\geq 1$  `HIGH` severity red flag is detected, score floor is bounded to $42$ (`CAUTION`).
197 |
198 | ---
199 |
200 | ## 8. Hackathon Demo Instructions
201 |
202 | 1. Launch the application (`python backend/app.py` or open `frontend/index.html`).
203 | 2. Click "Scan a Document" or navigate to `/scanner`.
204 | 3. Click the "Load Demo Policy" button in the toolbar. This loads the realistic "HyperNet Social App Terms", featuring various red flags.
205 | 4. Click "Start Security Scan".

```

```

206 | 5. Observe the 4-step animated scanning overlay (`Extracting clauses... -&gt; Classifying risks... -&gt; Calculating ris
207 | 6. View the Results Dashboard:
208 |     - Inspect the circular risk score gauge (~100/100 HIGH RISK).
209 |     - Review detected red flag cards with plain-English translations and recommendations.
210 |     - Switch to the "Interactive Policy Analysis" tab and click on highlighted spans to test the real-time Clause I
211 |     - Check the "Risk Breakdown" tab for category rankings.
212 | 7. Click "History" in the top navigation to verify persistent local logging.
213 |
214 | ---
215 |
216 | ## 9. Future AI Architecture & Roadmap
217 |
218 | ```
219 | Document / Policy Input
220 |     ↓
221 | Text Normalization & Cleaning
222 |     ↓
223 | Intelligent Clause Segmentation
224 |     ↓
225 | LLM / NLP Embedding & Zero-Shot Classification
226 | (e.g., Local Llama 3 / Gemma 2B via ONNX Runtime / Ollama)
227 |     ↓
228 | Cybersecurity Risk Taxonomy Alignment
229 |     ↓
230 | Plain-English Simplification & Defense Synthesis
231 |     ↓
232 | Interactive Results Dashboard
233 | ```
234 |
235 | The modular design of `ClauseAnalyzer` ensures that swapping regex rules with a HuggingFace Transformer pipeline or loca
236 |
237 | ---
238 |
239 | ## 10. Important Product Disclaimer
240 |
241 | &gt; Disclaimer: This tool provides automated risk analysis for informational and educational cybersecurity purposes
242 |
243 | ---
244 |
245 | ## 11. Team Information
246 |
247 | Team CorruptX
248 | - Vyom Khanna — Team Lead (Architecture & Full-Stack Engineering)
249 | - Nalin Shakher — Team Member (NLP Engine, Risk Taxonomy & Offline Pipeline)
250 | - Aayush Verma — Team Member (Cybersecurity UI/UX & Interactive Clause Inspector)

```